

Merged research report for the Surreal project

Polish Models of Omnific Arithmetic

Polish Presburger models for Glazer’s Question 2,
and Borel presentations and support barriers

Built from three manuscripts dated October 3, 2026

Abstract

Elliot Glazer’s *A Topological Tennenbaum Theorem* shows that an uncountable Polish model of $Q + I\text{Open}$ cannot have continuous addition and Borel multiplication, and asks (Question 2) whether some uncountable Polish space supports a model of Presburger arithmetic with continuous addition. Part I answers that question, as printed, affirmatively. The nonnegative cone $M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N})$ of the lexicographic group $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$, with the usual topology on the real coordinate and the discrete one on the integer coordinate, is a perfect, locally compact, uncountable Polish model of the full theory $\text{Th}(\mathbb{N}; 0, 1, +, <)$ whose addition is continuous. Replacing $\mathbb{R}$ by the lexicographically ordered group $\mathbb{Q}^{\mathbb{N}}$ with discrete coordinates gives a model homeomorphic to Baire space, with Type-2 computable addition and division by standard integers. Every definable relation is Δ_2^0 , and this is optimal for the order; the truncated subtraction and the predecessor are discontinuous exactly on explicit loci; no unital semiring multiplication exists on either model; the self-embeddings are classified (positive dilations in the real case, row-finite matrices with increasing positive pivots in the Baire case), and both models embed additively in Conway’s omnific integers $\mathbf{Oz}$ as $r\omega + n$ and $\sum_j a_j \omega^{1/(j+1)} + n$. Part I also isolates the coset-invariance hypothesis needed to carry the repository’s integer Cooper lemmas to nonstandard groups. Glazer himself speculated that the answer to Question 2 would be negative; the affirmative answer is given for the question as printed, and its priority has not been established.

Part II is the other side of the same boundary. For a countable linear order L , its well-ordered subsets form a Borel set exactly when L is scattered, and a Π_1^1 -complete set otherwise; the same condition decides whether a full real Hahn field $\mathbb{R}((t^{\Gamma}))$ has a standard Borel parametrization with Borel coefficient maps. The finite Puiseux principal-part ring $\mathcal{A}_{\text{fin}}$, of which $M_{\mathbb{R}}$ is one stratum, satisfies open induction and has a Polish presentation on which addition and multiplication are continuous off explicit closed nowhere dense cancellation loci; by Glazer’s Corollary 1 no Borel-compatible Polish recoding makes its addition continuous everywhere.

The report merges three independently written manuscripts; it states which source proved each result, prints shared results once, keeps every source’s limitations, and cites the collection’s earlier results instead of reprinting them as new. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized.

Sources: manuscripts 05 (the base text of Part I), 04 (Part I, marked [04]) and 03 (Part II) of batch 86, arrival commit 31fdc6571, placement commit 3d2177df4; pinned to repository commits a11efab09 (05), 3ad5f878c (04) and c5b4881fa (03). Their titles and contributions are in [section 1.2](#); the provenance of every section is in [section E](#), and the non-claims of every source are collected in [section F](#).

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1 Scope, sources and conventions

This report collects three manuscripts written on the same day, October 3, 2026, in answer to one research brief: where Elliot Glazer’s topological Tennenbaum programme meets the repository’s omnific arithmetic. Their common spine is the question which arithmetic operations of a Polish or standard Borel presentation of a substructure of $\mathbf{Oz}$ can be continuous. [Part I](#) (label prefix `pma:pr:`) constructs Polish models of Presburger arithmetic with continuous addition and studies them; its base text is manuscript 05, and the further results of manuscript 04 are printed as marked sections. [Part II](#) (label prefix `pma:bor:`) is manuscript 03: Borel presentations of Hahn fields, support barriers, and the finite principal-part ring on which Glazer’s theorem forbids continuous addition. The two parts meet in [section 26](#).

1.1 Glazer’s paper and its two questions

Glazer’s paper [1] is an eight-page preprint, dated November 20, 2023 on its first page and submitted to arXiv on November 22, 2023 as arXiv:2311.13699. On October 3, 2026 the arXiv record listed one version (v1) and no journal reference. It answers negatively a question of Enayat, Hamkins and Wcisło [2]: no model of Peano arithmetic on a perfect Polish space has continuous addition and multiplication (its Theorem 1). Its Theorem 2 says that if $(M, +, \cdot) \models Q + I\Delta_0 + \text{Exp}$ lives on an uncountable Polish space with Borel addition and multiplication, then neither operation is continuous. Glazer notes that the axiom Exp is used only to obtain the function $x \mapsto 2^x$, and only for the discontinuity of multiplication; this gives his Corollary 1: *there is no model of $Q + \text{IOpen}$ on an uncountable Polish space in which addition is continuous and multiplication is Borel*. The paper ends with two questions. Question 1 asks whether Theorem 2 is provable in ATR_0 . Question 2 is quoted in [section 2](#): does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

Glazer’s speculation. Immediately after Question 2, Glazer writes that he speculates that an affirmative answer to Question 1 “would lead to a negative answer to Question 2” [1, p. 8]. His reason is an analogy: removing the exponential function from the proof that multiplication is discontinuous would parallel removing multiplication from a proof that addition is discontinuous. Neither source of [Part I](#) mentions this sentence; it is recorded here because [Part I](#) goes the other way.

How this report reads it [write]. [Part I](#) answers Question 2 *as printed* affirmatively ([theorem 3.1](#)). The question asks only for an uncountable Polish space, a model of Presburger arithmetic and continuous addition; it does not ask that the topology be the order topology, and it mentions no multiplication. The speculation is a heuristic about proofs, not a stated implication, and nothing here bears on Question 1, which stays open; no conclusion about Question 1 is drawn from the affirmative answer. Three facts locate the difference. First, the affirmative models are not an artefact of a missing definability condition: every definable relation in them is Δ_2^0 ([theorem 7.1](#)), so the order and every other definable relation are Borel, as in the Borel models of Glazer’s theorems. Second, what fails in them is multiplication: no unital semiring multiplication exists on them at all ([theorem 10.4](#)), so Corollary 1 does not apply. Third, on a ring that does carry Borel multiplication and open induction, Corollary 1 does forbid continuous addition under every Borel-compatible Polish recoding ([theorem 25.2](#)), and [section 26](#) shows that the real model of [Part I](#) sits inside that ring. We do not know

whether Glazer intended Question 2 with a further hypothesis, for instance an expansion to a model of a stronger theory, under which his speculation could hold; that would be a different question.

Priority. Priority of the affirmative answer is **not established**. Source 05 searched for the exact question and combinations of its terms and found no earlier answer, and says that this does not prove that none exists; source 04 says that historical priority has not been established. The intake re-checked only that the arXiv record has one version and no journal reference (October 3, 2026). The answer is an elementary construction; an earlier publication of it, or a published variant of the question, may exist.

1.2 Three manuscripts, one report

Source	Manuscript (delivered pages)	Pin	Printed as
05	<i>Polish Presburger Arithmetic inside the Omnific Integers: An affirmative construction for Glazer’s Question 2, effective Baire-space models, and structural obstructions</i> (24, A4); archive <code>polish_presburger_glazer</code>	<code>a11efab09</code>	base text of Part I ; worked examples in section A
04	<i>Polish Presburger Arithmetic: A positive construction for Glazer’s question, an effective Baire-space model, and exact regularity boundaries</i> (25, letter); archive <code>Polish_Presburger_Glazer_ProveIt</code>	<code>3ad5f878c</code>	Part I , every statement marked [04]; second proofs; sections B and C
03	<i>Borel Presentations and Support Barriers in Omnific Arithmetic: Scattered exponent orders, Hahn fields, and the exact discontinuities of finite normal forms</i> (23, letter); archive <code>glazer_proveit_research</code>	<code>c5b4881fa</code>	Part II and section D

The numbers are the manuscript numbers of batch 86 of the repository’s intake (manuscripts 01 and 02 of that batch are unrelated OEIS reports). All three pins are ancestors of the placement commit, and the repository files the sources inspected are unchanged between their pins and the placement.

Why one report. 04 and 05 prove the same main theorem independently: they share 0.6% of their word 8-grams, essentially only Glazer’s quoted question and his bibliography entry, and have different pins, notation and supplementary theorems. 03 answers the same brief from the other side: its finite principal-part ring contains the real model of 04 and 05 as one stratum ([section 26](#)), and its Theorem 10.2 ([theorem 25.2](#)) is Glazer’s obstruction on that ring. No earlier report of the collection names Glazer, his questions, Polish models of Presburger arithmetic, or Borel presentations of Hahn fields.

Why base 05. On the shared theorems 05 has the weaker hypotheses: its Polish criterion ([theorem 5.2](#)) is an equivalence for every Polish group topology on the coefficient group, where 04’s transfer theorem ([theorem 5.4](#)) is a sufficient condition under the stronger hypothesis that the positive cone be open and division continuous; its Δ_2^0 theorem holds

under the same weaker hypothesis; and it treats the models explicitly as fragments of **Oz**. Results of 04 not in 05 are printed in full and marked [04]: continuity of fixed division under its hypothesis, the general closed-order obstruction and the invariant-metric obstruction, the reduction of all unital additive maps to the divisible part, the matrix classification of the self-embeddings of the Baire model with automatic continuity, and the Laurent-series model with its growth-order obstruction. Where 04 proves a result of 05 by a different argument, its proof is printed as a marked second proof.

Reading conventions. In [Part I](#), unmarked text is source 05's; text from source 04 is marked [04] (statements in their titles, inserted passages in the sentence that introduces them). In [Part II](#) all unmarked text is source 03's. Text written for this report is marked [merge] (a mathematical observation combining the sources, with proof) or [write] (editorial notes: provenance, citations of the collection, corrections). Phrases such as "this manuscript" or "this article" inside a source's text refer to that source. References to the collection's other reports give their labels in typewriter type, for example `odg:thm:floor`; these labels are not hyperlinks.

Labels. Every label carries the prefix `pma:`. Source 05's labels are `pma:pr:<label>`, source 03's are `pma:bor:<label>`. Source 04's labels are `pma:pr:<label>`, except the ten that coincide with labels of 05 (`eq:cone`, `eq:division`, `sec:algebra`, `sec:baire`, `sec:embeddings`, `sec:formalization`, `sec:multiplication`, `sec:questions`, `thm:baire`, `thm:qe`), which take the sub-prefix `pma:pr:s04:`. Of these, `eq:cone` and `eq:division` denote displays identical to 05's and are printed once, under 05's labels; source 04's displays `eq:divisionformula`, `eq:monuslocus`, `eq:surrealR` and `eq:surrealQ` are likewise printed once as 05's (4.2), (8.2), (9.1) and (9.2), and its Definition 2.1 (`def:split`) is 05's setup in [section 3](#). Where 04 states a theorem that 05 also states, 04's label is a second label of 05's statement (for example `pma:pr:thm:monuslocus` is [theorem 8.3](#)). Labels written for the merge have the bare prefix `pma:` or a part prefix.

1.3 Notation

The three sources use clashing symbols. The table fixes the convention of this report; renamed symbols are marked $\rightarrow$.

Notion	This report	In the sources
Presburger arithmetic	$\text{Th}(\mathbb{N}; 0, 1, +, <)$	05: the same; 04: $\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +)$, kept in 04's text. The two are definitionally equivalent: $x \leq y$ is $\exists z (x + z = y)$.
Lexicographic group, coefficient group D dominant	$D \xrightarrow{\times} \mathbb{Z} = G_D$	05: the same; 04: $D \oplus_{\text{lex}} \mathbb{Z} = G(D) \rightarrow G_D$
Nonnegative cone	$M_D; M_E$ for a subgroup E	04: $M(D), M(E) \rightarrow M_D, M_E$
Baire coefficient group	$D_B = \mathbb{Q}^{\mathbb{N}}$, lexicographic, each $\mathbb{Q}$ discrete	05: the same; 04: $D_{\omega} \rightarrow D_B$ (the same group, order and topology)
Baire model	$M_B = (D_B \xrightarrow{\times} \mathbb{Z})_{\geq 0}$	04: $M_{\omega} \rightarrow M_B$ (the same model)
Laurent model (04)	$M_L = M_{D_L}, D_L = \mathbb{Q}((t))$	04 only
Baire space	$\mathbb{N}^{\mathbb{N}}$	04's macro $\backslash B = \mathbb{N}^{\mathbb{N}}$; 03's $\backslash B = \mathcal{B}$ is defined but never used

Notion	This report	In the sources
Homeomorphisms with Baire space	05: $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$ (tag code); 04: $\Psi : M_B \rightarrow \mathbb{N}^{\mathbb{N}}$, charts $\Psi_{\geq}$, $\Psi_{>}$, integer code $\zeta : \mathbb{Z} \rightarrow \mathbb{N}$	04: $H, H_{\geq}, H_{>}, h \rightarrow \Psi, \Psi_{\geq}, \Psi_{>}, \zeta$. The two codings are different maps.
Lexicographic symbol	$\overrightarrow{X}$ (05); $\mathbb{Z}_{\text{lex}}^d$ for lexicographic powers (03)	04's subscript lex in $\oplus_{\text{lex}} \rightarrow \overrightarrow{X}$
Division by a standard m	$q_m(x) \in M$, $r_m(x) \in \{0, \dots, m-1\}$ (05); $Q_m(x) \in M$, $R_m(x) = (0, z \bmod m) \in M$ (04)	kept; R_m is r_m viewed as an element of the model
Congruence predicates	$D_m(t) \iff t \in mG$ (05); $u \equiv_m v \iff u - v \in mG$ (04)	kept; $u \equiv_m v \iff D_m(u - v)$
Coset invariance	" mG -invariant" (05) = " m -residue invariant" (04)	the same notion
Cofinal infinite element	"cofinal infinite additive unit" (05) = "arithmetically infinite order unit" (04)	the same notion; not a multiplicative unit
Truncated subtraction	$T(x, y)$ (05), $x \dot{-} y$ (04)	the same operation; $\text{Disc}(f)$ is the discontinuity set of f
Pivots, filtration (04)	$p(k), F_k$ in theorem 11.4 ; F_n in proposition 10.6	local; p elsewhere is the predecessor; 03's closed pieces are $F_{Z,k}$
Growth classes (04)	$\mathcal{A}(M)$	local to proposition 10.7 ; not the ring $\mathcal{A}_{\text{fin}}$
Finite principal-part ring (03)	$\mathcal{A}_{\text{fin}} = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{>0}} \mathbb{R}X^q$, cone $\mathcal{A}_{\text{fin}, \geq 0}$	03: $\mathcal{A}, \mathcal{A}_{\geq 0} \rightarrow \mathcal{A}_{\text{fin}}, \mathcal{A}_{\text{fin}, \geq 0}$. Not the collection's Oz , which the base of the omnific Diophantine report once called A .
Rational Hahn ring (03)	$\mathcal{H}_{\mathbb{Q}}$, (17.1)	03: $H \rightarrow \mathcal{H}_{\mathbb{Q}}$
Kleene–Brouwer embedding (03)	$\mu_{\text{KB}} : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (0, 1)$	03: $h \rightarrow \mu_{\text{KB}}$. Not the dyadic coding κ or the midpoint coding μ into $(1, 2)$ of the omnific-notations report.
Growth variable (03)	$X = t^{-1}$, read as ω	03 §2.1: increasing- t convention in sections 18 and 19 , reverse X -convention from section 22 on; $q \mapsto -q$ converts
Well-ordered subsets (03)	$\text{WO}(L) \subseteq 2^L$	not the well-orderings $\text{WO}_{\mu}(X)$ of type μ of the surreal-well-orders report
Local sets (03)	W (proof of theorem 23.2); $\mathcal{Z}$ (lemma 24.2); $D_+, D_{\times}$ (theorem 24.3)	03: $D \rightarrow W$ and $\mathcal{Z}$; $D_+, D_{\times}$ kept. In Part I , D is always the divisible coefficient group.
L	a countable linear order (03)	in 05's and 04's elimination proofs, a lower bound or a common multiple (local)
e_j, e	unit vectors of $\mathbb{Q}^{\mathbb{N}}$ (05, 04); computable enumerations $e, e_+, e_{\geq 0}$ (05, section 6)	as in 05
Omnific integers	Oz = $\mathbb{Z} \oplus \Pi$, growth notation ω^y	as in the collection's docs/NOTATION.md

1.4 Relation to the repository

The formal project `Logic/PresburgerArithmetic`. Both sources of [Part I](#) inspected this project at their pins, 05 its Lean file `Lean/PresburgerArithmetic/Cooper.lean` and 04 its Rocq file `Coq/Cooper.v`. The project decides Presburger sentences over the ordinary integers. Its periodicity lemmas assume a predicate on $\mathbb{Z}$ invariant under one standard shift; [example 4.4](#) and [proposition 4.6](#) show that this hypothesis cannot be carried verbatim to nonstandard $\mathbb{Z}$ -groups, and [lemma 4.7](#) and [theorem 4.8](#) give the model-uniform replacement. This is a porting requirement, not a defect of the integer theorems. [Section 13.5](#) names the declarations. **Nothing in this report is formalized**, and its placement in the Surreal collection, with a pointer from the formal project, confers no formal status.

The Surreal collection. Results already in the collection are cited, not reprinted as new:

- `exr:prop:division` and the paragraph after it in the `exponential-relations-over-omnific-integers` report: $\mathbf{Oz}$ is a $\mathbb{Z}$ -group, hence a model of Presburger arithmetic in its ordered-group presentation. [Proposition 4.2](#) and [theorem 4.9](#) prove the classical facts again for the split groups G_D .
- `isg:cf:main:presburger` in the `discrete-initial-subgroups-and-omnific-normalization` report: a set-sized $\mathbb{Z}$ -group has an initial realization in $\mathbf{Oz}$ if and only if it splits as $G^{\text{dv}} \xrightarrow{\sim} \mathbb{Z}$ with $G^{\text{dv}} = \bigcap_n nG$. The groups G_D of [Part I](#) are split by construction, with $\bigcap_m mG_D = D \times \{0\}$ ([corollary 4.3](#)), so that theorem applies to them [merge]; whether 05's maps $\iota_{\mathbb{R}}, \iota_{\mathbb{B}}$ are initial realizations in its sense is not checked here.
- `onot:neg:lem:kbtree`, `onot:neg:thm:pionesone` and `onot:neg:rem:secondroute` in the `omnific-notations` report: the lightface Π_1^1 -completeness of support validity. 03's [lemma 18.2](#) and [theorem 22.1](#) are their boldface counterparts (notes after those statements).
- `odg:thm:floor` in the `omnific-diophantine-geometry` report, proved in Lean for every surreal: 03's [lemma 23.1](#) is its restriction to the Puiseux field (note after it).
- `dsn:cor:openinduction` in the `definable-surreals-and-omnific-integers` report, and the Shepherdson paragraph of the `omnific-diophantine-geometry` report: the route of 03's [theorem 23.2](#) (note after it).

All these labels exist at the repository's current state and already existed at the three pins. Of them only `odg:thm:floor` is formalized (`docs/FORMALIZATION.md`, row "Full `odg:thm:floor`"; the others are recorded there as pending.

1.5 The status statements of the sources

The three manuscripts state their status as follows; these statements apply to the parts of this report taken from them.

Source 05: status and scope

The mathematical assertions designated as theorems are accompanied by proofs. The construction answers the cited question *as written*; independent bibliographic priority has not been established. Classical Presburger elimination and elementary topological-group facts are not claimed as new. This AI-assisted manuscript is not independently refereed or Lean-verified. The accompanying executable tests are finite checks, not a replacement for the proofs.

Source 04: from its abstract

The construction gives an affirmative answer to Question 2 as printed in Elliot Glazer’s 2023 paper *A Topological Tennenbaum Theorem*. The historical priority of this answer has not been established. The new results are supported here by conventional proofs and executable exact tests, not by a claimed Lean or Rocq kernel check.

Source 03: status and provenance

Status and provenance. Complete mathematical arguments are provided for the constructions and deductions, with classical inputs identified. ProveIt’s existing coanalytic support lower bound and omnific floor formula are credited, not claimed as new. The main contribution of this report is the presentation dichotomy and its algebraic consequences, together with the exact finite-support cancellation analysis. Global novelty and independent correctness review remain open; no established famous conjecture or Lean verification is claimed.

Every limitation and non-claim of the three sources is also listed, source by source, in [section F](#).

Part I**Polish Presburger models**

This part answers Glazer’s Question 2 as printed. Its base text is source 05; source 04’s further results and second proofs are marked [04]. Repository pins: 05 inspected commit `a11efab09`, 04 commit `3ad5f878c`; the files they name are unchanged at the placement.

2 The research overlap and the precise target

The relevant intersection is *topological and effective model theory of weak arithmetic*. Glazer’s work supplies a sharply formulated question; ProveIt supplies two concrete mathematical interfaces: an executable Cooper quantifier-elimination development and a normal-form construction of the omnific integers. This makes it possible to connect an external Polish topology, internal induction, and exact surreal coordinates without appealing to the repository’s unreviewed research claims.

In the final section of [1], Glazer asks:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

The source is Question 2 on page 8 of arXiv:2311.13699v1. We use precisely that formulation. The topology need not be the arithmetic order topology, and no multiplication is required. These two distinctions are essential.

Glazer’s obstruction for stronger arithmetic includes the impossibility of an uncountable Polish model of $\mathcal{Q} + I\text{Open}$ with continuous addition and Borel multiplication [1, Corollary 1]. The earlier study of Enayat, Hamkins, and Wcisło [2] motivates the broader problem of finding topological presentations of arithmetic. Our construction does not contradict those results: it supports the additive language, and its additive structure prevents a semiring expansion.

2.1 Repository interfaces actually inspected

The repository snapshot used here is

a11efab09a2d08f692a09829cb6d3440ddd1795a.

The Presburger README [14] describes a complete Lean integer quantifier eliminator and an independent Rocq/Coq proof of the normalized one-variable step. The inspected file PresburgerArithmetic/Cooper.lean [15] states `periodic_interval` and `periodic_has_residue` for predicates on the ordinary type `Int`. Their hypotheses are entirely appropriate there. They must not be transferred unchanged to a nonstandard ordered group; section 4.2 gives an explicit counterexample and the correct replacement.

The surreal README [16] records an omnific ring whose normal forms have nonnegative exponents and integer constant term. Only this normal-form interface and the standard arithmetic of its additive group are needed below. We do not assume that the present topology or any of this article’s theorems has already been formalized in that repository.

2.2 What is established, and what is not

The principal result is an explicit affirmative construction, rather than a claim that a difficult negative theorem has been overturned. The lexicographic algebra is classical. The contribution developed here is the precise topology and its application to the stated question, together with an effective Baire presentation and a collection of sharp structural consequences.

The literature search verified the question in the source PDF and checked the arXiv record; it did not establish the absence of an earlier solution. Consequently, “an affirmative answer to the stated question” is the appropriate conclusion, not an independently certified first solution. No answer to Glazer’s separate reverse-mathematical Question 1 is claimed.

All constructions and proofs take place in ordinary ZFC with set-sized carriers. The notation $\mathbf{Oz}$ is used only to identify where explicit set-sized images lie. No topology on an unrestricted proper class, and no set of all surreal numbers, is used.

2.3 Source 04’s framing

Source 04 opens with the same question and the same two repository connections (its Section 1). It adds three points, in its own words; “this article” and “this package” there mean source 04.

The interpretation of the signature matters. Throughout this article, Presburger arithmetic is

$$\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +),$$

with its definable order, and with no multiplication of two variables. Continuity of addition does *not* mean continuity of every definable function. Indeed, the difference between these requirements is central to our result. Adding order and fixed congruence predicates to the language does not change which first-order sets are definable, but the predicates need not be open or closed in the topology.

The characterization of Presburger groups as discretely ordered abelian groups with divisible quotient by the standard integers, and their quantifier elimination after naming congruences, are classical. A convenient modern reference is Jeřábek's preliminaries [4, Section 2]. We include proofs of the facts needed here so the construction can be audited independently of that characterization.

The specific contribution is the topological application to the printed question, strengthened by an effective model on literal Baire space and an analysis of definable operations and self-embeddings. The finite-residue argument also identifies an exact hypothesis needed when transporting a standard-integer proof to arbitrary models. It is not a claim that the repository's integer proof is defective.

This is an unrefereed research manuscript. The statements called theorems below have proofs in the manuscript; the Python program is an independent finite diagnostic and a usable stream implementation. No new theorem in this package is represented as machine-checked. The literature check verified the wording of Glazer's question and the relevant background; it did not establish that the affirmative construction has never appeared elsewhere. Thus an answer to the specified question is claimed at the mathematical level, but first-publication priority is not claimed.

3 The construction in one theorem

We include 0 in $\mathbb{N}$. For an ordered abelian group D , let

$$G_D = D \overrightarrow{\times} \mathbb{Z}$$

mean the direct-product additive group with the D coordinate more significant in the order. Thus

$$(d, n) < (e, m) \iff d < e \text{ or } (d = e \text{ and } n < m). \quad (3.1)$$

Write $\mathbf{1} = (0, 1)$ and let

$$M_D = (G_D)_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (3.2)$$

The ambient product topology, whenever specified, is separate from (3.2)'s arithmetic order.

Theorem 3.1 (Polish and effective Presburger models). *The following assertions hold.*

- (1) *With $D = \mathbb{R}$ in its usual topology, $M_{\mathbb{R}}$ is an uncountable, locally compact Polish space without isolated points. Coordinatewise addition is continuous, and*

$$(M_{\mathbb{R}}; 0, \mathbf{1}, +, <) \models \text{Th}(\mathbb{N}; 0, 1, +, <).$$

- (2) *With $D = \mathbb{Q}^{\mathbb{N}}$, ordered by the first differing coordinate and with each rational coordinate discrete, the space $M_{\mathbb{B}} = M_D$ is homeomorphic to $\mathbb{N}^{\mathbb{N}}$. There is an explicit such homeomorphism under which addition, successor, and division by each positive standard integer are Type-2 computable.*
- (3) *In both models, every first-order definable relation, with parameters, is Δ_2^0 . The order is neither open nor closed. In the Baire presentation, each formula has a computable bound on the number of changes in a convergent truth approximation, relative to its parameter codes.*

- (4) Both models embed as ordered additive submonoids of $\mathbf{Oz}$, fixing 0 and 1. Neither admits any unital semiring multiplication with its given addition, zero, and unit.

The proofs occupy [sections 4 to 7, 9 and 10](#). The first assertion alone answers the source question. The later assertions explain how much regularity is possible and where it necessarily stops.

For orientation, an element $(r, -17)$ with $r > 0$ is a *positive* element of $M_{\mathbb{R}}$. Indeed it exceeds every standard numeral $(0, k)$. Nevertheless,

$$(1/j, 0) \longrightarrow (0, 0) \quad (j \longrightarrow \infty). \quad (3.3)$$

in the chosen topology. Arithmetic size and topological proximity are different notions. The construction would fail if one silently demanded that this topology be the order topology.

Source 04's statement [04]. Source 04 (its Section 1.2) sets up the same real model:

Let $\mathbb{Z}$ have the discrete topology and put

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (3.4)$$

Set $0 = (0, 0)$, $1 = (0, 1)$, and

$$(r, z) + (s, w) = (r + s, z + w).$$

Order the ambient group $\mathbb{R} \times \mathbb{Z}$ lexicographically, with the real coordinate dominant. Then $M_{\mathbb{R}}$ is exactly its nonnegative cone.

Remark 3.2 (Source 04's statement of [theorem 3.1](#) [04]). The structure $(M_{\mathbb{R}}; 0, 1, +)$ is a model of $\text{Pr}_{\mathbb{N}}$. With the topology inherited from $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$, its universe is an uncountable perfect locally compact Polish space, and addition is jointly continuous. There is also a model of $\text{Pr}_{\mathbb{N}}$ on $\mathbb{N}^{\mathbb{N}}$ with jointly continuous addition. In both models, quotient and remainder by each fixed positive standard integer are continuous, and every first-order definable relation with parameters is Δ_2^0 .

The algebraic assertion is proved in [section 4](#) and [section 4.3](#); the topology and the explicit Baire-space realization are proved in [sections 5.2, 5.4 and 6.3](#). The proof uses familiar $\mathbb{Z}$ -groups, not an alteration of what Presburger arithmetic means.

[merge] The two statements agree. Source 04's adds continuity of quotient and remainder by each fixed standard integer, which it proves for every admissible coefficient group ([theorem 5.4](#)) and which 05 proves for its two models ([proposition 8.1](#)); source 05's adds the optimal order rank, the finite-mind-change truth approximation and the omnific realization. Both prove the no-semiring and self-embedding results.

4 Full Presburger arithmetic, not merely a few axioms

4.1 Discretely ordered groups with standard residues

Definition 4.1. A $\mathbb{Z}$ -group in this article is a linearly ordered abelian group G with a distinguished least positive element $\mathbf{1}$, such that for every positive standard integer m and every $x \in G$ there exist $q \in G$ and $r \in \{0, \dots, m-1\}$ satisfying

$$x = mq + r\mathbf{1}. \quad (4.1)$$

Here mq denotes repeated addition by a *standard* integer, not a new multiplication operation on G .

Ordered abelian groups are torsion-free. The quotient and remainder in (4.1) are unique: if $m(q - q') = (r' - r)\mathbf{1}$ and $q > q'$, then $q - q' \geq \mathbf{1}$, whereas $r' - r < m$; the other inequality is similar. We suppress $\mathbf{1}$ in standard integer constants when no confusion is possible.

Proposition 4.2 (The split lexicographic group). *If D is a divisible ordered abelian group, then $D \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group. Its standard division algorithm is*

$$(d, n) = m(d/m, \lfloor n/m \rfloor) + (0, n \bmod m). \quad (4.2)$$

If $(d, n) \geq 0$, the displayed quotient is nonnegative.

Proof. Coordinatewise addition preserves the lexicographic order. A positive (d, n) either has $d > 0$, in which case it exceeds $(0, 1)$, or has $d = 0$ and $n \geq 1$. Thus $(0, 1)$ is the least positive element. Divisibility and torsion-freeness of D give the unique element d/m . Ordinary Euclidean division of n now proves (4.2). If $d > 0$, then $d/m > 0$, regardless of the integer quotient. If $d = 0$ and $n \geq 0$, the integer quotient is nonnegative. $\square$

Source 04 proves the same statement (its Lemma 2.2) with the same explicit quotient and the same nonnegativity clause; its uniqueness argument compares coordinates instead of using least positivity.

Second proof (source 04). If $(d, z) > 0$ and $d > 0$, then $(d, z) > (0, 1)$. If $d = 0$, positivity says $z \geq 1$. This proves discreteness. Existence in (4.1) follows from divisibility of D and ordinary integer Euclidean division, including for negative z .

For uniqueness, equality of $mq + r\mathbf{1}$ and $mq' + r'\mathbf{1}$ first gives equality of the D -coordinates of q, q' . Equality of the integer coordinates then gives ordinary uniqueness of quotient and remainder. Finally, if $d > 0$, then $d/m > 0$. If $d = 0$ and $z \geq 0$, then $\lfloor z/m \rfloor \geq 0$. $\square$

Corollary 4.3 ([04]). *For $m \geq 1$,*

$$mG_D = D \times m\mathbb{Z}, \quad (d, z) \equiv (e, w) \pmod{mG_D} \iff z \equiv w \pmod{m}.$$

Moreover,

$$\bigcap_{m \geq 1} mG_D = D \times \{0\}. \quad (4.3)$$

Proof. Every D -coordinate is an m -multiple. The only ordinary integer divisible by every positive integer is zero. $\square$

The standard subgroup $\mathbb{Z}\mathbf{1}$ is convex, and $G_D/\mathbb{Z}\mathbf{1}$ is the ordered divisible group D . This gives the usual quick recognition of G_D as a Presburger group. For the precise model-theoretic conclusion, we next give a direct elimination argument.

The implication from these axioms to *full* Presburger arithmetic is important enough to prove. We use the classical elimination architecture of Presburger and Cooper; [3] supplies historical and algorithmic context. The proof below makes explicit which algebraic hypotheses are needed in a nonstandard group.

4.2 Coset invariance versus standard-shift periodicity

For $m \geq 1$, write $mG = \{mh : h \in G\}$. A predicate P is mG -invariant if

$$y - z \in mG \implies (P(y) \leftrightarrow P(z)). \quad (4.4)$$

This is stronger than invariance under the single standard shift $m\mathbf{1}$.

Example 4.4 (A false generalization of an integer lemma). In $G = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$, let $P(d, n)$ mean $d > 0$. For every positive standard m ,

$$P(x + m\mathbf{1}) \leftrightarrow P(x)$$

holds. Yet P has witnesses and is false at every standard residue $0, 1, \dots, m-1$. Thus

$$(\exists x P(x)) \not\iff \bigvee_{0 \leq r < m} P(r\mathbf{1}).$$

It also fails the naive finite-interval reduction with lower bound 0 and upper bound $(1, 0)$. No flaw in the repository's Int theorem is involved: its domain is the ordinary integers, where every multiple is obtained by a standard finite iteration.

The general counterexample [04]. Source 04 phrases coset invariance as a definition and gives the counterexample for every nonzero divisible D , with the predicate $d \neq 0$ in place of $d > 0$. In its text G is a linearly ordered abelian group with a least positive element $\mathbf{1}$ and unique division by positive standard integers as in (4.1), that is, a $\mathbb{Z}$ -group.

Definition 4.5 ([04]). A predicate $P \subseteq G$ is m -residue invariant if

$$u - v \in mG \implies (P(u) \iff P(v)).$$

No definability hypothesis is imposed in this definition.

The condition is stronger than invariance under the single translation $x \mapsto x + m\mathbf{1}$. This distinction is invisible over the ordinary group $\mathbb{Z}$, but is essential over nonstandard groups.

Proposition 4.6 (One-step periodicity is insufficient [04]). If $D \neq 0$, the external predicate

$$P(d, z) \iff d \neq 0$$

on G_D satisfies $P(x + m\mathbf{1}) \leftrightarrow P(x)$ for every standard $m \geq 1$. Nevertheless P is nonempty and is false at every standard residue $0, 1, \dots, m-1$.

Proof. A standard integer translation does not change the D -coordinate. A nonzero D -coordinate exists, whereas every standard integer has D -coordinate zero. In particular P is not m -residue invariant: $(d, 0) - (0, 0) \in mG_D$ for every d . $\square$

Lemma 4.7 (The coset-invariant finite search). Let G be a $\mathbb{Z}$ -group, $m \geq 1$ standard, and P an mG -invariant predicate. For $L, U \in G$,

$$\begin{aligned} \exists y (L \leq y \leq U \wedge P(y)) \\ \iff \bigvee_{0 \leq r < m} (L + r \leq U \wedge P(L + r)). \end{aligned} \quad (4.5)$$

With only an upper bound, candidates $U - r$ suffice; with only a lower bound, candidates $L + r$ suffice. With neither bound, the candidates are $0, \dots, m-1$.

Proof. Given a witness $y \geq L$, write $y - L = mh + r$ with $0 \leq r < m$. One has $h \geq 0$: otherwise $h \leq -1$, and then $mh + r \leq -m + (m - 1) = -1$, contradicting $y - L \geq 0$. Consequently

$$L \leq L + r \leq y \leq U.$$

The difference $y - (L + r) = mh$ lies in mG , so $P(L + r)$ holds. Conversely every displayed candidate is already a witness. The same argument applied to $U - y$ proves the upper-bound version. Without bounds, use the standard remainder of y itself. $\square$

Congruence predicates have exactly the stronger invariance (4.4). If M is a common multiple of finitely many standard moduli, any Boolean combination of congruences with those moduli is MG -invariant. This remains true with arbitrary nonstandard parameters: adding Mh to a variable changes an affine congruence term by an element of each relevant modulus subgroup.

4.3 Second route [04]: finite witnesses at a boundary

Source 04's form of lemma 4.7 works with strict bounds and an explicit finite candidate set. It is the same theorem with a different normalization of the bounds, and it is the step that 04 uses in its elimination proof below. Here G is a $\mathbb{Z}$ -group, and a standard integer r also denotes $r1$.

Theorem 4.8 (Finite boundary criterion [04]). *Let A, B be finite lists of elements of G , let $m \geq 1$, and let P be m -residue invariant. Put*

$$\Phi(x) := P(x) \wedge \bigwedge_{a \in A} (a < x) \wedge \bigwedge_{b \in B} (x < b).$$

Define a finite candidate set C by

$$C = \begin{cases} \{a_* + 1, \dots, a_* + m\}, & A \neq \emptyset, \quad a_* = \max A, \\ \{b_* - 1, \dots, b_* - m\}, & A = \emptyset, B \neq \emptyset, \quad b_* = \min B, \\ \{0, \dots, m - 1\}, & A = B = \emptyset. \end{cases} \quad (4.6)$$

Then

$$(\exists x \in G) \Phi(x) \iff \bigvee_{c \in C} \Phi(c). \quad (4.7)$$

Proof. The reverse direction is immediate. Suppose first that $A \neq \emptyset$ and x is a witness. Divide $x - a_* - 1$ by m to obtain

$$x - a_* = mq + s, \quad 1 \leq s \leq m.$$

Here q is an element of G , not necessarily a standard integer. If $q < 0$, discreteness gives $q \leq -1$, whence $mq + s \leq 0$, contradicting $x > a_*$. Thus $q \geq 0$ and $c = a_* + s \leq x$. This c is larger than every lower bound and smaller than every upper bound because $c \leq x < b$. Also $x - c = mq \in mG$, so $P(c)$ holds.

If A is empty and B is nonempty, divide $b_* - x - 1$ instead. The result is $b_* - x = mq + s$ with $q \geq 0$ and $1 \leq s \leq m$. The candidate $c = b_* - s$ satisfies $x \leq c < b_* \leq b$ for every upper bound and is in the same residue class as x . Finally, when both lists are empty, take the unique standard representative of x modulo mG . $\square$

This proof uses no induction through a nonstandard interval. Its only finite searches have standard lengths fixed by the formula being eliminated. Weak bounds can be converted to strict ones using discreteness: $a \leq x$ is $a - 1 < x$, and $x \leq b$ is $x < b + 1$.

4.4 A self-contained elimination argument

Expand the ordered-group language by predicates

$$D_m(t) \iff t \in mG \quad (m \geq 2),$$

and regard D_1 as always true. All of these are definable by existential formulas in the group language.

Theorem 4.9 (Classical Presburger elimination over $\mathbb{Z}$ -groups). *In the expanded language*

$$\{0, 1, +, -, \leq, (D_m)_{m \geq 2}\},$$

the axioms of $\mathbb{Z}$ -groups eliminate quantifiers and form a complete theory. In particular every $\mathbb{Z}$ -group is elementarily equivalent to $(\mathbb{Z}; 0, 1, +, -, \leq, (m \mid \cdot)_m)$.

Proof. It suffices to eliminate one existential quantifier from a quantifier-free formula. Rewrite the formula in disjunctive normal form. Negated inequalities are replaced using discreteness, for example $\neg(s \leq t) \leftrightarrow t + 1 \leq s$. An equality is the conjunction of two inequalities; a disequality is a disjunction of strict inequalities. Finally, standard residue exhaustiveness gives

$$\neg D_m(t) \iff \bigvee_{r=1}^{m-1} D_m(t - r).$$

Thus it is enough to eliminate $\exists x$ from a conjunction of affine inequalities and positive divisibility atoms, along with conjuncts not involving x .

Let A be a positive common multiple of every nonzero absolute coefficient of x occurring in that conjunction; take $A = 1$ if there are none. Set $y = Ax$ and add the condition $D_A(y)$. An atom

$$ax + t \leq 0, \quad a \neq 0,$$

can be multiplied by the positive standard integer $k = A/|a|$, giving

$$\text{sgn}(a)y + kt \leq 0.$$

It is now an upper or lower bound on y . For a congruence, torsion-free cancellation gives the equivalence

$$D_m(ax + t) \iff D_{km}(\text{sgn}(a)y + kt). \quad (4.8)$$

Indeed $kz \in kmG$ if and only if $z \in mG$: one direction multiplies a witness, and the other cancels k from $kz = kmh$. The additional atom $D_A(y)$ is exactly what ensures that every candidate y comes from a unique x .

After normalizing signs, we have finitely many lower bounds $L_i \leq y$, finitely many upper bounds $y \leq U_j$, and a conjunction $P(y)$ of congruence conditions. Let M be a positive common multiple of all their moduli, including A . Then P is MG -invariant.

If lower bounds exist, choose their maximum L . By [lemma 4.7](#), a witness exists exactly when one of $L, L + 1, \dots, L + M - 1$ satisfies all the normalized conjuncts. The maximum need not be introduced as a new function symbol: take a finite disjunction over the indices i , require $L_h \leq L_i$ for every h , and substitute $L_i + r$ for y . This produces a quantifier-free formula in the parameters. If there are upper bounds but no lower bounds, choose a minimal U_j and substitute $U_j - r$. If no bounds exist, substitute the standard residues r . Conjuncts independent of x are retained throughout.

This eliminates one existential quantifier. Induction on formula complexity, replacing universal quantification by negated existential quantification, eliminates all quantifiers.

A quantifier-free sentence uses only standard integer multiples of 1. Their order and their residues are the same in every $\mathbb{Z}$ -group, by least positivity and uniqueness of standard remainders. Therefore all $\mathbb{Z}$ -groups satisfy the same sentences. The ordinary integers are one such group, proving completeness and the final assertion. $\square$

4.5 Second route [04]: normalization and elimination

Source 04 proves [theorem 4.9](#) by a differently organized argument: it keeps negated congruences instead of rewriting them through residues, substitutes away equalities in the eliminated variable, and uses strict bounds with [theorem 4.8](#). Its statement is uniform over all groups G of [section 4.3](#).

Use the group language

$$L_{Pr} = \{0, 1, +, -, <, (\equiv_m)_{m \geq 2}\},$$

where $u \equiv_m v$ means $u - v \in mG$. Affine terms have integer coefficients. The congruence predicates are definable by existential quantification in the underlying additive language.

Theorem 4.10 (Quantifier elimination, with a uniform proof [04]). *Every L_{Pr} formula is equivalent, uniformly in all groups G satisfying the hypotheses above, to a quantifier-free Boolean combination of affine inequalities, affine equalities, and congruences. Consequently every such G is elementarily equivalent to $(\mathbb{Z}; 0, 1, +, -, <)$ in the reduct language.*

Proof. It is enough to eliminate one existential quantifier from a conjunction of literals, after writing the quantifier-free part in disjunctive normal form. Negated equality is split into two strict inequalities, and negated inequalities are converted by discreteness. Negated congruences can be kept as negated congruences.

Let x be the variable to eliminate. If it occurs with no nonzero coefficient, the quantifier is vacuous. Otherwise let L be the least common multiple of the positive absolute values of its nonzero coefficients. Introduce $y = Lx$ and impose $y \in LG$. There is a unique x for every such y because G is torsion-free.

For an atom with term $ax + t$, $a \neq 0$, put $B = L/|a|$. Multiplying an inequality or equality by the positive standard integer B replaces its x -coefficient by $\text{sgn}(a)$ on y . For divisibility use

$$ax + t \in nG \iff \text{sgn}(a)y + Bt \in nBG. \quad (4.9)$$

Indeed the forward implication multiplies a witness by B , and the reverse implication cancels B in the torsion-free group. Atoms with zero x -coefficient remain independent conditions.

The resulting conjunction has only coefficient 1 or -1 on y in its inequalities, equalities, and congruences. An equality involving y permits substitution. In the absence of such an equality, the inequalities form finite strict lower and upper lists. The remaining congruences and negated congruences form a predicate $P(y)$ invariant modulo mG , where m is the least common multiple of their moduli, including L . For example, if $n \mid m$, replacing y by $y + mg$ cannot change membership of $\pm y + t$ in nG .

Apply [theorem 4.8](#). The maximum or minimum appearing there is not an extra function symbol: take a finite disjunction over choices of a largest lower bound, or a smallest upper bound, and add the finitely many inequalities that express that choice. Substitution of

the finite candidates leaves a quantifier-free formula in the remaining variables. Negation handles universal quantifiers, so induction on the external formula tree proves elimination.

Finally a closed quantifier-free formula has the same truth value in every such G as in $\mathbb{Z}$. Its terms are standard numerals. Their equality and order are fixed, and their congruences have the usual residues by division. Eliminating all quantifiers therefore proves elementary equivalence. $\square$

Corollary 4.11 (The entire induction scheme). *For every divisible ordered abelian group D ,*

$$(M_D; 0, 1, +, <) \models \text{Th}(\mathbb{N}; 0, 1, +, <).$$

In particular it satisfies induction for every first-order formula of this additive language, with arbitrary parameters from M_D .

Proof. By [proposition 4.2](#) and [theorem 4.9](#), $G_D \equiv \mathbb{Z}$. The nonnegative cone is uniformly definable by $x \geq 0$. Relativizing every quantifier to that cone translates any sentence about M_D into a sentence about G_D . The same translation in $\mathbb{Z}$ gives the ordinary nonnegative integers. Each universally closed induction instance is therefore true in M_D . $\square$

Proposition 4.12 (Definable least elements). *Every nonempty definable subset of M_D , allowing parameters, has a least element.*

Proof. Eliminate quantifiers and take a finite disjunction of conjunctions as in [theorem 4.9](#). Include $x \geq 0$ explicitly. For each conjunction, the substitution $y = Ax$ introduces a lower bound $y \geq 0$. Hence a nonempty conjunction has a least solution among the finitely many valid candidates at its maximum lower bound; the residue argument shows that one of these candidates lies below every witness. Dividing by A is order-preserving. Finally take the minimum of the least solutions of the nonempty disjuncts. $\square$

This also gives a direct induction check after elimination: a nonempty set of counterexamples to an induction instance would have a least member. It cannot be 0, and its predecessor would contradict the induction step. The external standard part $\{(0, n) : n \in \mathbb{N}\}$ is a proper inductive subset when $D \neq 0$, but is not definable, even with parameters. Induction quantifies over definable sets, not over every external subset.

Source 04 proves [corollary 4.11](#) (its Corollary 3.5, for $\text{Pr}_{\mathbb{N}}$) by the same relativization, and adds [04]:

The same argument shows that every definable subset of M_D^k is a Boolean combination of restrictions of group affine inequalities, equalities, and standard congruences. This observation will control its Borel complexity without an induction over projections of Borel sets.

5 Polishness and a locally compact example

A Polish space is a separable space whose topology is induced by some complete metric. A particular convenient metric may be incomplete without the space ceasing to be Polish. This distinction matters for open rays.

5.1 A useful exact criterion

We first record the standard subspace criterion with a proof, so that the completeness issue is visible.

Lemma 5.1 (Polish subspaces). *A subspace A of a Polish space X is Polish if and only if A is a G_δ subset of X .*

Proof. Fix a compatible complete metric d on X . If $A = \bigcap_j U_j$ with U_j open, consider on A the continuous functions $f_j(x) = 1/d(x, X \setminus U_j)$; when $U_j = X$, set $f_j = 0$. The graph of $x \mapsto (f_j(x))_j$ is closed in $X \times \mathbb{R}^\mathbb{N}$. Indeed a limit outside some U_j would force f_j to tend to infinity, not to a real number. Projection identifies this closed graph homeomorphically with A . Countable products and closed subspaces of complete separable metric spaces are complete and separable.

Conversely, let δ be a complete metric inducing the subspace topology on A . For each n , let W_n be the union of all ambient open sets U that meet A and satisfy

$$\text{diam}_d(U) < 2^{-n}, \quad \text{diam}_\delta(U \cap A) < 2^{-n}.$$

These unions cover A . If $x \in \bigcap_n W_n$, choose $x \in U_n$ and $a_n \in U_n \cap A$. Then $a_n \rightarrow x$ in d . For each fixed n , all sufficiently late a_k belong to U_n , so (a_k) is δ -Cauchy. Its δ -limit lies in A , and agreement of the two topologies identifies that limit with x . Thus $A = \bigcap_n W_n$. $\square$

Theorem 5.2 (The positive-cone criterion). *Let D be a divisible ordered abelian group equipped with a Polish topological-group topology, not necessarily its order topology. Give $\mathbb{Z}$ the discrete topology. Then the subspace $M_D \subseteq D \times \mathbb{Z}$ is Polish if and only if $D_{>0}$ is G_δ in D . When this holds, $D_{>0}$ is in fact Δ_2^0 , and addition on M_D is continuous.*

Proof. If $D_{>0}$ is G_δ , so is its union with the closed singleton $\{0\}$. On each clopen integer slice, the carrier (3.2) is either $D_{>0}$ or $D_{\geq 0}$. A countable topological sum of G_δ subspaces is G_δ in the corresponding sum: its complement is a countable union of F_σ subsets of the clopen slices. Apply lemma 5.1.

Conversely, the slice with integer coordinate -1 is closed in M_D and homeomorphic to $D_{>0}$. If M_D is Polish, so is that slice, and lemma 5.1 makes $D_{>0}$ a G_δ subset of D . Negation is a homeomorphism on D , so $D_{<0}$ is G_δ as well. The complement of $D_{>0}$ is $D_{<0} \cup \{0\}$, also G_δ . Thus $D_{>0}$ is both F_σ and G_δ .

Finally the cone is closed under coordinatewise addition. Restricting continuous addition on $D \times \mathbb{Z}$ to a subspace with its image in that subspace gives a continuous map $M_D^2 \rightarrow M_D$. $\square$

5.2 Admissible coefficient groups and continuous division [04]

Source 04 proves a sufficient version of theorem 5.2 under a stronger hypothesis, and obtains in return the continuity of division by standard integers for every such D ; source 05 proves that continuity only for its two explicit models (proposition 8.1). The Polish clause of theorem 5.4 is the case of theorem 5.2 in which $D_{>0}$ is open.

Definition 5.3 ([04]). *An admissible topological divisible group is a divisible linearly ordered abelian group D with a Polish group topology such that $D_{>0}$ is open and each map $d \mapsto d/m$, $m \geq 1$, is continuous. The specified topology is additional data; it is not silently identified with the order topology of G_D or M_D .*

Theorem 5.4 (Transfer [04]). *Let D be an admissible topological divisible group, and give G_D the product topology with discrete $\mathbb{Z}$. Then M_D , with its subspace topology, is Polish. Addition on M_D is jointly continuous. The functions*

$$Q_m(d, z) = (d/m, \lfloor z/m \rfloor), \quad R_m(d, z) = (0, z \bmod m)$$

are continuous maps from M_D to itself for each fixed $m \geq 1$.

Proof. The ambient group is Polish. Its complement of M_D is

$$(D_{<0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{<0}).$$

The first set is open and the second is closed. In a metrizable space every open set is F_σ , so this complement is F_σ and M_D is G_δ . A G_δ subspace of a Polish space is Polish. Equivalently, the integer-coordinate slices are a countable disjoint sum of copies of the Polish spaces $D_{\geq 0}$ and $D_{>0}$.

The set M_D is closed under addition; restriction of the continuous ambient addition is therefore continuous as a map into the subspace. The displayed formulas for Q_m, R_m are continuous in the ambient coordinates: division on D is continuous, and every function on the discrete coordinate is continuous. Their images belong to M_D by [proposition 4.2](#). $\square$

5.3 The real-coordinate model, including a complete metric

For $D = \mathbb{R}$, the positive cone is open. More explicitly, the slice

$$X_n = \{r : (r, n) \in M_{\mathbb{R}}\}$$

is $[0, \infty)$ when $n \geq 0$ and $(0, \infty)$ when $n < 0$. Hence

$$M_{\mathbb{R}} \cong \bigsqcup_{n \geq 0} [0, \infty) \sqcup \bigsqcup_{n < 0} (0, \infty). \quad (5.1)$$

This proves separability, local compactness, and absence of isolated points directly.

For a fully explicit compatible complete metric, define

$$d_n(r, s) = \begin{cases} \min\{1, |r - s|\}, & n \geq 0, \\ \min\{1, |\log r - \log s|\}, & n < 0, \end{cases}$$

and set

$$d((r, n), (s, m)) = \begin{cases} d_n(r, s), & n = m, \\ 2, & n \neq m. \end{cases} \quad (5.2)$$

Every Cauchy sequence is eventually contained in one slice. The closed ray with the first metric is complete; the open ray with the logarithmic metric is isometric, before truncation, to $\mathbb{R}$. Thus (5.2) is complete. It induces exactly the topology in (5.1), which is the original product-subspace topology. Continuity of addition follows from that topology, not from a claim of uniform continuity in this metric.

The map $r \mapsto (r, 0)$ on $r > 0$ establishes uncountability. Together with [corollary 4.11](#), this completes the simplest proof of the affirmative answer.

Remark 5.5 (Why the arithmetic order topology is not used). The least positive element is **1**. In G_D , the order interval $(x - 1, x + 1)$ is the singleton $\{x\}$. The positive cone has the same discreteness at its minimum. Consequently its arithmetic order topology is discrete, and an uncountable such topology cannot be Polish. The topology of (5.1) is different, as (5.2) and the convergence in (3.3) explicitly demonstrate.

5.4 The real-coordinate model, source 04's account [04]

Source 04 gives the same slice decomposition and the same complete metric as (5.2), except that distinct slices are at distance 1; it adds two arithmetic illustrations and a remark.

Take $D = \mathbb{R}$ with the usual order and topology. It is admissible, so [corollary 4.11](#) and [theorems 5.4](#) and [7.2](#) apply to (3.4).

Topology, perfection, and a complete metric. For $z \geq 0$, the slice of integer coordinate z is a copy of $[0, \infty)$; for $z < 0$, it is a copy of $(0, \infty)$. Each slice is clopen in $M_{\mathbb{R}}$. Thus $M_{\mathbb{R}}$ is uncountable, locally compact, and has no isolated points. In particular every standard numeral $(0, n)$ is nonisolated.

It is important not to assert that the inherited Euclidean metric is complete: $(0, \infty)$ with that metric is not. Here is a compatible complete metric, with different integer slices distance 1 apart:

$$\rho((r, z), (s, w)) = \begin{cases} 1, & z \neq w, \\ \min\{1, |r - s|\}, & z = w \geq 0, \\ \min\{1, |\log r - \log s|\}, & z = w < 0. \end{cases} \quad (5.3)$$

The last line only occurs for positive r, s . A Cauchy sequence is eventually in one slice. On a nonnegative slice use completeness of $[0, \infty)$; on a negative slice use the homeomorphism $r \mapsto \log r$ onto $\mathbb{R}$. The truncated metrics are metrics, and the inter-slice distance preserves the triangle inequality. Countable dense subsets in the slices give a countable dense subset of the whole space.

Arithmetically huge elements converging to zero. Put $a_j = (1/j, 0)$ for $j \geq 1$. Then

$$a_j > n\mathbf{1} \quad \text{for every standard } n, \quad a_j \longrightarrow 0 \quad \text{topologically.}$$

There is no contradiction. The arithmetic order and the chosen topology are distinct structures. The arithmetic order topology on M_D would be discrete, since every nonzero point has a predecessor and a successor, and 0 is isolated by its immediate successor. That discrete topology is *not* the topology used in the construction.

For a concrete division example, in $M_{\mathbb{R}}$ one has

$$(2, -7) = 3(2/3, -3) + (0, 2).$$

The quotient $(2/3, -3)$ is positive because its real coordinate is positive. This illustrates why restricting the integer coordinate to nonnegative values at all positive real coordinates would destroy the desired arithmetic: the negative integer sheets are necessary.

Remark 5.6 (Why a Cartesian product of two positive cones is not used [04]). The universe is not $\mathbb{R}_{\geq 0} \times \mathbb{N}$. The latter does not have the required predecessor property at points $(r, 0)$ with $r > 0$. By contrast, in $M_{\mathbb{R}}$ every positive (r, z) has predecessor $(r, z - 1)$, except that the ordinary zero remains the least element as required.

6 An explicit Baire-space realization

Let

$$D_B = \mathbb{Q}^{\mathbb{N}}$$

with coordinatewise addition. Each factor $\mathbb{Q}$ is discrete, *not* given its Euclidean subspace topology. Two distinct sequences are compared at the least index at which they differ. This lexicographic order makes D_B a divisible ordered abelian group. Its product topology is Polish and zero-dimensional; for example,

$$d_D(a, b) = 2^{-\min\{j: a_j \neq b_j\}} \quad (a \neq b), \quad d_D(a, a) = 0$$

is a compatible complete metric.

The sets

$$C_{k,q} = \{a : a_0 = \dots = a_{k-1} = 0, a_k = q\}, \quad k \in \mathbb{N}, q \in \mathbb{Q}_{>0},$$

are clopen cylinders, and

$$(D_B)_{>0} = \bigsqcup_{k,q>0} C_{k,q}. \quad (6.1)$$

No restriction is placed on later coordinates. Each cylinder is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$. The nonnegative cone $(D_B)_{\geq 0}$ is closed, since the negative cone is open. [theorem 5.2](#) already proves that

$$M_B = (D_B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$$

is Polish. We now identify its homeomorphism type without relying on a space-classification theorem.

6.1 The zero branch and its explicit tree code

Fix computable bijections

$$e : \mathbb{N} \rightarrow \mathbb{Q}, \quad e_+ : \mathbb{N} \rightarrow \mathbb{Q}_{>0}, \quad e_{\geq 0} : \mathbb{N} \rightarrow \mathbb{Q}_{\geq 0},$$

with computable inverses. A code $s \in \mathbb{N}^{\mathbb{N}}$ determines a sequence $a \in (D_B)_{\geq 0}$ as follows. As long as all previously chosen coefficients are zero, set $a_j = e_{\geq 0}(s_j)$. Once the first positive coefficient has appeared, use $a_j = e(s_j)$ at all later stages. This is a bijection: the inverse uses the corresponding enumeration at each step. A finite output prefix depends on exactly a finite input prefix, in both directions. Thus it is a homeomorphism

$$\mathbb{N}^{\mathbb{N}} \cong (D_B)_{\geq 0}. \quad (6.2)$$

The all-zero rational sequence is not omitted; it is the branch that continues choosing zero indefinitely.

Theorem 6.1 (Baire-space carrier). *There is an explicit homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$. In particular M_B is uncountable, has no isolated points, and is nowhere locally compact.*

Proof. Partition M_B into the following countably many clopen blocks. For each $n \geq 0$, take the entire slice $(D_B)_{\geq 0} \times \{n\}$. For $n < 0$, partition that slice further into $C_{k,q} \times \{n\}$ using (6.1). By (6.2), every nonnegative slice is homeomorphic to Baire space; every negative-slice block is so by coding its unrestricted rational tail. A countable disjoint sum of copies of $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to $\mathbb{N}^{\mathbb{N}}$: use the first natural coordinate to choose a block and the remaining coordinates to choose a point in it.

For completeness, no nonempty basic cylinder of $\mathbb{N}^{\mathbb{N}}$ is relatively compact. Its projection to the next unrestricted discrete coordinate has infinite image, whereas the continuous image of a compact set in a discrete space is finite. Every neighborhood contains such a cylinder. The other two assertions follow from the elementary properties of the product $\mathbb{N}^{\mathbb{N}}$. $\square$

6.2 A total code for every sequence of natural numbers

Here is a fully specified block indexing, also used in the accompanying program. Let $\langle a, b \rangle$ be Cantor's pairing function. The first digit of a Baire code is a tag. An even tag $2n$ specifies the nonnegative integer coordinate n and the tree code (6.2). An odd tag

$$2\langle a, \langle k, b \rangle \rangle + 1 \quad (6.3)$$

specifies integer coordinate $-a - 1$, first positive rational coordinate k , and value $e_+(b)$ there. The remaining digits code the unrestricted rational tail using e .

Every natural tag is of exactly one of these forms. Every infinite natural sequence therefore codes a valid element; no hidden domain test is required. Conversely, a point with negative integer coordinate has a first positive rational coefficient, so its tag and tail are uniquely recoverable. This describes h and h^{-1} explicitly.

Theorem 6.2 (Computable continuous arithmetic). *Under this presentation, the operations*

$$x + y, \quad x + 1, \quad q_m(x) = \lfloor x/m \rfloor, \quad r_m(x) = x \bmod m$$

are Type-2 computable for every positive standard integer m . Here $q_m(x)$ is the unique quotient in the model's Euclidean division by m , not a real-valued floor operation. The remainders have their finite discrete topology.

Proof. A Type-2 computation reads finitely many input digits to produce each requested output digit. Decode the integer coordinates n, z of the two inputs. Each rational coefficient $a_j + b_j$ is computable from finite prefixes. If $n + z \geq 0$, output tag $2(n + z)$ and encode the coefficient sequence by the nonnegative-cone tree. The tree decisions only inspect finite prefixes.

If $n + z < 0$, at least one input has a negative integer coordinate and hence a strictly positive coefficient sequence. Its tag supplies an index k of a positive leading coefficient. Since both coefficient sequences are lex-nonnegative, their sum is strictly positive and its first nonzero coordinate occurs at or before k . Search these finitely many coordinates, output the negative tag (6.3), and then encode the tail. Thus the tag itself is computable; the algorithm does not attempt to decide whether an arbitrary sequence is identically zero.

For division use

$$q_m(a, n) = (a/m, \lfloor n/m \rfloor), \quad r_m(a, n) = n \bmod m.$$

If the quotient's integer coordinate is negative, the input integer coordinate was negative as well, and the same known leading index can be used. Otherwise the nonnegative tree code applies. Constants and successor are special cases of these finite-prefix procedures. $\square$

The adjective "computable" here concerns operations on infinite sequence codes. It does not assert that every element is computable, or that equality and order of arbitrary streams can be decided in finite time. The distinction will be quantified next.

6.3 Second route [04]: explicit charts

Source 04 builds the same space: its M_ω is M_B , with the same group $\mathbb{Q}^{\mathbb{N}}$, the same discrete coordinates and the same lexicographic order. Its charts differ from the tag coding of [theorem 6.1](#): they go from the model to Baire space and code the nonnegative and the negative integer slices separately. Source 04's homeomorphism H , its charts $H_{\geq}$, $H_{>}$ and its integer coding h are written Ψ , $\Psi_{\geq}$, $\Psi_{>}$ and ζ here ([section 1.3](#)).

6.3.1 The lexicographic rational product

Let

$$D_B = \mathbb{Q}^{\mathbb{N}}$$

with coordinatewise addition, the product topology in which *each copy of $\mathbb{Q}$ is discrete*, and lexicographic order by the first unequal coordinate. A nonzero sequence has a least nonzero index, so the order is defined on all sequences, not merely on finite-support ones. It is a linear order compatible with addition. Division by standard integers is coordinatewise.

The metric

$$d(u, v) = \begin{cases} 0, & u = v, \\ 2^{-k}, & k = \min\{i : u_i \neq v_i\}, \end{cases}$$

is a complete compatible ultrametric. Finite-support rational sequences form a countable dense set. The positive cone is

$$(D_B)_{>0} = \bigcup_{k \in \mathbb{N}} \bigcup_{q \in \mathbb{Q}_{>0}} \{d : d_0 = \dots = d_{k-1} = 0, d_k = q\},$$

an open union of clopen cylinders. Hence D_B is admissible. Write

$$M_B = M_{D_B}.$$

It is already an uncountable Polish Presburger model with continuous addition. We now identify its space explicitly with $\mathbb{N}^{\mathbb{N}}$.

6.3.2 Two elementary coding homeomorphisms

Fix bijections

$$c : \mathbb{Q} \longrightarrow \mathbb{N}, \quad b : \mathbb{Q}_{>0} \longrightarrow \mathbb{N}_{>0}, \quad a : \mathbb{N} \times \mathbb{Q}_{>0} \longrightarrow \mathbb{N}, \quad \zeta : \mathbb{Z} \longrightarrow \mathbb{N}.$$

Since their domains and codomains are discrete, these are homeomorphisms between discrete spaces.

For $d \in (D_B)_{\geq 0}$ define $\Psi_{\geq}(d) \in \mathbb{N}^{\mathbb{N}}$ as follows. Send the zero sequence to the zero sequence. If k is the first nonzero index of d , then $d_k > 0$, and set

$$\Psi_{\geq}(d)_i = \begin{cases} 0, & i < k, \\ b(d_k), & i = k, \\ c(d_i), & i > k. \end{cases} \quad (6.4)$$

Every nonzero element of $\mathbb{N}^{\mathbb{N}}$ has a first positive entry, so this coding has an evident inverse. Both it and its inverse preserve information in finite initial segments: to determine the first n coordinates, only the first n coordinates are needed. It is therefore a homeomorphism, including at the zero sequence.

For $d \in (D_B)_{>0}$, let k again be its first nonzero index and put

$$\Psi_{>}(d)_0 = a(k, d_k), \quad \Psi_{>}(d)_{j+1} = c(d_{k+1+j}) \quad (j \geq 0). \quad (6.5)$$

The pair (k, d_k) identifies a clopen positive cylinder; within that cylinder the tail is an arbitrary rational sequence. This is another homeomorphism onto $\mathbb{N}^{\mathbb{N}}$.

Theorem 6.3 (A concrete Baire homeomorphism [04]). *The map*

$$\Psi : M_B \longrightarrow \mathbb{N}^{\mathbb{N}}, \quad \Psi(d, z) = \begin{cases} \zeta(z) \frown \Psi_{\geq}(d), & z \geq 0, \\ \zeta(z) \frown \Psi_{>}(d), & z < 0, \end{cases} \quad (6.6)$$

is a homeomorphism. Here $\zeta(z) \frown u$ means prepend the one entry $\zeta(z)$ to the sequence u .

Proof. The integer slices of M_B are clopen. Their images are exactly the clopen cylinders of $\mathbb{N}^{\mathbb{N}}$ specified by the first coordinate. On each slice the map is one of the two homeomorphisms just proved. These slice maps assemble to a bijection whose map and inverse are continuous. $\square$

Transporting addition by Ψ gives a binary operation on the *literal* set $\mathbb{N}^{\mathbb{N}}$:

$$u \oplus v = \Psi(\Psi^{-1}(u) + \Psi^{-1}(v)). \quad (6.7)$$

This proves the Baire-space assertion of [remark 3.2](#); it does not merely produce a Borel-isomorphic universe. The space is perfect and nowhere locally compact. For the latter assertion, every nonempty basic cylinder in $\mathbb{N}^{\mathbb{N}}$ has infinitely many disjoint next-coordinate cylinders covering it, with no finite subcover. If a compact neighborhood existed, a closed basic cylinder inside it would be compact, a contradiction.

6.3.3 Effective operations on infinite streams

A map on Baire space is *Type-2 computable* here if an algorithm, given oracle access to the input sequences, returns every requested output coordinate after reading finitely many input coordinates. The number of queries may depend on the input. This is a statement about computations with infinite input oracles, not an enumeration of an uncountable model by ordinary natural-number indices.

Choose the bijections above effectively. For example, index positive rationals by the binary tree starting at $1/1$, with children

$$\frac{p}{q} \mapsto \frac{p}{p+q}, \quad \frac{p+q}{q}.$$

Every positive rational in lowest terms has a unique finite path: repeatedly subtract the smaller numerator or denominator from the larger. The sum strictly decreases until $(1, 1)$ is reached. Recording whether the larger entry was the numerator or denominator and reversing the record recovers the path. This proves both bijectivity and effective invertibility. Enumerate $\mathbb{Q}$ by interleaving positive and negative rationals and adding zero. Combine this enumeration with an effective pairing of two natural numbers to obtain a and with a signed-integer coding to obtain ζ .

Theorem 6.4 (Effective strengthening [04]). *The homeomorphism Ψ and its inverse can be chosen Type-2 computable. With this choice, the addition $\oplus$ in (6.7), and quotient and remainder by each fixed positive integer, are Type-2 computable on $\mathbb{N}^{\mathbb{N}}$.*

Proof. For $\Psi_{\geq}$, a requested coordinate i requires only the first $i + 1$ coefficients: either all are zero, the first nonzero is at i , or it occurred earlier. These are precisely the three cases in (6.4). The same finite-prefix procedure computes its inverse. No decision whether an entire infinite sequence is zero is made.

For $\Psi_{>}$, search for the first nonzero coefficient. The search terminates because this chart has a positive, not merely nonnegative, domain. Once found, all output coordinates are

computable; its inverse reads the encoded pair in the first coordinate and then the tail. Reading the integer coordinate chooses the appropriate chart of Ψ and Ψ^{-1} .

Coordinatewise rational addition and division are computable, as are integer addition, quotient, and remainder. Compose these algorithms with Ψ and Ψ^{-1} . Whenever the output has negative integer coordinate its D_B -coordinate is strictly positive, so the positive-chart search is guaranteed to terminate. For addition, for instance, a negative sum of integer coordinates requires at least one negative input integer coordinate; that input has strictly positive D_B -coordinate, and the other has nonnegative D_B -coordinate. Their sum is strictly positive. $\square$

The companion `verification.py` implements these maps as memoized indexed streams. A caller supplies a function returning the i th natural number of a sequence. The module exports `baire_add`, `baire_quotient`, and `baire_remainder`. Its round-trip checks include long initial zero blocks, which exercise the boundary chart at the zero coefficient sequence.

7 Definability, effective truth, and the optimal order rank

A subset of a metrizable space is Δ_2^0 when it is both F_σ and G_δ . Finite Boolean combinations of such sets are again Δ_2^0 . Closed sets, open sets, and clopen sets belong to this class.

Theorem 7.1 (All definable relations have low Borel rank). *Under the hypotheses of [theorem 5.2](#), every first-order definable subset of M_D^k , with parameters, is Δ_2^0 .*

Proof. In $G_D = D \times \mathbb{Z}$, the strict positive cone is

$$(D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{>0}).$$

It is Δ_2^0 , as is its complement. Affine group terms are continuous maps to G_D , so their inequalities have Δ_2^0 truth sets. Equalities have closed truth sets because the ambient group is Hausdorff.

Divisibility is especially simple. Since D is divisible,

$$mG_D = D \times m\mathbb{Z}. \tag{7.1}$$

This is clopen. Thus congruence atoms have clopen truth sets, including when their affine terms contain parameters. Translate a cone formula into the group and apply [theorem 4.9](#). The result is a finite Boolean combination of these atoms. Finally restrict to M_D^k . $\square$

Second route [04]. Source 04 proves [theorem 7.1](#) for its admissible groups ([section 5.2](#)), a special case of the hypotheses of [theorem 5.2](#), by an explicit description of the order relation:

Theorem 7.2 (Low complexity of all definable relations [04]). *For every admissible D , every first-order definable relation with parameters on M_D , in every finite arity, is simultaneously F_σ and G_δ .*

Proof. It suffices by [theorem 4.10](#) to consider affine atoms. Affine group maps are continuous, so an affine equality defines a closed set. A congruence condition depends only on finitely many discrete integer coordinates and is clopen, by [corollary 4.3](#).

For the order, write two group elements as $(d, z), (e, w)$. The strict comparison is

$$(d, z) < (e, w) \iff e - d > 0 \quad \text{or} \quad (e = d \text{ and } z < w). \tag{7.2}$$

The first condition is open; the second is closed because equality of D -coordinates is closed and the integer comparison is clopen. Thus the order relation is F_σ . Its complement has the identical form with the opposite strict comparison and weak integer comparison, hence is also F_σ . Pulling back by affine maps preserves both classes. Finite Boolean combinations of Δ_2^0 sets remain Δ_2^0 , which proves the claim. $\square$

This is stronger than merely obtaining a Borel presentation. The proof also explains why quantification does not raise the complexity in this particular presentation: its apparent projections are replaced by finite residue searches and finite Boolean operations.

7.1 A formula-by-formula bound on changes of mind

Theorem 7.3 (Effective finite-mind-change truth). *For each first-order Presburger formula $\varphi(\bar{x}; \bar{p})$, one can compute a natural number K_φ and an algorithm producing*

$$A_s(\bar{x}; \bar{p}) \in \{0, 1\} \quad (s \in \mathbb{N})$$

from finite prefixes of the Baire codes, such that

$$\lim_s A_s(\bar{x}; \bar{p}) = \mathbf{1}_{M_B \models \varphi(\bar{x}; \bar{p})}.$$

For every input tuple the displayed sequence changes value at most K_φ times. Parameter codes are supplied as oracles; they are not assumed computable.

Proof. Compute a quantifier-free group formula equivalent to the relativized cone formula, by [theorem 4.9](#). An affine group term has the form (c, n) , where n is an exactly computable integer and each rational coefficient c_j is computable from finite prefixes of the inputs.

To approximate the sign of (c, n) , first use the sign of n . At stage s , inspect $c_0, \dots, c_{s-1}$. If a nonzero coefficient has been found, use the sign of the first such coefficient instead. If $c = 0$, the initial answer was correct. Otherwise the answer becomes correct when the first nonzero coefficient is reached and never changes again. An equality atom can be handled in the same way: it starts with the answer determined by n and becomes permanently false if a nonzero coefficient is found. A congruence atom is decided immediately from n by [\(7.1\)](#).

Evaluate the fixed Boolean combination of these approximations at each stage. Its value can change only when an equality or inequality atom changes. The number of those atom occurrences is a computable bound K_φ . Each stage reads only finitely many digits. The truth set is effectively F_σ because it is

$$\bigcup_N \bigcap_{s \geq N} \{\bar{x} : A_s(\bar{x}; \bar{p}) = 1\},$$

where the stage sets are effectively clopen relative to the parameter codes. The same expression with 0 proves effective G_δ status. $\square$

No effective stopping time is asserted: a later rational coordinate may be the first nonzero one. Deciding sentences and deciding arbitrary parameter instances are different problems. The former is ordinary Presburger decidability; the latter is represented here by a convergent, uniformly bounded-change process.

7.2 Why rank one is impossible for the order

Theorem 7.4 (Open or closed arithmetic order forces discreteness). *Let X carry a model of Presburger arithmetic and a topology in which successor $x \mapsto x + 1$ is continuous. If either of the relations $<$ or $\leq$ is open or closed in X^2 , then X is discrete. Consequently, in any uncountable Polish model with continuous addition, the arithmetic order is neither open nor closed.*

Proof. If $\leq$ is open, then for fixed a both sets $\{x : x \leq a\}$ and $\{x : a \leq x\}$ are open. Their intersection is $\{a\}$, so every point is isolated.

If $<$ is closed, taking the complement and reversing coordinates makes $\leq$ open. If $<$ is open, the discrete-order equivalence

$$x \leq y \iff x < y + 1$$

and continuity of successor make $\leq$ open. Finally, if $\leq$ is closed, then

$$x < y \iff x + 1 \leq y$$

makes $<$ closed, reducing to the previous case. An uncountable discrete space is not separable, hence not Polish. $\square$

Together, [theorems 7.1](#) and [7.4](#) make the Δ_2^0 order bound optimal. This is a statement about the order relation, not about every individual definable set: many congruence sets are clopen. It also rules out a continuous Boolean-valued order test on the Baire model, despite its computable addition.

7.3 A general obstruction from closed order [04]

In a Presburger model M , define truncated subtraction by

$$x \dot{-} y = \begin{cases} z, & y + z = x \text{ for some } z \in M, \\ 0, & x < y. \end{cases}$$

The witness in the first case is unique. Equivalently this is $\max\{x - y, 0\}$ in the ordered group completion.

Theorem 7.5 (Closed order forces discreteness [04]). *Let a model of $\text{Pr}_{\mathbb{N}}$ carry a Hausdorff topology. If its weak order relation is closed in M^2 , then every point of M is isolated. If the topology is also separable, then M is countable.*

Proof. By totality, the complement of the converse weak order is strict order. Thus $<$ is open in M^2 . Fixing either argument gives open order rays. For $x > 0$, let $x - 1$ be its predecessor. Discreteness of the arithmetic order gives

$$\{x\} = \{y : x - 1 < y < x + 1\}.$$

At zero one has $\{0\} = \{y : y < 1\}$. Each singleton is open, so the topology is discrete. A dense subset of a discrete space must contain every point; separability therefore implies countability. $\square$

Corollary 7.6 (No continuous monus on an uncountable separable model [04]). *If truncated subtraction on a Hausdorff Presburger model is jointly continuous, the topology is discrete. In particular no uncountable separable Hausdorff Presburger model has continuous truncated subtraction.*

Proof. The relation $x \leq y$ is exactly $x \dot{-} y = 0$. A singleton is closed in a Hausdorff space, so continuity would make weak order closed. Apply [theorem 7.5](#). $\square$

No continuity hypothesis on addition was needed for this obstruction. Consequently, extending the signature of our examples by their definable monus operation destroys the assertion that all named operations are continuous. Definitional equivalence of first-order theories does not by itself preserve this topological property.

Remark 7.7 (Two incomparable order obstructions [merge]). [Theorem 7.4](#) (source 05) and [theorem 7.5](#) (source 04) both show that the order of a nondiscrete model cannot be too simple, under incomparable hypotheses. 05 assumes that successor is continuous, assumes no separation axiom, and treats four cases ($<$ or $\leq$, open or closed). 04 assumes only that the weak order $\leq$ is closed and the topology Hausdorff, with no continuity at all. Neither implies the other: 04 covers a closed $\leq$ in a topology where successor is discontinuous, 05 covers open orders and topologies that are not Hausdorff. Both are therefore printed. For the metrizable models of this part, where successor is continuous, both give the same conclusion, and so does [corollary 7.6](#) for truncated subtraction, whose exact discontinuity locus is [theorem 8.3](#).

8 Which definable operations are continuous?

The coordinate topology makes several arithmetic operations exceptionally regular, but not every definable operation. This section identifies exact boundaries in the two explicit models.

Proposition 8.1 (Euclidean self-similarity). *In $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$, for each positive standard integer m the map*

$$x \mapsto (q_m(x), r_m(x)) \quad (8.1)$$

is a homeomorphism from the model onto $M \times \{0, \dots, m-1\}$, where the finite factor is discrete. Its inverse is $(q, r) \mapsto mq + r\mathbf{1}$.

Proof. Uniqueness and existence are [proposition 4.2](#). In both chosen coefficient topologies, division of d by m is continuous. The integer maps $n \mapsto \lfloor n/m \rfloor$ and $n \mapsto n \bmod m$ are continuous on a discrete space. Hence (8.1) is continuous into the product. Repeated addition and addition of a fixed standard numeral make its inverse continuous. $\square$

The same formula also explains why all standard congruences ignore the entire coefficient sequence. Two elements have the same remainder modulo every positive standard integer if and only if their integer coordinates agree. The forward direction uses the elementary fact that an ordinary integer divisible by every positive integer is zero. Their infinite coordinates can still be arbitrary and different.

Proposition 8.2 (The total predecessor has exactly one discontinuity). *Define*

$$p(0) = 0, \quad p(x) = x - \mathbf{1} \quad (x > 0).$$

In either explicit model, p is continuous exactly at the points other than 0.

Proof. Away from 0, this is a restriction of translation by $-\mathbf{1}$ in the ambient topological group. Since $\{0\}$ is closed, the nonzero domain is open, and translation has its image in the cone.

In the real model let $t_j = 1/j$; in the Baire model let $t_j = e_j$, the rational sequence with 1 in coordinate j and zeros elsewhere. In either case $t_j > 0$ and $t_j \rightarrow 0$ in D . Then

$$(t_j, 0) \rightarrow 0, \quad p(t_j, 0) = (t_j, -1).$$

The latter sequence does not converge to $p(0) = 0$, because its integer coordinate is always -1 and integer coordinates are discrete. $\square$

Source 04 proves [proposition 8.2](#) (its Proposition 7.4), treating the numerals $(0, n)$, $n \geq 1$, separately:

Second proof (source 04). At points with positive D -coordinate, $p(d, z) = (d, z - 1)$ locally. At an ordinary numeral $(0, n)$ with $n \geq 1$, the same formula holds on its integer-coordinate slice and maps into M_D , so it is continuous there. At zero, the sequence $(s_j, 0) \rightarrow 0$ used above has $p(s_j, 0) = (s_j, -1)$, which cannot converge to zero. $\square$

Theorem 8.3 (Exact discontinuity locus of truncated subtraction). *For $M = M_{\mathbb{R}}$ or $M_{\mathbb{B}}$, let*

$$T(x, y) = \max\{x - y, 0\}$$

using the arithmetic order in the ambient group. Its discontinuity locus is exactly

$$\{((d, n), (d, z)) \in M^2 : n \neq z\}. \quad (8.2)$$

Proof. Both positive and negative cones in the coefficient group D are open for our two examples. If the coefficient difference $d - e$ is nonzero, its sign is locally fixed. Thus T locally agrees either with the continuous ambient difference or with the constant zero map.

Suppose instead that $x = y$. Along inputs sufficiently near this pair, the integer difference is zero. The output is either their ambient difference or zero. For any neighborhood V of zero in the ambient group, both possible outputs lie in V when the inputs are sufficiently near (x, x) . This proves continuity there.

It remains to consider equal coefficient coordinates and $k = n - z \neq 0$. Use the positive null sequence t_j from [proposition 8.2](#). If $k > 0$, replace y by $y + (t_j, 0)$. These inputs converge to (x, y) , but their truncated differences are zero, whereas $T(x, y) = (0, k)$. If $k < 0$, replace x by $x + (t_j, 0)$. The outputs are (t_j, k) , which cannot converge to the original value 0 because their integer coordinate is the nonzero integer k . All perturbed inputs remain in the positive cone. This proves exactly (8.2). $\square$

Source 04 states [theorem 8.3](#) as its Theorem 7.3, for the same two models and with the same locus, written $\text{Disc}(\dot{-}) = \{((d, z), (e, w)) : d = e \text{ and } z \neq w\}$, and adds [04]: *this locus is a closed nowhere dense subset of M^2 , and truncated subtraction is continuous on its dense open complement*. Its proof covers the addition.

Second proof (source 04), with the nowhere-density clause. Suppose first that $d \neq e$. The sign of $d - e$ is locally constant because both strict cones of D are open. On a neighborhood the monus operation is therefore either the continuous ambient difference or the constant zero map. Its values lie in the positive cone, so it is continuous as a map to that subspace.

Next let $d = e$ and $z = w$. In a sufficiently small neighborhood the two integer coordinates stay equal to these fixed integers. Monus has integer coordinate zero and D -coordinate $\max\{u - v, 0\}$. This positive-part map is continuous at zero in both groups under consideration. Over $\mathbb{R}$ this follows from the usual absolute-value estimate. Over $D_{\mathbb{B}}$, if $u - v$ has its first n coordinates zero, so does its positive part: it is either $u - v$ or zero. These finite-prefix neighborhoods prove continuity at the point.

For the converse, choose $s_j > 0$ in D with $s_j \rightarrow 0$: use $1/j$ in $\mathbb{R}$ or the j th unit vector in $D_{\mathbb{B}}$. Let $x = (d, z)$, $y = (d, w)$ and $z \neq w$. If $z < w$, replace x by $x_j = (d + s_j, z) \in M_D$. Then $x_j \rightarrow x$ but

$$x_j \dot{-} y = (s_j, z - w), \quad x \dot{-} y = 0.$$

The output integer coordinate is constantly negative and cannot converge to the zero integer coordinate. If $z > w$, instead replace y by $y_j = (d + s_j, w)$. Then $x \dot{-} y_j = 0$ for every j , whereas $x \dot{-} y = (0, z - w) \neq 0$. This proves discontinuity in both cases, including at the boundary $d = 0$.

The set in (8.2) is closed: equality of the D -coordinates is closed and inequality of the discrete integer coordinates is clopen. It has empty interior. At any of its points, perturbing the first D -coordinate upward by a sufficiently small positive element stays in M_D^2 , lies in any prescribed neighborhood, and makes the two D -coordinates unequal. Thus the closed set is nowhere dense. $\square$

The standard numerals form a closed, relatively discrete subset of both models, yet zero is not isolated in the whole space. This causes no conflict with [proposition 8.2](#). Nor does continuity of standard division imply continuity of every definable piecewise operation: the branch boundary is decisive.

Source 04 continues [04]:

The failure is not an indiscriminate loss of regularity: the graph of monus is still Δ_2^0 by [theorem 7.2](#), because monus is definable. Its ordinary predecessor specialization has a smaller exceptional set.

8.1 Polish completeness need not be translation-invariant [04]

Proposition 8.4 (A complete-metric obstruction [04]). *If zero is nonisolated in a metrizable Presburger model, there is no compatible complete metric invariant under all additive translations. More generally, for any compatible complete metric, the successor map cannot be a uniform homeomorphism onto its image.*

Proof. Successor is a bijection from the model onto $M \setminus \{0\}$. If it were a uniform homeomorphism onto this subspace, every Cauchy sequence in the image would pull back to a Cauchy sequence in the complete domain and would therefore converge in the image. Thus the image would be a complete subspace of a metric space, hence closed. Since zero is nonisolated, $M \setminus \{0\}$ is not closed, a contradiction. A translation-invariant metric would make successor an isometry, which is a special case. $\square$

This applies to both main examples. Their Polishness and continuous addition therefore cannot be replaced by a claim of completeness in the uniform structure of an invariant additive metric.

Structure or operation	Real-coordinate model	Baire-coordinate model
Underlying space	Perfect, locally compact, Polish	Homeomorphic to $\mathbb{N}^{\mathbb{N}}$
Addition	Continuous	Type-2 computable
Fixed quotient/remainder	Continuous	Type-2 computable
All definable relations	Δ_2^0	Δ_2^0
Monus	Discontinuous exactly on (8.2)	Same exact locus
Total predecessor	Discontinuous only at 0	Discontinuous only at 0
Arithmetic order topology	Discrete; not used	Discrete; not used

9 Explicit set-sized fragments of the omnific integers

We use Conway normal-form notation

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma,$$

where S is reverse well-ordered and all coefficients are real. In the omnific ring the occurring exponents are nonnegative, and the coefficient at exponent zero is an ordinary integer. This is the interface recorded in the repository's surreal development [16]. Addition is coefficientwise, and a nonzero normal form has the sign of its leading coefficient.

Theorem 9.1 (One-scale and countable-scale realizations). *There are injective ordered additive maps, fixing 0 and 1,*

$$\iota_{\mathbb{R}} : \mathbb{R} \overrightarrow{\times} \mathbb{Z} \longrightarrow \mathbf{Oz}, \quad \iota_{\mathbb{R}}(r, n) = r\omega + n, \quad (9.1)$$

$$\iota_{\mathbb{B}} : \mathbb{Q}^{\mathbb{N}} \overrightarrow{\times} \mathbb{Z} \longrightarrow \mathbf{Oz}, \quad \iota_{\mathbb{B}}(a, n) = \sum_{j=0}^{\infty} a_j \omega^{1/(j+1)} + n. \quad (9.2)$$

Their positive images, equipped with the transported coefficient topologies, are respectively the Polish models $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$.

Proof. The first formula is a finite normal form. For the second, the fixed exponents

$$1, \frac{1}{2}, \frac{1}{3}, \dots$$

strictly decrease and are positive. Every nonempty subset of these exponents has a largest element, namely the one with least index. Thus every possible support in (9.2) is reverse well-ordered, and adjoining exponent zero preserves that property. The expression is a legitimate normal form, even when infinitely many coefficients are nonzero. All positive-exponent coefficients are real, and the constant term is the integer n , so the value is omnific.

Uniqueness of normal forms gives injectivity. Addition of two such expressions is coordinatewise on a common fixed support, giving additivity. The leading nonzero coefficient is the first nonzero a_j ; if all are zero, the sign is that of n . Therefore the map preserves and reflects the lexicographic order. The same reasoning proves the one-scale case. Transporting a topology through a bijection onto its image preserves all the asserted topological properties. $\square$

9.1 What this realization does and does not preserve

These are ordered additive embeddings, *not* ring embeddings. Both images contain ω , but neither contains ω^2 . Their multiplicative nonclosure is therefore immediate from the normal forms. The stronger theorem in [section 10](#) will show that replacing surreal multiplication by some other multiplication cannot repair the problem while retaining the same carrier, addition, and unit.

The topologies in [theorem 9.1](#) are precisely specified coefficient topologies. The real coordinate is Euclidean in (9.1); the rational coordinates are discrete and combined by a product topology in (9.2). Neither is the discrete arithmetic order topology. No assertion that they are a canonical topology on all omnific integers is intended.

The ordinary constant-term map becomes simply

$$\text{ct}(\iota(d, n)) = n.$$

It is continuous into the discrete integers for these models. Standard congruence data see only this coordinate, matching (7.1). The standard part itself remains externally specified and nondefinable in the Presburger structure.

9.2 Why no proper-class difficulty arises

The source sets $\mathbb{R} \times \mathbb{Z}$ and $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ are ordinary sets. The normal-form constructions above are definable assignments on those sets. Replacement therefore gives set-sized images. The set of birthdays of their elements has an ordinal supremum; choosing a larger ordinal places both images inside a set-sized birthday cutoff of **No**. No explicit optimal cutoff is needed for any theorem here.

Accordingly, one can conduct the entire argument in the product-group models and use [theorem 9.1](#) only as a final realization theorem. This is also the most economical formalization strategy. It avoids making a class-sized object the carrier of a Polish space or a first-order structure in ordinary set-based semantics.

Source 04's account [04]. Source 04 proves [theorem 9.1](#) as its Proposition 10.1 with the same proof; its second map is written ι_{ω} there and ι_B here. It adds the following remarks on the normal-form rule and on the transported topologies.

The construction is purely set-sized, but it has a natural expression in the surreal normal-form language. We use only the classical normal-form rule: a reverse-well-ordered set of surreal exponents with real coefficients defines a surreal sum; addition is coefficientwise, and order is decided by the greatest exponent with nonzero coefficient. Omnific integers have positive growth exponents and an ordinary integer constant term. This is the normal-form description relevant to the ordered-group study in [5] and to the repository's omnific development [19].

We transfer the topologies from the coefficient models to these images. This is not a claim that the whole proper class **Oz** has a Polish topology, nor that a canonical global surreal topology is being restricted. For example, ω/j converges to zero in the transported real-coefficient topology on the first image, despite each term being arithmetically infinite. Likewise $\omega^{1/(k+1)}$ converges to zero in the second coefficient topology.

These are additive subgroups, not subrings. Squaring ω yields ω^2 , which belongs to neither displayed image. The absence of inherited multiplicative closure is consistent with the stronger nonexistence result [corollary 10.5](#) for *any* compatible semiring multiplication on these positive cones.

For foundations, the domains $\mathbb{R} \times \mathbb{Z}$ and $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ are sets. One may work entirely with their normal-form codes in ZFC and interpret the notation inside a sufficiently large set-sized portion of the surreal construction. Alternatively one may use a class theory for the ambient surreal field while retaining these set-sized domains. None of the proofs quantifies over a set of all surreals, a set of all class maps, or a Polish space whose points form a proper class.

10 An absolute obstruction to multiplication

A Borel obstruction would not rule out a wildly discontinuous multiplication. For the models constructed here, the obstruction is stronger and purely algebraic.

Definition 10.1. In a Presburger model M , a positive element u is a *cofinal infinite additive unit* if

$$u > n\mathbf{1} \quad \text{for every standard } n,$$

and for every $x \in M$ there is a positive standard integer N with $x \leq Nu$. The word “unit” here refers only to cofinality under repeated addition; it is not a multiplicative unit.

Source 04 splits the hypothesis of [theorem 10.4](#) into two notions and proves a positivity lemma first:

Definition 10.2 ([04]). An element h in an ordered additive monoid is an *order unit* if for every x there is a positive standard integer N with $x \leq Nh$. It is *arithmetically infinite* if $h > n\mathbf{1}$ for every standard n .

Lemma 10.3 (Positivity is automatic in a semiring expansion [04]). *Suppose a Presburger monoid is expanded to a unital semiring with the same addition, zero, and one. Then the product of two positive elements is positive, and multiplication by a positive element is strictly increasing on either side.*

Proof. If $x, y > 0$, write $x = 1 + a$, $y = 1 + b$ with $a, b \geq 0$. Distributivity and the unit laws give $xy = 1 + a + b + ab \geq 1$. If $x > y$, write $x = y + c$ with $c > 0$. For $z > 0$, one has $xz = yz + cz > yz$ and similarly on the other side. Here the order is the additive natural order of the Presburger model, so no separate order axiom for the multiplication has been assumed. $\square$

Theorem 10.4 (Cofinal-infinite-element obstruction). *A Presburger model with a cofinal infinite additive unit cannot be expanded to a unital semiring with the same addition, zero, and multiplicative unit $\mathbf{1}$. No continuity, measurability, commutativity of multiplication, or induction involving multiplication is assumed.*

Proof. Suppose such a multiplication exists. Cofinality applied to u^2 gives a positive standard N such that $u^2 \leq Nu$. Since u exceeds every standard numeral, it has a decomposition

$$u = (N + 1)\mathbf{1} + v \quad (v \in M).$$

Distributivity and the unit law give

$$u^2 = (N + 1)u + vu \geq (N + 1)u > Nu,$$

a contradiction. The first inequality uses only that vu belongs to the nonnegative carrier. The last one follows from $u > 0$ and additive cancellation. In any semiring expansion of this additive monoid, its natural order $x \leq y \leftrightarrow \exists z (x + z = y)$ is already the Presburger order, so no additional order-compatibility axiom was hidden in the argument. $\square$

Corollary 10.5. *Neither $M_{\mathbb{R}}$ nor $M_{\mathbb{B}}$ admits a unital semiring multiplication with the displayed addition and unit.*

Proof. In $M_{\mathbb{R}}$ take $u = (1, 0)$. It exceeds every standard numeral, and an integer $N > r$ gives $(r, n) < Nu$. In $M_{\mathbb{B}}$ take $u = (e_0, 0)$, where e_0 has first coefficient 1 and all later coefficients zero. For any (a, n) choose an integer $N > a_0$; comparison at coordinate zero gives $(a, n) < Nu$. Apply [theorem 10.4](#). $\square$

In their omnific realizations, this cofinal element is ω . The missing larger scale ω^2 is not merely a defect of one chosen multiplication: every possible semiring multiplication would have to produce an element growing faster than every standard multiple of ω , which the additive carrier cannot contain.

For a different uncountable Polish Presburger model that does allow a semiring expansion algebraically, Glazer's cited Borel obstruction for $Q + I\text{Open}$ would still matter. Our argument is not a replacement for that general theorem; it identifies a stronger obstruction in these particular additive structures.

An arithmetically infinite order unit in 04's sense is exactly a cofinal infinite additive unit in 05's sense, so [theorem 10.4](#) is 04's Theorem 9.3 and [corollary 10.5](#) its Corollary 9.4. Source 04's proof routes the strict inequality through [lemma 10.3](#):

Second proof of theorem 10.4 (source 04). Suppose such a multiplication exists and let h be the order unit. Choose a standard $N \geq 1$ with $h^2 \leq Nh$. Since $h > N1$, [lemma 10.3](#) gives $h^2 > Nh$, a contradiction. $\square$

This algebraic argument is specific to the supplied models and does not replace the general topological theorem about stronger arithmetic. Presburger models need not be additive reducts of models of Peano arithmetic.

10.1 A variation with no largest growth class [04]

Let $D_L = \mathbb{Q}((t))$ be the additive group of formal Laurent series

$$f = \sum_{k \geq k_0} a_k t^k, \quad k_0 \in \mathbb{Z}, \quad a_k \in \mathbb{Q},$$

ordered by the sign of the coefficient at the least nonzero exponent. Use the t -adic topology, with rational coefficients discrete. Its basic neighborhoods at zero are

$$F_n = \{f : v(f) \geq n\}, \quad n \in \mathbb{N},$$

where $v(0) = +\infty$. A bounded compatible metric is $\min\{1, 2^{-v(f-g)}\}$ for $f \neq g$.

Proposition 10.6 (A Laurent-series Polish construction [04]). *The group D_L is admissible, and $M_L = M_{D_L}$ is a Polish Presburger model homeomorphic to Baire space, with continuous addition and fixed quotient and remainder. Unlike the two main models, M_L has no arithmetically infinite order unit.*

Proof. A Cauchy sequence of Laurent series eventually has the same entire negative part, because its pairwise differences eventually lie in F_0 . That common negative part is finite. Every subsequent coefficient stabilizes in turn, giving a Laurent-series limit. Thus the metric is complete. Laurent polynomials with rational coefficients form a countable dense set. The leading sign is determined on a basic neighborhood, so the strict positive cone is open. Addition and division by fixed positive integers are coefficientwise continuous.

For the asserted homeomorphism, decompose D_L into the clopen cosets $p + \mathbb{Q}[[t]]$, where p is a finite negative-power rational polynomial. There are countably many such cosets, and each is a rational product. If $p \neq 0$, the sign on its coset is constant, determined by its leading coefficient. The coset $p = 0$ contributes respectively $(D_B)_{>0}$ or $(D_B)_{\geq 0}$ to the positive or nonnegative cone. Thus both Laurent cones are countable disjoint sums of spaces explicitly homeomorphic to $\mathbb{N}^{\mathbb{N}}$, by the codings in [section 6.3](#). Prepending an index codes such a sum by $\mathbb{N}^{\mathbb{N}}$. The integer slices of M_L are another such sum, proving the claim.

Finally, for $f > 0$ with least exponent k , the element $(t^{k-1}, 0)$ is larger than every standard multiple of (f, z) : its leading exponent is strictly earlier. Standard finite elements also fail to be order units. $\square$

The absence of an order unit does not establish the existence of a semiring expansion. In this example another simple obstruction decides that issue negatively.

10.1.1 A growth-order obstruction

For positive arithmetically infinite elements, define

$$x \asymp y \iff (\exists m, n \in \mathbb{N}_{>0})(x \leq my \wedge y \leq nx).$$

Let $\mathcal{A}(M)$ be the set of these Archimedean growth classes, ordered so that $[x] < [y]$ means $nx < y$ for every positive standard n . These are external set-sized classes of elements of our set-sized model; no first-order definability is being asserted.

Proposition 10.7 (Growth classes must embed into their strict final segments [04]). *If a Presburger monoid M has a unital semiring expansion, then for every arithmetically infinite h , the map*

$$[x] \mapsto [hx]$$

is an order embedding of $\mathcal{A}(M)$ into its strict final segment above $[h]$.

Proof. Multiplication by $h > 0$ preserves and reflects every comparison $x \leq ny$. Preservation follows from distributivity and positivity. For reflection, if $x > ny$ then strict monotonicity gives $hx > n(hy)$. It follows that multiplication induces an order embedding on growth classes. Because x is infinite, $x > n\mathbf{1}$ for every standard n , so $hx > nh$ for every n . Thus its class lies strictly above $[h]$. $\square$

Corollary 10.8 ([04]). *The Laurent-series model M_L also has no unital semiring expansion.*

Proof. Its infinite growth classes are determined exactly by the least Laurent exponent, with increasing growth corresponding to decreasing exponent. Hence $\mathcal{A}(M_L)$ has order type $\mathbb{Z}$. A strict final segment of $\mathbb{Z}$ is well-ordered, while $\mathbb{Z}$ contains an infinite strictly decreasing sequence. There is no order embedding of $\mathbb{Z}$ into such a final segment. Apply [proposition 10.7](#). $\square$

The obstruction is not a statement that every split Presburger group forbids multiplication. For example, take $D = X\mathbb{Q}[X]$ ordered by its highest-degree nonzero coefficient, and let the constant term be an ordinary integer. The ordered ring

$$\mathbb{Z} + X\mathbb{Q}[X]$$

has this split additive group, least positive element 1, and a positive cone closed under its polynomial multiplication. Its growth classes have order type $\mathbb{N}_{>0}$, which does embed in each

of its strict final segments. This countable comparison makes the scope of the obstruction explicit.

11 Endomorphisms, elementary submodels, and a contrast

The two models have markedly different self-embedding behavior. All homomorphisms in the next theorem preserve addition, zero, and **1**; continuity and injectivity are *not* assumed.

11.1 The real model: endomorphisms and elementary submodels

Theorem 11.1 (Rigidity of the real-coordinate model). *Every unital additive endomorphism $f : M_{\mathbb{R}} \rightarrow M_{\mathbb{R}}$ has the form*

$$f(r, n) = (cr, n) \quad (c \in \mathbb{R}_{>0}). \quad (11.1)$$

Conversely each such formula is an endomorphism. All of them are homeomorphisms and Presburger automorphisms. In particular $M_{\mathbb{R}}$ has no proper elementary self-embedding.

Proof. The group completion of the cancellative monoid $M_{\mathbb{R}}$ is $G = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$. Thus f extends uniquely to an additive group map $F : G \rightarrow G$ fixing **1**. Inside G ,

$$\bigcap_{m \geq 1} mG = \mathbb{R} \times \{0\}.$$

One inclusion follows from real divisibility; the other follows because the integer coordinate of an element of the intersection is divisible by every positive integer. Every additive group map preserves this intersection. Hence

$$F(r, 0) = (h(r), 0), \quad F(0, n) = (0, n)$$

for some additive map $h : \mathbb{R} \rightarrow \mathbb{R}$.

Since f maps the positive cone into itself, $h(r) \geq 0$ whenever $r \geq 0$. Thus h is monotone. If $c = h(1)$, rational comparisons and additivity give $h(r) = cr$ for every real r : sandwich r between rational numbers and use the corresponding inequalities for h . Here $c \geq 0$. But $c = 0$ would send the positive element $(1, -1)$ to $(0, -1)$, outside the cone. Thus $c > 0$.

Formula (11.1) and its inverse with scalar $1/c$ preserve the cone, addition, order, and unit, and are continuous in the product coordinates. This proves all assertions. $\square$

Theorem 11.2 (All elementary submodels of the real model). *The elementary submodels of $M_{\mathbb{R}}$ are exactly*

$$M_E = (E \overrightarrow{\times} \mathbb{Z})_{\geq 0}, \quad E \text{ a } \mathbb{Q}\text{-linear subspace of } \mathbb{R}. \quad (11.2)$$

Among them, the only ones Polish in the inherited topology are the standard copy $\mathbb{N}$ and the whole model $M_{\mathbb{R}}$. Every nonzero proper choice of E gives a dense, non-Polish elementary submodel.

Proof. If E is a $\mathbb{Q}$ -linear subspace, then $E \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group. Its inclusion in $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ preserves and reflects the expanded quantifier-free language, including all congruences, since a standard modulus depends only on the integer coordinate. Quantifier elimination makes this inclusion elementary. Relativizing to positive cones proves $M_E \preccurlyeq M_{\mathbb{R}}$.

Conversely, let $S \preccurlyeq M_{\mathbb{R}}$, and let $H = S - S$ be its group completion inside G . If $a, b \in S$ and $a - b \geq 0$, the unique solution of $b + x = a$ lies in S by elementarity. Therefore

$S = H \cap M_{\mathbb{R}}$. Since H contains every standard integer, subtracting the integer coordinate from any $(r, n) \in H$ shows that $H = E \times \mathbb{Z}$ for an additive subgroup $E \leq \mathbb{R}$. For $r \in E_{\geq 0}$, the model's unique division of $(r, 0)$ by a positive standard m has remainder zero and quotient $(r/m, 0)$. Elementarity puts that quotient in S . Applying the same observation to negatives in H shows E is divisible, hence $\mathbb{Q}$ -linear.

If $E \neq 0$, it contains all rational multiples of a nonzero real and is dense in $\mathbb{R}$. Thus M_E is dense in every slice of $M_{\mathbb{R}}$. Suppose M_E is also Polish in the inherited topology. Its closed slice of integer coordinate zero is homeomorphic to $E \cap [0, \infty)$. By [lemma 5.1](#), this is G_δ in $\mathbb{R}$. Its negative is also G_δ , so their union E is G_δ in $\mathbb{R}$. For any $r \in \mathbb{R}$, the dense G_δ sets E and $r + E$ intersect by the Baire category theorem. Hence $r \in E - E = E$, giving $E = \mathbb{R}$. Finally $E = 0$ gives the relatively discrete standard copy $\mathbb{N}$, which is Polish, and $E = \mathbb{R}$ gives the whole model. $\square$

For example, $E = \mathbb{Q}$ gives a countable dense elementary submodel. Its inherited topology is not Polish; there is no inconsistency between its density and the complete metric on the ambient model.

11.2 Reduction to the divisible part [04]

This section considers embeddings in the arithmetic language containing $0, 1, +$. Naming 1 is explicit. Order is already definable, so its preservation is not an extra independent assumption.

Proposition 11.3 (All unital additive maps [04]). *Let D, E be divisible linearly ordered abelian groups. Every map $f : M_D \rightarrow M_E$ preserving $0, 1, +$ is injective and has the unique form*

$$f(d, z) = (T(d), z), \quad (11.3)$$

where $T : D \rightarrow E$ is a $\mathbb{Q}$ -linear order embedding. Conversely every such T defines a map of this kind. Every one of these maps is an elementary embedding of Presburger models.

Proof. If $x > y$, then $x = y + 1 + u$ for some $u \in M_D$. Thus $f(x) = f(y) + 1 + f(u) > f(y)$, so f strictly preserves order and is injective. Every group element is a difference of nonnegative elements. The rule

$$\bar{f}(x - y) = f(x) - f(y)$$

therefore extends f uniquely to an additive order embedding of group completions. It is well defined because $x - y = x' - y'$ implies $x + y' = x' + y$, and f preserves that equality.

For each $d \in D$ and every standard $m \geq 1$, the element $(d, 0)$ is an m -multiple. Its image is consequently in every mG_E , whose intersection is $E \times \{0\}$ by (4.3). Hence $\bar{f}(d, 0) = (T(d), 0)$. Since $f(1) = 1$, the map fixes every standard integer, giving (11.3). Divisibility and torsion-freeness make T $\mathbb{Q}$ -linear; order preservation and injectivity are inherited. The converse follows directly from the formula.

Such a group map preserves and reflects order, equality, and all congruences, the last because integer coordinates are unchanged. Quantifier elimination in [theorem 4.10](#) makes it elementary on the groups. Relativizing to the nonnegative cones makes f elementary as well. $\square$

Source 04 proves [theorem 11.1](#) (its Theorem 8.2) from [proposition 11.3](#):

Second proof (source 04). By [proposition 11.3](#), it remains to classify increasing additive maps $T : \mathbb{R} \rightarrow \mathbb{R}$. Put $c = T(1) > 0$. For rational q one has $T(q) = cq$. For a real r , compare r with rational numbers from below and above and use monotonicity. The resulting inequalities force $T(r) = cr$. The inverse dilation is $f_{1/c}$, and both dilations are continuous in the specified product topology. $\square$

Here 1 in $T(1)$ denotes the real vector $1 \in D$, not the arithmetic unit $(0, 1)$. The distinction prevents a mistaken conclusion that c must equal one. The model is co-Hopfian for unital additive maps but is not rigid.

11.3 The Baire model: all self-embeddings [04] and clopen self-copies

Source 04 classifies every unital additive self-map of M_B ; source 05's shift theorem follows.

For $d \in D_B \setminus \{0\}$ define

$$v(d) = \min\{k : d_k \neq 0\}.$$

Let e_k be the rational sequence with a single 1 in coordinate k , and put

$$F_k = \{d : d_0 = \dots = d_{k-1} = 0\}.$$

Thus $F_0 = D_B$; the subgroups F_k form a neighborhood basis at zero.

Theorem 11.4 (Complete matrix classification [04]). *Every $\mathbb{Q}$ -linear order embedding $T : D_B \rightarrow D_B$, without any assumed continuity, has the form*

$$(Td)_i = \sum_{k:p(k) \leq i} a_{ik} d_k, \quad (11.4)$$

where $p : \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing and

$$a_{ik} = 0 \ (i < p(k)), \quad a_{p(k),k} > 0. \quad (11.5)$$

The coefficients below each pivot, a_{ik} with $i > p(k)$, are arbitrary rationals. Conversely every such matrix defines an order embedding. All the sums in (11.4) are finite. Every such map is continuous and has closed image. It is surjective exactly when $p(k) = k$ for all k .

Proof. Let T first be an arbitrary $\mathbb{Q}$ -linear order embedding. Since $Te_k > 0$, its first nonzero coefficient is positive. Define $p(k) = v(Te_k)$. The inequalities

$$ne_{k+1} < e_k \quad (n = 1, 2, \dots)$$

remain true after applying T . If $p(k+1) < p(k)$, the image on the left already dominates the right for $n = 1$. If the pivots were equal, a large enough standard n would make the positive leading rational coefficient on the left larger than the one on the right. Both are impossible. Thus $p(k+1) > p(k)$, and in particular $p(k) \geq k$.

Every $d \in F_k$ satisfies $|d| < Ne_k$ for some standard integer $N > 0$: choose $N > |d_k|$, with a further unit of margin. Order preservation implies $|Td| < NTe_k$, hence

$$T(F_k) \subseteq F_{p(k)} \subseteq F_k. \quad (11.6)$$

This already proves continuity at zero, and therefore everywhere by additivity. It also excludes noncontinuous linear maps arising from a choice of a vector-space basis.

Let $a_{ik} = (Te_k)_i$. To compute coordinate i of Td , subtract the finite initial sum $\sum_{k=0}^i d_k e_k$. The remainder belongs to F_{i+1} , whose image has zero coordinates through i by (11.6). Thus $(Td)_i = \sum_{k=0}^i a_{ik} d_k$. Terms with $p(k) > i$ vanish, yielding (11.4) and (11.5).

Conversely, a strictly increasing p satisfies $p(k) \geq k$, so each row in (11.4) has finite support. The formula defines a linear continuous map on all rational sequences. If $d \neq 0$ has first nonzero index k , then all rows before $p(k)$ vanish and

$$(Td)_{p(k)} = a_{p(k),k} d_k.$$

Its sign is the sign of d_k . This proves order preservation and injectivity, including comparison of arbitrary pairs by their difference.

To describe the range, for any proposed target sequence y recursively recover

$$d_k = \frac{y_{p(k)} - \sum_{j < k} a_{p(k),j} d_j}{a_{p(k),k}}. \quad (11.7)$$

Each recovered coordinate is a continuous function of finitely many coordinates of y . The target belongs to the range exactly when, for every nonpivot row i ,

$$y_i = \sum_{k:p(k) < i} a_{ik} d_k. \quad (11.8)$$

These are closed, indeed clopen, finite-coordinate conditions. Their intersection is closed. If every row is a pivot, the recovery solves the system for every target. This happens exactly when $p(k) = k$. If a row j is omitted, no image has its first nonzero coordinate equal to j , so surjectivity fails. $\square$

Corollary 11.5 (Automatic continuity and the topology of self-images [04]). *Every unital additive self-map of M_B is elementary, continuous, and a homeomorphism onto a closed image. Its image is clopen when the complement of $p(\mathbb{N})$ is finite, and closed nowhere dense when that complement is infinite. The automorphisms are exactly the infinite lower triangular rational matrices with strictly positive diagonal, acting on the divisible coordinate and fixing the integer coordinate.*

Proof. Combine proposition 11.3 and theorem 11.4. The inverse on the image is continuous by the coordinate-recovery formula. The group image is $T(D_B) \times \mathbb{Z}$, and its intersection with M_B is exactly the monoid image, because T reflects order.

When finitely many rows are not pivots, (11.8) gives a finite intersection of clopen conditions, so the group and monoid images are clopen. When infinitely many rows are not pivots, they occur arbitrarily far out. At an image point whose D_B -coordinate is positive, keep an initial segment containing its leading coordinate and change a later nonpivot coordinate so that its constraint fails. The result remains positive and lies in any sufficiently specified cylinder. At a point $(0, z)$ in the image, necessarily $z \geq 0$. In any given zero-prefix cylinder first insert a positive pivot far beyond that prefix, using a small-in-topology image vector Te_k . Then violate a still later nonpivot constraint. Positivity is unchanged and the resulting point is outside the image. Thus the image has empty interior even at the boundary. It is closed, hence nowhere dense. $\square$

Example 11.6 (Two contrasting proper elementary copies [04]). The shift

$$S(d)_0 = 0, \quad S(d)_{k+1} = d_k$$

has pivots $p(k) = k + 1$ and produces a proper clopen elementary self-copy of M_B . The insertion map

$$E(d)_{2k} = d_k, \quad E(d)_{2k+1} = 0$$

has pivots $p(k) = 2k$ and produces a closed nowhere dense elementary self-copy. Both are homeomorphisms onto their images. Their order-theoretic and model-theoretic behavior is identical at the level of elementary embedding, while their image interiors are different.

Theorem 11.7 (Clopen elementary self-copies of the Baire model). *Let*

$$\sigma((a_0, a_1, a_2, \dots), n) = ((0, a_0, a_1, a_2, \dots), n).$$

Then $\sigma : M_B \rightarrow M_B$ is a proper elementary embedding and a homeomorphism onto a clopen subspace. Its iterates satisfy

$$\bigcap_{k \geq 0} \sigma^k(M_B) = \{(0, n) : n \in \mathbb{N}\}.$$

Thus a descending sequence of proper clopen elementary self-copies has intersection exactly the standard elementary submodel.

Proof. Inserting an initial zero preserves and reflects the lexicographic order, addition, and every standard congruence. It maps the divisible coefficient group into the subgroup with first coordinate zero. Quantifier elimination gives elementarity, first for the groups and then for their cones. The image is exactly the subset with $a_0 = 0$, which is clopen because the first rational coordinate is discrete. Deleting that zero is a continuous inverse on the image. It is proper, since elements with $a_0 > 0$ are omitted. The k -th image has its first k rational coordinates zero. Membership in all images therefore means that every rational coordinate is zero; positivity then forces $n \geq 0$. $\square$

These clopen submodels are external subsets, not Presburger-definable subsets, even with parameters: a proper definable subset containing zero and closed under successor would violate induction. Topological simplicity and internal definability are different properties.

[merge] The map σ of [theorem 11.7](#) (source 05) is the shift S of [example 11.6](#) (source 04), with pivots $p(k) = k + 1$. Source 05 adds the intersection of the iterated images; source 04 adds the insertion map E , whose image is closed and nowhere dense.

12 A compactness obstruction independent of multiplication

The positive answer on Baire space does not extend to every familiar uncountable Polish space. In particular compact carriers are impossible. The underlying fact is the standard compact-cancellative-monoid argument, which we include in its commutative form.

Lemma 12.1 (Compact cancellation produces inverses). *Every compact Hausdorff commutative cancellative topological monoid is a group algebraically.*

Proof. Write the identity as 0 and fix a . Let S be the closure of $\{na : n \geq 1\}$. Continuity of addition makes S a nonempty compact subsemigroup. By the compact intersection property and Zorn's lemma, S has a minimal nonempty compact subsemigroup K .

For $x \in K$, the set $x + K$ is a nonempty compact subset of K and is a subsemigroup: for $u, v \in K$,

$$(x + u) + (x + v) = x + (x + u + v) \in x + K.$$

Minimality gives $x + K = K$. Since $x \in K$, there is $y \in K$ with $x + y = x = x + 0$, and cancellation gives $y = 0$. Thus $0 \in K \subseteq S$. But $a + X$ is compact and hence closed, and contains every positive multiple of a . Therefore it contains S and in particular 0 . So $a + b = 0$ for some $b \in X$. $\square$

Corollary 12.2 (No compact positive-cone model). *No model of Presburger arithmetic has a compact Hausdorff topology with continuous addition. In particular Cantor space cannot be a carrier, although Baire space can.*

Proof. Its addition is a commutative cancellative monoid. The lemma would give an additive inverse of $\mathbf{1}$, but no nonnegative element can sum with $\mathbf{1}$ to zero. $\square$

This obstruction uses neither Borel definability nor multiplication. It is separate from [theorem 10.4](#): the real model is locally compact but not compact, and the Baire model is nowhere locally compact.

13 Formalization and reproducibility

13.1 A minimal theorem dependency graph

The construction naturally splits into independently checkable layers. The core algebra does not require real analysis or surreal numbers. First define a discretely ordered abelian group with standard residue representatives. Prove the coset-invariant finite-search lemma. Then prove coefficient normalization, elimination, completeness, and the positive-cone interpretation. Only afterward supply the real or rational-sequence carrier and its topology. The omnific realization can be verified last, as an independent bridge.

A practical Lean development can use the following *proposed* modules. These names are a plan, not existing checked files:

Proposed module	Required mathematical content
<code>ZGroup.Basic</code>	Least positive element, standard residue witnesses, uniqueness, and order properties of the quotient.
<code>ZGroup.CosetCooper</code>	Full mG -invariance and the finite-search criterion; the counterexample to weaker periodicity.
<code>ZGroup.Elimination</code>	Affine-term semantics over an arbitrary $\mathbb{Z}$ -group, coefficient scaling, quantifier elimination.
<code>PolishPresburger.Real</code>	The positive cone of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ and its complete slice metric.
<code>PolishPresburger.Baire</code>	The rational-sequence cone and the explicit block/tree homeomorphism.
<code>PolishPresburger.Structure</code>	Order-rank obstruction, absence of semiring expansions, and self-embedding theorems.
<code>PolishPresburger.Omnific</code>	The fixed-support normal-form maps and their additive/order laws.

13.2 The nonstandard Cooper boundary must be explicit

The inspected repository theorem `periodic_interval` has an ordinary-integer predicate and a one-step periodicity hypothesis [15]. Its proof can iterate by any ordinary integer

quotient. Over a general $\mathbb{Z}$ -group, the quotient in $x - L = mq + r$ is an element of that group, not necessarily a standard numeral. A proof by ordinary natural-number induction cannot supply invariance under that shift for an arbitrary external predicate.

The correct interface is therefore a predicate on the quotient G/mG , or equivalently the hypothesis (4.4). Affine congruence atoms satisfy it by algebra, and Boolean combinations preserve it. This makes the generalized arithmetic lemma reusable and prevents an incorrect formal transfer from being hidden by notation. The original integer theorem should be retained as an instance; it does not need to be weakened or retracted.

The existing decision procedure's semantics are over integers. Reusing its syntax and transformations is plausible, but proving soundness over all $\mathbb{Z}$ -groups is an additional obligation. The repository's existence of an executable integer eliminator, by itself, does not certify [corollary 4.11](#) for a new carrier.

13.3 Executable material included with this manuscript

`polish_presburger.py` implements exact finite-rank rational arithmetic and the Baire codec of [section 6](#). The codec uses a computable positive-rational enumeration and Cantor pairing, supplies addition and standard division on infinite-sequence oracles, and gives a one-change comparison approximation. Every Baire input code is valid. The lower-level encoding routine takes a positive-cone promise, documented in its interface; the public arithmetic operations preserve that promise.

`verify.py` performs deterministic tests using exact fractions.Fraction arithmetic and a fixed seed. The executed run produced **17,930 successful checks**. They cover pairing and rational-enumeration round trips, additive and order identities, Euclidean division, cone closure, coset-invariant finite candidates, Baire-code prefix round trips, arithmetic output prefixes, the comparison change bound, and the predecessor boundary witness. The deliberately false weak-periodicity generalization is tested as a counterexample, not asserted as a theorem.

Reproduction

With Python 3.10 or later, run `python verify.py`. No external Python package is needed. The script writes `verification_results.json`. Compile the article twice with `pdflatex -interaction=nonstopmode -halt-on-error polish_presburger_glazer.tex` to resolve references.

Tests on finite-rank elements do not quantify over arbitrary elements of $\mathbb{Q}^{\mathbb{N}}$, and finite prefixes do not decide equality of infinite streams. Polishness, induction, and universal semantic statements are established by the proofs, not by the test count. No new Lean or Rocq formalization is included in this archive.

[write] The box above gives source 05's delivery names. In this report 05's programs are `code/05-omnific-presburger-polish_presburger.py` and `code/05-omnific-presburger-verify.py`, its recorded run is `data/05-omnific-presburger-verification_results.json`, and its manuscript is merged into `this_article.tex`. The runner imports `polish_presburger` by its delivery name and writes its record beside itself, so it is rerun on a copy with the delivery names restored (the report's README gives the commands). The intake reran it on such a copy with Python 3.14.4: 17,930 checks passed, and the record differed from the delivered one only in the interpreter version and line endings.

13.4 The formalization interface, source 04's audit [04]

Source 04 inspected the Rocq file of the formal project rather than the Lean file, at its own pin.

13.4.1 A pinned, limited source audit

The repository files used in this manuscript were checked at commit

3ad5f878c26d2d82c092190af2f3edc50d6feb06.

The relevant Presburger source is `Logic/PresburgerArithmetic/Coq/Cooper.v`; the project description is `Logic/PresburgerArithmetic/README.md`. The latter describes the Lean pipeline through syntax, normal form, one-variable elimination, and sentence decision [17]. The present manuscript does not claim to have rebuilt that repository or audited its unrelated mathematical projects.

In the inspected Coq source, the predicate hypothesis is defined over *ordinary* integers by one-step translation invariance. The lemma `periodic_shift` then proves invariance under arbitrary integer multiples, using induction on ordinary positive and negative integer representations. This supports `periodic_has_residue` and the interval criterion [18]. That is sound for its stated standard-integer domain.

For a model-uniform version, replacing the type of integers by an arbitrary Presburger group while leaving the predicate hypothesis unchanged would be invalid: [proposition 4.6](#) is a counterexample. The quotient in $x = mq + r$ may have a nonzero divisible coordinate, and iteration through standard integer shifts does not reach mq . The repair is conceptual, not a missing induction tactic: use [definition 4.5](#). Congruence literals satisfy that stronger property directly, so [theorem 4.8](#) supplies the needed uniform lemma.

13.4.2 A concrete implementation sequence

A formalization can be divided into independently reviewable stages. The following are proposed new components, not assertions about existing repository declarations.

Abstract arithmetic semantics. Define a class of linearly ordered abelian groups with a named least positive unit and a standard finite quotient–remainder principle. Interpret the existing affine syntax in an arbitrary group of this class. Keep the syntactic integers as *coefficients*; they must not be confused with the semantic carrier.

Finite boundary theorem. Formalize m -residue invariance as invariance under differences in mG . Prove [theorem 4.8](#), separating existence of a finite candidate from any decidability hypothesis on its predicate. Then specialize to a Boolean evaluator when the residue conditions are decidable. This keeps the mathematical statement valid even when the carrier has undecidable equality or order in a computational representation.

Model-uniform elimination. Reuse the finite syntax transformations, with a fresh semantic correctness proof for coefficient normalization, cancellation in (4.9), bound selection, and finite substitution. The correctness endpoint should quantify over all groups satisfying the abstract assumptions. Specialization to $\mathbb{Z}$ must recover the current integer semantics rather than replace it.

Split models and topological structure. Construct G_D and its nonnegative cone, prove explicit division, and obtain the complete Presburger theory by relativization. Next formalize the G_δ argument, continuity of restricted addition, and the coordinate formulas for fixed quotient and remainder. For D_B , construct the two coding homeomorphisms rather than invoking a black-box classification of zero-dimensional Polish spaces.

Self-embeddings. First prove extension from the positive cone to the group completion and preservation of the divisible subgroup. For D_B , prove the filtration lemma (11.6) before introducing matrices. This exposes the reason for automatic continuity and avoids an unproved assumption that an arbitrary linear map is determined by its values on the coordinate vectors. Those vectors are *not* a vector-space basis of the full product; the finite-row conclusion requires the order argument.

13.4.3 Trust and proof-status ledger

Component	Status in this package	Dependence
Presburger characterization and QE	Classical; self-contained argument supplied	Ordered groups and finite syntax
Affirmative Polish construction	Conventional proof supplied	Explicit split group and topology
Baire homeomorphism and stream operations	Proof and executable implementation	Effective rational coding
Monus and embedding theorems	Conventional proofs supplied	Earlier lemmas in this manuscript
Finite arithmetic diagnostics	Executed successfully	Python standard library, exact fractions
New Lean/Rocq kernel certification	Not supplied	Proposed work above
Historical priority of the answer	Not established	Further literature and expert review

The distinction is especially important here because the repository already contains machine-checked material. Reusing its mathematical strategy does not automatically confer its certification status on the new semantic carrier or on the topology.

13.5 Relation to the formal project `Logic/PresburgerArithmetic` [write]

At the placement, the project's Lean development (namespace `PresburgerArithmetic`) and its Rocq file are unchanged from the two pins. The declarations the sources discuss are:

- Lean, `Lean/PresburgerArithmetic/Cooper.lean`: `periodic_shift_of_mod`, `periodic_interval`, `periodic_has_residue`, `periodic_unbounded_above`, `periodic_unbounded_below` and `cooper_finite_criterion`, all for a predicate P on `Int` with the one-step hypothesis $\forall x (P(x + m) \leftrightarrow P(x))$;
- Lean, `Lean/PresburgerArithmetic/Decision.lean`: the decision procedure `PresburgerArithmetic.Formula.presburgerArithmeticDecidable` and its correctness theorem `Formula.decideSentence_eq_true_iff`;

- Rocq, Coq/Cooper.v: `periodic`, `periodic_shift`, `periodic_has_residue`, `periodic_interval`, `periodic_unbounded_above`, `periodic_unbounded_below`, `cooper_finite_criterion` and `cooper_step_decidable`, all over $\mathbb{Z}$.

These statements are about the ordinary integers, where they are correct. [Example 4.4](#) and [proposition 4.6](#) show that their periodicity hypothesis cannot be read over a nonstandard $\mathbb{Z}$ -group, and [lemma 4.7](#) and [theorem 4.8](#) state the coset-invariant replacement that a model-uniform version would need. Neither source changed the project, and neither reports a defect in it. **No statement of this report is formalized:** the proposed modules of [section 13](#) and the stages of [section 13.4](#) are plans, and the collection's docs/FORMALIZATION.md maps no label of this report.

14 Further research questions

Sources 05 and 04 each end with a list of questions (ten and eleven). Four questions occur in both lists and are merged in [section 14.1](#); the remaining six of 05 and seven of 04 follow, so that this part has seventeen questions. Source 05 introduces its list as follows, and source 04 its own list as in the second paragraph. None of these questions is answered in this report; [Part II](#) bears on [question 14.4](#) and on the last questions of both lists ([section 26](#)).

The following questions are proposed directions arising from the proved results. Except where explicitly linked to the source question, their status is not presented as a certification that the literature has no answer.

The construction answers the printed existence question, but it leaves several sharper problems. The questions below distinguish implementation work, structural problems, and possible new obstructions. No solution to them is asserted merely by listing them.

14.1 Questions asked by both sources

Question 14.1 (A formal certificate for the affirmative answer [05, 04]). *Source 05:* Can the one-scale model, the full $\mathbb{Z}$ -group elimination theorem, and its Polish topology be formalized with a compact audit boundary in ProveIt? The key initial milestone is the generic coset-invariant Cooper lemma, rather than merely a finite test of arithmetic identities. *Source 04:* Can the complete proof of [remark 3.2](#), including the Baire homeomorphism and the full induction scheme via model-uniform elimination, be added to ProveIt with no new axioms? The first milestone should be the finite boundary lemma with residue invariance, followed by the semantic specialization checks described in [section 13.4](#).

Question 14.2 (Weak metatheory [05, 04]). *Source 05:* What weak foundational theory suffices to certify the affirmative construction, including the chosen representation of Polishness? The real model has an explicit complete metric, and the Baire model has an explicit sequence code, so a general model-existence theorem is unnecessary. This is distinct from Glazer's Question 1 concerning the strength needed for his stronger-arithmetic obstruction. *Source 04:* What is the weakest useful subsystem of second-order arithmetic in which the Baire-space construction, its stream algorithms, and its formula-by-formula model correctness can be coded and proved? Distinguish coding all points of Baire space from asserting a set of all such points. This question concerns the positive construction, not a resolution of the separate reverse-mathematical problem posed in Glazer's Question 1 [1].

Question 14.3 (Connected carriers [05, 04]). *Source 05:* What additional obstruction prevents or permits a continuous-addition model on a connected Polish space? Our locally compact

model has countably many ray components, and the arithmetic order cannot be open or closed. A proof for a prescribed connected carrier must use more than the order-topology observation alone. *Source 04*: Can a connected uncountable Polish space support a Presburger model with continuous addition? Our examples do not decide this: their integer coordinate is a continuous map onto a countable discrete space. Requiring all fixed remainders to be continuous would already force nontrivial clopen partitions, so the connected problem must separate continuity of addition from continuity of these definable functions.

Question 14.4 (Multiplication and growth orders [05, 04]). *Source 05*: Can one give an equally explicit Polish continuous-addition Presburger model whose underlying additive structure admits a unital semiring expansion? Any candidate must avoid a cofinal infinite additive unit. If one also requires $Q + IOpen$, the multiplication cannot be Borel by the cited obstruction. Separating algebraic expandability from Borel expandability is the central issue. *Source 04*: Find admissible uncountable topological groups D whose infinite growth order embeds into each of its strict final segments, thereby passing [proposition 10.7](#). For their positive cones, determine whether any semiring multiplication exists, and, separately, which descriptive regularity it can have. Glazer's obstruction to Borel multiplication under stronger arithmetic remains relevant, but it does not by itself classify bare semiring expansions.

[merge] For [question 14.4](#): the real model is the additive monoid of one stratum of 03's ring $\mathcal{A}_{fin, \geq 0}$, which does carry a semiring multiplication, Borel and continuous off a closed nowhere dense set ([theorem 24.3](#)); but $\mathcal{A}_{fin, \geq 0}$ is not a model of Presburger arithmetic, and on it continuous addition is impossible ([theorem 25.2](#)). 03's [question 28.1](#) asks the multiplicative counterpart on $\mathcal{A}_{fin, \geq 0}$.

14.2 Further questions of source 05

Question 14.5 (Prescribed zero-dimensional locally compact carriers). Which uncountable zero-dimensional locally compact Polish spaces support continuous-addition Presburger models? Compact spaces are excluded and Baire space is realized, but Baire space is nowhere locally compact. This leaves a meaningful topological region not settled by our examples.

Question 14.6 (Classifying admissible coefficient groups). Which divisible ordered Polish abelian groups have G_δ positive cone? [Theorem 5.2](#) turns this into an exact criterion for the split construction. Can their possible underlying topologies, order ranks, and continuous order-preserving endomorphisms be classified under useful additional hypotheses?

Question 14.7 (Continuity of a richer definable language). How much of Presburger definable choice can be made continuous in one uncountable Polish presentation? Standard quotients and remainders are continuous here, but total predecessor and truncated subtraction are not everywhere continuous. Is there a natural maximal collection of definable operations admitting simultaneous continuity?

Question 14.8 (Optimal finite-mind-change complexity). For a given Presburger formula, what is the smallest possible uniform change bound in the Baire presentation? The proof gives the number of order/equality atom occurrences after elimination. That bound need not be optimal and may be much larger than necessary. Relating the minimal bound to the finite difference hierarchy or to formula structure would refine the Δ_2^0 result.

Question 14.9 (Polish elementary-submodel lattices). Which lattices of closed or clopen elementary submodels can occur in a Polish continuous-addition Presburger model? The real model has only two Polish elementary submodels with inherited topology, whereas

the Baire model has infinitely many proper clopen elementary self-copies with standard intersection. What intermediate behaviors can be realized?

Question 14.10 (Canonical surreal cutoffs and topologies). Find useful explicit birthday bounds for the two fixed-support omnific images and formalize the coefficient-topology transport inside those cutoffs. More generally, which support sets admit Polish coefficient topologies with continuous addition and full Presburger induction on the resulting positive cone? Such a classification should keep ring closure separate from additive closure.

14.3 Further questions of source 04

The last question below and the last question of [section 14.2](#) (canonical surreal cutoffs and topologies) overlap in their second halves: both ask which set-sized supports or subgroups of $\mathbf{Oz}$ carry Polish coefficient topologies with continuous addition. They are printed separately because their first halves differ. [Theorem 25.2](#) gives a negative datum for one ring-closed family ([section 26](#)).

Question 14.11 (Beyond split topological models). Which Polish presentations of Presburger groups admit a continuous integer-coordinate retraction and a topological decomposition $D \twoheadrightarrow \mathbb{Z}$ of the kind used here? What can occur when the algebraic extension or the topology does not split? Any characterization must separate the standard subgroup, the divisible subgroup, and the full system of residues modulo standard integers.

Question 14.12 (Rigidity with continuous addition). Is there an uncountable rigid Presburger model with a Polish topology making addition continuous and order Borel? Rigid models are studied by Jeřábek [4]; our split examples are not rigid, because positive rational scaling of their divisible part gives nontrivial automorphisms. One must therefore move beyond the present easy symmetry.

Question 14.13 (Exact discontinuity profiles of definable functions). For each definable function $f : M_D^k \rightarrow M_D$, can its discontinuity set be computed from a quantifier-free description of its graph? For the two main models, determine whether every such set admits a finite description in terms of equality of divisible-coordinate affine expressions and residue conditions. Keep graph complexity separate from target-valued Baire class; they are not interchangeable statements for arbitrary target spaces.

Question 14.14 (Embedding categories for other coefficient groups). Extend [theorem 11.4](#) to products of ordered coefficient groups other than $\mathbb{Q}$, and to countable support systems with more complicated growth orders. Which hypotheses force every order embedding to be continuous? Which image topologies replace the finite-versus-infinite omitted-pivot dichotomy?

Question 14.15 (Effective truth versus effective operations). The Baire presentation has computable addition and fixed division, but not a computable test for every definable relation on arbitrary input streams. Give sharp oracle bounds for evaluation of formulas in this presentation, including uniform bounds under parameter coding and quantifier elimination. Distinguish the decidability of closed Presburger sentences from the discontinuous evaluation of an atomic relation on arbitrary infinite names.

Question 14.16 (Metrics compatible with arithmetic translations). What useful uniform continuity properties of addition can be retained by compatible complete metrics after the invariant-metric obstruction in [proposition 8.4](#)? Determine whether addition can be made jointly uniformly continuous, or all translations uniformly continuous in one direction,

and quantify the unavoidable distortion of their inverses. This would clarify completion arguments for nonclosed positive cones.

Question 14.17 (Coefficient topologies inside surreal arithmetic). Classify set-sized omnific subgroups carrying Polish coefficient topologies with continuous addition, and determine how those topologies behave under changes of normal-form support. The natural comparison is between coefficient topology, arithmetic order, and model-theoretic elementary embedding, not between a Polish topology and the whole proper class of surreals.

15 Conclusions of Part I

The affirmative construction can be summarized by one change of viewpoint: use the lexicographic order to interpret arithmetic, and a coefficient topology to interpret continuity. Divisibility of the leading-coordinate group supplies all standard residues; quantifier elimination then supplies full Presburger induction. Neither countability of the standard model nor discreteness of the arithmetic order forces the external presentation to be countable or discrete.

The real model gives a locally compact answer, while the rational-sequence model gives a genuinely effective answer on Baire space. Their order relations have the smallest possible nontrivial Borel rank, yet they exhibit different elementary self-embedding behavior. Their realization inside **Oz** connects these results directly to the repository's normal-form arithmetic. Finally, the cofinal-infinite-element argument shows exactly why these additive examples cannot be mistaken for topological models of full arithmetic.

Source 04's conclusion [04]. The separation needed for the construction is simple: place a divisible ordered group above a discrete integer coordinate algebraically, while retaining the ordinary or product topology on that divisible group. The arithmetic becomes discretely ordered without forcing the topology to be discrete. Unique standard residues give the whole Presburger theory, not just a fragment.

The real-coordinate construction supplies a perfect locally compact example; the rational product supplies literal Baire space and effective stream operations. Their regularity is sharply delimited: every definable relation has Borel rank at most two, whereas monus is discontinuous exactly on an explicitly described geometric boundary. Their self-embeddings range from unavoidable surjective dilations to closed nowhere dense elementary copies, but all are automatically continuous.

For Provell, the most immediate next step is precise and finite: generalize the Cooper residue lemma with the correct subgroup-invariance hypothesis, then connect the already familiar syntax to these new semantics. That would turn the conventional answer presented here into a kernel-checked one without changing the mathematical construction.

Part II

Borel presentations and support barriers

This part is source 03, pinned to repository commit c5b4881fa. It keeps 03's definitions, in particular coefficient observability ([definition 17.1](#)) and its two exponent conventions: $X = t^{-1}$ is read as ω , and the change $q \mapsto -q$ converts the increasing t -convention into the

reverse X -convention (section 17). Four symbols are renamed (section 1.3): 03's ring $\mathcal{A}$ is $\mathcal{A}_{\text{fin}}$, its rational Hahn ring H is $\mathcal{H}_{\mathbb{Q}}$, its Kleene–Brouwer embedding h is μ_{KB} , and two local sets called D are W and $\mathcal{Z}$. Several of 03's results restate results already in the collection; they keep their proofs here, because theorem 25.2 needs their exact hypotheses, and a note after each names the collection's statement. In 03's text, "this report" means source 03.

16 The intersection and the questions answered

Glazer's paper *A Topological Tennenbaum Theorem* studies the interaction between arithmetic, Borel definability, and complete separable topologies. The part used here is particularly well matched to omnific arithmetic: an uncountable Polish model of Robinson arithmetic plus open induction cannot have continuous addition and Borel multiplication [1, Corollary 1]. This is weaker than the hypothesis needed in his separate multiplication obstruction; we preserve that distinction throughout.

The relevant ProveIt material is the surreal-number project, especially the reports on omnific notations and computer algebra, and its construction of the omnific ring and floor. The inspected repository guides explicitly distinguish research drafts, finite verification programs, and statements mapped to Lean proofs [20, 21]. That distinction is retained here. The repository already reports a Π_1^1 classification of unrestricted support validity, using Kleene–Brouwer coding. Thus merely repeating a well-founded-tree reduction would not constitute a new advance on this material.

Instead, we ask for an exact boundary and a positive arithmetic counterpart. For which countable exponent orders is support validity Borel? Does the answer also determine whether the whole Hahn field can be given a standard Borel presentation with readable coefficients? Can a concrete uncountable integer part have arithmetic that is continuous almost everywhere in the category sense, even though Glazer rules out global continuity? Precisely where does continuity fail?

16.1 Main theorem map

The report proves the following connected package.

1. **Support and field-presentation dichotomy.** For countable L , $\text{WO}(L)$ is Borel if and only if L contains no copy of $\mathbb{Q}$. Otherwise it is coanalytic-complete (Theorem 18.4). For countable ordered abelian Γ , the same condition characterizes coefficient-observable standard Borel presentations of $\mathbb{R}((t^\Gamma))$ (Theorem 20.2).
2. **Exact benchmark complexities and a root-closure obstruction.** The support predicates for $\mathbb{Z}$ and $\mathbb{Z}_{\text{lex}}^d$, $d \geq 2$, are respectively Σ_2^0 -complete and Π_3^0 -complete. A nontrivial countable exponent group in a square-root-closed full real Hahn field cannot lie on the Borel side (Theorem 19.1, Corollary 20.4).
3. **Uniform bounds and localization.** Every Borel coefficient-observable family of valid series has its support order types bounded below one countable ordinal. Arbitrarily large countable support types, and coanalytic-complete validity, already occur in a fixed 0–1 coefficient slice inside $[X, 2X)$ (Theorems 21.2 and 22.1).
4. **A positive model with exact negative boundaries.** The finite principal-part model $\mathcal{A}_{\text{fin}, \geq 0}$ satisfies open induction and has a Polish presentation with explicit closed

nowhere dense discontinuity sets for both operations. Yet no Polish topology with the same Borel sets makes addition continuous everywhere (Theorems 23.2, 24.3, and 25.2).

The order-theoretic dichotomy is deduced from classical Hausdorff theory and well-founded-tree completeness. The uniform bound is an application of classical analytic boundedness, for which we include a proof. The integer-part model is of Shepherdson type, not a newly discovered model of arithmetic. The report's claims of contribution concern the explicit combined classification, the observable-field criterion, the benchmark calculations, and the exact topological analysis. These are proved here; bibliographic priority is not asserted.

16.2 A precise source boundary

The source audit used selected repository guides at commit

c5b4881fa603339c7ff90f5e1098ede0651243d8.

The inspected paths are recorded in Appendix D. This was not an exhaustive audit of the repository, nor a rebuild of its Lean development. No new theorem below depends on an unreviewed repository theorem: the required normal-form and integer-part arguments are supplied independently. The repository is used to identify the research overlap and to avoid claiming its previously recorded results as new.

17 Conventions and the meaning of a presentation

17.1 Two opposite exponent conventions

For an ordered abelian group Γ , a Hahn series is written

$$f = \sum_{\gamma \in \Gamma} a_{\gamma} t^{\gamma}, \quad a_{\gamma} \in \mathbb{R},$$

with $\text{supp}(f) = \{\gamma : a_{\gamma} \neq 0\}$ well-ordered in the increasing order of Γ . Its sign is the sign of the coefficient at the least exponent. Thus t is infinitesimal when $\Gamma = \mathbb{Q}$ and $t = t^1$.

For the omnific applications we instead write $X = t^{-1}$, so X is positive infinite. In

$$f = \sum_{q \in \mathbb{Q}} b_q X^q,$$

the support must be *reverse* well-ordered: every nonempty subset has a greatest exponent. The change $q \mapsto -q$ converts one convention into the other. All descriptive-set-theoretic statements in Sections 18 and 19 use the increasing convention. Section 22 explicitly reverses it.

The ring

$$\mathcal{H}_{\mathbb{Q}} = \left\{ n + \sum_{q \in S} b_q X^q : n \in \mathbb{Z}, S \subseteq \mathbb{Q}_{>0} \text{ reverse well-ordered} \right\} \quad (17.1)$$

is a set-sized rational-exponent omnific ring. Replacing X^q by the Conway monomial ω^q identifies it with the corresponding normal-form subring of **Oz**. Hahn addition and multiplication agree with those normal-form operations. The normal-form viewpoint and its ordered-algebraic context are standard; see [11, 5].

17.2 Borel spaces, completeness, and observability

A Polish space is a separable topological space admitting a compatible complete metric. A standard Borel space is a measurable space isomorphic to a Borel subset of a Polish space. We use the usual boldface pointclasses: Σ_2^0 consists of countable unions of closed sets, Π_3^0 of countable intersections of Σ_2^0 sets, and Π_1^1 denotes coanalytic sets. An analytic set is a projection of a Borel set in a product of Polish spaces. Completeness is with respect to continuous reductions from zero-dimensional Polish spaces. Standard references are [8, 9].

Definition 17.1. Let K be a set of series over a countable exponent alphabet. A *coefficient-observable standard Borel parametrization* of K consists of a standard Borel space Y and a surjection $\theta : Y \rightarrow K$ such that, for every exponent γ , the map

$$y \mapsto [t^\gamma]\theta(y)$$

is Borel as a map to $\mathbb{R}$. Repetitions of names are allowed.

No Borel structure on K is presupposed in this definition. Since the alphabet is countable, coefficient observability says exactly that the composite code map $Y \rightarrow \mathbb{R}^\Gamma$ is Borel. This is stronger than the mere existence of a set-theoretic bijection with the reals, and different from an abstract Borel presentation that is allowed to make every canonical coefficient unobservable. Our negative results explicitly retain the observability hypothesis.

Definition 17.2. For a countable linear order L , put

$$\text{WO}(L) = \{S \in 2^L : (S, <_L) \text{ is well-ordered}\}.$$

The empty support is allowed. The order L is *scattered* if it contains no order-isomorphic copy of $\mathbb{Q}$.

A subset of 2^L is a set of codes, not a new collection of proper-class objects. In particular, asking whether $\text{WO}(L)$ is Borel is an ordinary set-sized descriptive-set-theoretic question.

18 The exact support dichotomy

18.1 A complete obstruction on a dense countable alphabet

Lemma 18.1. *For every countable linear order L , $\text{WO}(L)$ is coanalytic.*

Proof. A support S fails to be well-ordered exactly when there exists $(x_n)_{n \in \mathbb{N}} \in L^\mathbb{N}$ with $x_n \in S$ and $x_{n+1} <_L x_n$ for every n . The relation on $(S, (x_n))$ specified by these countably many coordinate conditions is closed, with L discrete. Its projection is analytic. Its complement is $\text{WO}(L)$. $\square$

Let Tr be the closed space of trees $T \subseteq \mathbb{N}^{<\mathbb{N}}$, where trees are closed under initial segments, and let $\text{WF} \subseteq \text{Tr}$ be the well-founded trees. Recall briefly why WF is coanalytic-complete. If A is coanalytic in Baire space, its complement is the projection of the body of a tree R on pairs of finite sequences. For $x \in \mathbb{N}^\mathbb{N}$, let

$$T_x = \{s : (x \upharpoonright |s|, s) \in R\}.$$

Membership of a fixed s depends on finitely many coordinates of x , so $x \mapsto T_x$ is continuous, and $x \in A$ exactly when T_x has no branch. This is the standard universal-tree proof [9, Section 16.B].

Lemma 18.2 (Kleene–Brouwer coding). *There is a fixed injection $\mu_{\text{KB}} : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (0, 1)$ such that*

$$T \in \text{WF} \iff \mu_{\text{KB}}[T] \in \text{WO}(\mathbb{Q}).$$

Consequently $\text{WO}(\mathbb{Q})$ is Π_1^1 -complete.

Proof. Order the finite sequences by the Kleene–Brouwer order: a proper extension precedes its prefix, and otherwise the first differing entry determines the order, with the ordinary order on $\mathbb{N}$. This is a countable linear order. Embed it in $\mathbb{Q} \cap (0, 1)$ as follows. Enumerate its elements. At each finite stage, put the next element at a rational strictly between its nearest already assigned lower and upper neighbors, using 0 and 1 as external bounds when needed. A midpoint works. The resulting μ_{KB} is one fixed order embedding of the entire tree alphabet, not an embedding selected separately for each T .

If T has a branch, its successively longer prefixes form a strictly descending Kleene–Brouwer sequence. If T is well-founded, well-founded induction over its nodes shows that the Kleene–Brouwer order on each descendant subtree is a well-order: it is the increasing $\mathbb{N}$ -indexed sum of the child-subtree orders, followed by the root. Therefore $\text{KB}(T)$ is a well-order exactly when T is well-founded.

The map $T \mapsto \mu_{\text{KB}}[T]$ is continuous. At an exponent in the range of μ_{KB} , its characteristic function reads exactly one tree-membership bit; outside the range it is constantly zero. Thus it reduces WF to $\text{WO}(\mathbb{Q})$. Lemma 18.1 supplies membership in the pointclass. $\square$

[write] Lemma 18.2 and part (ii) of theorem 18.4 below are boldface counterparts of results already in the collection’s omnific-notations report: `onot:neg:lem:kbtree` (a tree is well-founded exactly when its Kleene–Brouwer order is a well-order), `onot:neg:thm:pionesone` (Π_1^1 -completeness, for recursive trees, of the validity of decidable dyadic supports in $[1, 2)$ with coefficient one) and `onot:neg:rem:secondroute` (the midpoint codings of that report’s sources 08 and 09, which are the construction of μ_{KB} up to the target interval). Source 03 reproves the reduction to make the continuous (boldface) form explicit, as its contribution ledger says (section D). What is new in section 18 is the dichotomy between scattered and nonscattered orders.

18.2 Why scattered alphabets are different

We use the classical countable Hausdorff characterization: starting with finite linear orders, closing under $\mathbb{Z}$ -indexed sums and finite sums through countable ordinal stages produces precisely the countable scattered orders. A precise formulation is recalled in [10, Definition 1]. Only this characterization, not its more advanced rank consequences, is needed below.

Lemma 18.3. *Suppose $L = \sum_{i \in \mathbb{Z}} L_i$ is a lexicographic sum and each $\text{WO}(L_i)$ is Borel. Then $\text{WO}(L)$ is Borel.*

Proof. For $S \subseteq L$, let $S_i = S \cap L_i$ and $I(S) = \{i : S_i \neq \emptyset\}$. Then

$$S \in \text{WO}(L) \iff (\forall i \in \mathbb{Z} \ S_i \in \text{WO}(L_i)) \ \wedge \ (\exists N \in \mathbb{Z} \ \forall i < N \ S_i = \emptyset). \quad (18.1)$$

Necessity of the second condition follows because unboundedly negative occupied indices yield an infinite descending sequence, choosing one element from each of a decreasing sequence of such indices. Conversely, when the condition holds, S is a well-ordered sum of well-ordered fibers over a subset of an integer tail.

Each restriction map $S \mapsto S_i$ is continuous. The fiber-empty conditions are closed. The right-hand side of (18.1) is therefore Borel. For a finite sum the analogous criterion has only finitely many fiber conditions and no additional obstruction. $\square$

Theorem 18.4 (Support dichotomy). *For every countable linear order L :*

- (i) *if L is scattered, $\text{WO}(L)$ is Borel;*
- (ii) *if L is not scattered, $\text{WO}(L)$ is Π_1^1 -complete.*

In particular, Borelness of $\text{WO}(L)$ is equivalent to scatteredness of L .

Proof. For a finite order the support predicate is the whole finite code space. Induction through the Hausdorff construction, using Lemma 18.3 and the finite-sum observation, proves (i).

For (ii), fix an order embedding $j : \mathbb{Q} \rightarrow L$. Extending a support by zero outside $j[\mathbb{Q}]$ gives a continuous map $S \mapsto j[S]$ from $2^{\mathbb{Q}}$ to 2^L , preserving and reflecting well-orderedness. Compose this map with Lemma 18.2, and use Lemma 18.1. A coanalytic-complete set is not Borel, completing the equivalence. $\square$

Remark 18.5. The quantification here is over *all subsets of one fixed countable order*. It is not a statement about the Borel complexity of the class of all scattered orders, nor about finding a Hausdorff decomposition uniformly from an arbitrary code for L . The fixed decomposition used in the positive proof need not be computable.

19 Exact benchmarks: integers, finite lexicographic powers, and rationals

The dichotomy gives a qualitative boundary. On useful discrete alphabets one can calculate the precise Borel level.

Theorem 19.1. *With the product topology on the corresponding binary support spaces:*

- (i) $\text{WO}(\mathbb{Z})$ is Σ_2^0 -complete;
- (ii) for every integer $d \geq 2$, $\text{WO}(\mathbb{Z}_{\text{lex}}^d)$ is Π_3^0 -complete;
- (iii) every well-ordered subset of $\mathbb{Z}_{\text{lex}}^d$ has order type at most ω^d , and this bound is attained.

Proof. A subset of $\mathbb{Z}$ is well-ordered if and only if it is bounded below. Hence

$$\text{WO}(\mathbb{Z}) = \bigcup_{N \in \mathbb{Z}} \{S : \forall n < N (n \notin S)\},$$

a countable union of closed sets. The continuous map $A \subseteq \mathbb{N} \mapsto \{-n - 1 : n \in A\}$ reduces the familiar Σ_2^0 -complete set Fin of finite subsets of $\mathbb{N}$ to $\text{WO}(\mathbb{Z})$. This proves (i).

For completeness of the benchmark being used, Fin is countable and dense in $2^{\mathbb{N}}$. Were it G_δ , it would be comeager, while it is also meager; the Baire theorem rules this out. The usual completeness criterion for Borel sets in zero-dimensional Polish spaces then gives its Σ_2^0 -completeness [9, Section 16.A].

For $d = 2$, view the order as $\sum_{i \in \mathbb{Z}} \mathbb{Z}$. Formula (18.1) is the intersection of a Π_3^0 condition (every fiber is bounded below) and a Σ_2^0 condition (the occupied outer indices are bounded

below). The result is Π_3^0 . For $d > 2$, the same formula and induction work: countable intersections of Π_3^0 sets remain Π_3^0 .

For hardness, use

$$\text{Fin}^{\mathbb{N}} = \{a \in 2^{\mathbb{N} \times \mathbb{N}} : \forall m \{n : a(m, n) = 1\} \text{ is finite}\}.$$

This set is Π_3^0 -complete. Indeed, any Π_3^0 subset of a zero-dimensional Polish space is a countable intersection of Σ_2^0 sets; reduce each of these sets to Fin and take the product of the reductions. Map a to the support

$$S_a = \{(m, -n - 1) : a(m, n) = 1\} \subseteq \mathbb{Z}_{\text{lex}}^2. \quad (19.1)$$

The outer indices are automatically bounded below by zero. A fiber is well-ordered exactly when its row is finite. Thus (19.1) is a continuous reduction. Embedding $\mathbb{Z}_{\text{lex}}^2$ into $\mathbb{Z}_{\text{lex}}^d$ gives hardness for all $d \geq 2$.

Finally, a well-ordered subset of $\mathbb{Z}$ has type at most ω . Inductively, a well-ordered support in $\mathbb{Z}^d$ is a sum of at most ω fibers of type at most ω^{d-1} , and so has type at most $\omega^{d-1} \cdot \omega = \omega^d$. The subset $\mathbb{N}^d$, with the same lexicographic convention (first coordinate most significant), has type exactly ω^d . $\square$

Exponent alphabet	Valid-support complexity	Support-type bound
$\mathbb{Z}$	Σ_2^0 -complete	$\leq \omega$
$\mathbb{Z}_{\text{lex}}^d, d \geq 2$	Π_3^0 -complete	$\leq \omega^d$
A fixed countable scattered order	Borel	Some countable bound
$\mathbb{Q}$, or any countable nonscattered order	Π_1^1 -complete	Unbounded in ω_1

The last two support-type assertions follow also from Theorem 21.2 below. An important feature of the table is that the finite lexicographic rank can grow while the exact Borel level stays at Π_3^0 . Support order type and Borel complexity measure different things. Conversely, passing from $\mathbb{Z}$ to its divisible hull $\mathbb{Q}$ is a genuine jump out of the entire Borel hierarchy.

20 An observable-presentation dichotomy for full Hahn fields

Fix a countable ordered abelian group Γ . Let

$$V_\Gamma = \{a \in \mathbb{R}^\Gamma : \text{supp}(a) \in \text{WO}(\Gamma)\}.$$

As a set, V_Γ is the canonical coefficient presentation of the full Hahn field $K_\Gamma = \mathbb{R}((t^\Gamma))$.

Lemma 20.1. *If Γ is scattered, V_Γ is Borel. If Γ is nonscattered, V_Γ is Π_1^1 -complete.*

Proof. The support map

$$\sigma : \mathbb{R}^\Gamma \rightarrow 2^\Gamma, \quad \sigma(a)(\gamma) = 1 \iff a_\gamma \neq 0$$

is Borel. Thus the scattered case follows from Theorem 18.4. In all cases, failure of validity is witnessed by an infinite descending sequence of nonzero coordinates, an analytic condition, so V_Γ is coanalytic. The binary-coefficient inclusion $2^\Gamma \hookrightarrow \mathbb{R}^\Gamma$ is continuous and reduces $\text{WO}(\Gamma)$ to V_Γ , proving completeness in the remaining case. $\square$

Theorem 20.2 (Observable Hahn-field criterion). *For countable ordered abelian Γ , the following are equivalent:*

- (i) Γ is scattered as a linear order;
- (ii) V_Γ is a Borel subset of $\mathbb{R}^\Gamma$;
- (iii) K_Γ admits a coefficient-observable standard Borel parametrization, allowing repeated names;
- (iv) K_Γ admits a standard Borel presentation with Borel coefficient maps, Borel addition and multiplication, Borel inversion on the nonzero elements, and Borel order.

Proof. Lemma 20.1 gives (i) $\Leftrightarrow$ (ii). Assume (ii), and use V_Γ itself as the standard Borel domain. Addition and negation are coordinatewise and therefore Borel. For $\Gamma = \{0\}$ all assertions reduce to those for $\mathbb{R}$. Otherwise, for multiplication enumerate $\Gamma = \{\delta_0, \delta_1, \dots\}$ without repetition. At a fixed exponent γ , the product coefficient is

$$(ab)_\gamma = \lim_{N \rightarrow \infty} \sum_{j < N} a_{\delta_j} b_{\gamma - \delta_j}. \quad (20.1)$$

For valid Hahn supports, only finitely many terms are nonzero. To see the finiteness, infinitely many contributing δ would contain a strictly increasing sequence in the well-ordered support of a ; the corresponding $\gamma - \delta$ would be a strictly decreasing sequence in the support of b . This is impossible. Thus the sums in (20.1) eventually stabilize. They are continuous on the ambient product, and their pointwise limit on the Borel domain is Borel.

The standard Hahn-field theorem supplies closure under multiplication and the existence of multiplicative inverses; its well-ordered-support lemma is the usual Neumann lemma [11]. The graph

$$\{(a, b) \in (V_\Gamma \setminus \{0\}) \times V_\Gamma : ab = 1\}$$

is Borel and has exactly one point in each vertical section. The Lusin–Novikov theorem gives Borel inversion. Equivalently, for an open set U the inverse image under inversion is the projection of this graph intersected with $V_\Gamma \times U$, whose sections have at most one point.

Positivity is Borel because it is expressed by

$$\exists \gamma \in \Gamma (a_\gamma > 0 \wedge \forall \delta < \gamma \ a_\delta = 0).$$

There are only countably many coordinates. Order follows by subtraction. This proves (iv), which immediately implies (iii).

Assume (iii). The coefficient code map from its standard Borel parameter space to $\mathbb{R}^\Gamma$ is Borel, and its image is exactly V_Γ . Consequently V_Γ is analytic. For nonscattered Γ , Lemma 20.1 makes it coanalytic-complete and hence nonanalytic. Thus Γ must be scattered, proving (i). $\square$

Remark 20.3 (What the negative direction does and does not exclude). The theorem excludes even a surjective system of Borel-observable names, not just unique or canonical names. It does not exclude abstract Borel models in which the coefficient functions fail to be Borel. Nor does it exclude useful Borel subfields of K_Γ , including real closed subfields. The word *full* is essential.

20.1 Root closure forces the non-Borel side for full fields

Corollary 20.4. *Suppose $\Gamma \neq \{0\}$ is countable and every positive element of the full Hahn field $\mathbb{R}((t^\Gamma))$ has a square root in that field. Then Γ is nonscattered, and the field admits no coefficient-observable standard Borel parametrization. In particular, this holds whenever the full Hahn field is real closed.*

Proof. For every $\gamma \in \Gamma$, the positive monomial t^γ has a root f with $f^2 = t^\gamma$. Taking least exponents gives $2 \min \text{supp}(f) = \gamma$. Thus Γ is 2-divisible. For $a < b$ in Γ , the element $a + (b - a)/2$ lies strictly between them. A nontrivial ordered group has no endpoints, so Γ is a countable dense linear order without endpoints and is order-isomorphic to $\mathbb{Q}$. Apply Theorem 20.2. $\square$

This is a necessary-condition argument; it does not use a classification of which Hahn fields are real closed. It also explains the source of the jump: arbitrary well-ordered supports over a densely divisible scale are the problem, not the mere availability of algebraic roots. The Puiseux field in Section 23 supplies the contrasting real closed Borel subfield.

21 Every observable family has a countable support bound

We now strengthen the nonexistence statement into a restriction on every individual Borel system of names. The essential input is the classical boundedness theorem for analytic families of well-founded trees [8]. The following proof makes the particular form we need explicit.

Lemma 21.1 (Analytic boundedness, with proof). *If $\mathcal{T} \subseteq \text{WF}$ is analytic, there is a countable ordinal β bounding the root ranks of all trees in $\mathcal{T}$.*

Proof. The empty tree is assigned rank zero and can be discarded. If no trees remain, take $\beta = 1$. Otherwise write the remaining analytic family $\mathcal{T}$ as the projection of a closed set $C \subseteq \mathbb{N}^{\mathbb{N}} \times \text{Tr}$. Fix a binary coding of trees using an enumeration of $\mathbb{N}^{<\mathbb{N}}$. Construct a countable tree U whose level- k nodes are triples (s, u, v) with $s, u \in \mathbb{N}^k$, $v \in 2^k$, for which some $(y, T) \in C$ satisfies

$$u = y \upharpoonright k, \quad v = \text{code}(T) \upharpoonright k, \quad s \in T.$$

Taking initial segments preserves these conditions, so U is a tree.

If U had an infinite branch, its three coordinates would give $b, y \in \mathbb{N}^{\mathbb{N}}$ and a binary code z . For each k choose a witness $(y_k, T_k) \in C$ for the level- k node. The witnesses converge to (y, T) , where T has code z : for every fixed number of coordinates the witnesses agree with (y, z) once k is large. Since C and Tr are closed, $(y, T) \in C$. For every j , the node $b \upharpoonright j$ belongs to T_k for all $k \geq j$, and hence belongs to T . Thus $T \in \mathcal{T}$ has a branch, a contradiction. Therefore U is well-founded.

For each $(y, T) \in C$, the assignment

$$s \mapsto (s, y \upharpoonright |s|, \text{code}(T) \upharpoonright |s|)$$

embeds the extension tree T into U , taking children to children. Induction on node ranks gives $\text{rk}(T) \leq \text{rk}(U)$. A well-founded countable tree has countable rank. Taking $\beta = \text{rk}(U) + 1$ gives a strict bound. $\square$

Theorem 21.2 (Uniform support bound). *Let L be countable, Y standard Borel, and $c : Y \rightarrow \mathbb{R}^L$ Borel. Suppose every $\text{supp}(c(y))$ is well-ordered in L . Then there is $\alpha < \omega_1$ such that*

$$\text{otp}(\text{supp}(c(y))) < \alpha \quad \text{for every } y \in Y.$$

The same conclusion holds for reverse well-ordered supports, with the reverse support order used to compute the type.

Proof. For $S \subseteq L$, form the descending-sequence tree D_S . Its root is the empty sequence, and its other nodes are finite strictly decreasing sequences of members of S . The map $S \mapsto D_S$ is continuous on binary support codes; the composite $y \mapsto D_{\text{supp}(c(y))}$ is Borel, since each nonzero coefficient test is Borel. Its image is an analytic family of well-founded trees. Apply Lemma 21.1.

For a well-ordered S of type ξ , the root rank of D_S is exactly ξ . Indeed, the rank of a node ending in $s \in S$ is the order type of the initial segment of S below s ; induction verifies the recursion. The root rank is the supremum of these ranks plus one, namely ξ . Reversing L proves the last assertion. $\square$

Corollary 21.3. *For a countable linear order L , the following are equivalent:*

- (i) L is scattered;
- (ii) the order types of all well-ordered subsets of L have a common countable upper bound;
- (iii) $\text{WO}(L)$ is analytic;
- (iv) $\text{WO}(L)$ is Borel.

Proof. Theorem 18.4 gives (i) $\Leftrightarrow$ (iv) and excludes (iii) in the nonscattered case. If (iv) holds, apply Theorem 21.2 to the identity binary-coefficient coding of $\text{WO}(L)$ to obtain (ii). If L is nonscattered, it contains $\mathbb{Q}$, and every countable ordinal embeds in $\mathbb{Q}$ by the finite-stage rational insertion construction used in Lemma 18.2. Its well-ordered suborders therefore have unbounded countable types, contradicting (ii). $\square$

The bound is uniform over the entire parameter space, even if it is uncountable. It is not asserted to be computable from an arbitrary code for the Borel map. In particular, this is a boldface descriptive theorem, not an unqualified statement that every effective system has a computable ordinal bound.

22 The obstruction already lies in one omnific interval

Return to the X -convention. Let $I = \mathbb{Q} \cap (1/3, 2/3)$ and consider the compact space of candidate coefficient arrays

$$C = \left\{ X + \sum_{q \in I} \varepsilon_q X^q : \varepsilon_q \in \{0, 1\} \right\}. \quad (22.1)$$

Here the displayed sums are only *candidate codes*. A member of C denotes a Hahn or surreal normal form only when its nonzero support is reverse well-ordered.

Theorem 22.1 (Bounded-interval localization). *The valid-code subset C_{valid} of (22.1) is Π_1^1 -complete. Every valid value lies in the omnific interval $[X, 2X)$. The reverse support order types in this slice are unbounded in ω_1 .*

Proof. Take the fixed increasing Kleene–Brouwer embedding μ_{KB} from Lemma 18.2 and put

$$e(s) = \frac{2}{3} - \frac{\mu_{\text{KB}}(s)}{3} \in I.$$

This reverses the Kleene–Brouwer order. Define the candidate code

$$z_T = X + \sum_{s \in T} X^{e(s)}. \quad (22.2)$$

The map $T \mapsto z_T$ is continuous in the coefficient product topology. The support below the leading exponent 1 is reverse well-ordered exactly when $\text{KB}(T)$ is well-ordered, exactly when $T \in \text{WF}$. Appending the single larger exponent 1 changes none of these equivalences. Coanalytic membership follows from the descending-sequence characterization applied to the reverse exponent order.

For every valid code, the lower part is nonnegative because all its nonzero coefficients are positive, while $2X - z_T$ has greatest exponent 1 and coefficient 1. Hence $X \leq z_T < 2X$; the lower equality occurs when the lower support is empty.

Every countable ordinal ξ embeds into I in its reverse order, by the same rational-insertion principle. Taking that binary support gives total reverse support type $1 + \xi$. These types are unbounded below ω_1 . $\square$

Corollary 22.2. *There is no coefficient-observable standard Borel parametrization onto C_{valid} , even with repeated names. More generally, no such parametrization can cover a family whose support types are unbounded in ω_1 .*

Proof. The general statement is the contrapositive of Theorem 21.2. Apply it to Theorem 22.1. $\square$

This localization removes several possible explanations for the obstruction. It is not due to large coefficients, an unbounded range of leading degrees, or a need to represent the proper class of all surreals. The leading monomial is fixed, the other coefficients are bits, the remaining exponents lie in one fixed rational band, and all actual values lie in one bounded omnific interval.

Remark 22.3 (Relation to ProveIt’s existing lower bound). The use of well-founded trees, and the fact that general support validation is coanalytic-hard, are already recorded in the omnific-notations report [21]. The present localization is included as an explicit corollary with a fixed coefficient cylinder. The more substantial extensions are the scattered/nonscattered classification and its observable-field and uniform-family formulations. None of these descriptions establishes global priority for a result outside the inspected material.

[write] Theorem 22.1 and the collection’s `onot:neg:thm:pionesone` are the same phenomenon in two slices, as the remark above says: there, lightface completeness for recursive trees, with decidable dyadic supports in $[1, 2)$ and coefficient one; here, boldface completeness with the fixed leading monomial X , bits on the band $(1/3, 2/3)$ and values in $[X, 2X)$. The unboundedness of the support types and corollary 22.2 are not in the omnific-notations report.

23 A real closed Borel subfield and its finite omnific integer part

The preceding obstructions do not make exact non-Archimedean algebra impossible. A particularly useful positive model separates finite support, bounded-denominator infinite support, and arbitrary Hahn support.

Let

$$\mathcal{P} = \bigcup_{n \geq 1} \mathbb{R}((t^{1/n})), \quad X = t^{-1}. \quad (23.1)$$

The union is a field: any two denominators have a common multiple. Newton–Puisseux says that the corresponding complex field $\bigcup_n \mathbb{C}((t^{1/n}))$ is algebraically closed [12, 13]. It is exactly $\mathcal{P}(i)$, and $\mathcal{P}$ is ordered by the leading coefficient. Hence $\mathcal{P}$ is real closed.

Define

$$\mathcal{A}_{\text{fin}} = \left\{ n + \sum_{q \in S} a_q X^q : n \in \mathbb{Z}, S \subseteq \mathbb{Q}_{>0} \text{ finite}, a_q \in \mathbb{R} \right\}. \quad (23.2)$$

Zero coefficients may be removed to obtain a unique finite normal form. The notation $\mathbb{Z} \oplus \bigoplus_{q>0} \mathbb{R}X^q$ describes an additive direct sum. It does not mean that $\mathcal{A}_{\text{fin}}$ is an $\mathbb{R}$ -algebra: its constant subring is $\mathbb{Z}$, not $\mathbb{R}$.

Lemma 23.1 (Exact principal-part floor). *The ring $\mathcal{A}_{\text{fin}}$ is an integer part of $\mathcal{P}$. More explicitly, write uniquely*

$$f = p + r + \varepsilon,$$

where p contains the positive powers of X , $r \in \mathbb{R}$ is the constant term, and ε has only negative powers of X . Then p is finite, ε is infinitesimal, and the unique $a \in \mathcal{A}_{\text{fin}}$ with $a \leq f < a + 1$ is

$$a = p + \begin{cases} \lfloor r \rfloor - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ \lfloor r \rfloor, & \text{otherwise.} \end{cases} \quad (23.3)$$

Proof. For $f \in \mathbb{R}((t^{1/n}))$, its exponents are bounded below in the lattice $\frac{1}{n}\mathbb{Z}$. Only finitely many are negative in the t -convention, so the positive- X principal part p is finite. Every remaining nonconstant term has negative X -exponent; if the tail is nonzero, its greatest exponent is negative, so it is smaller in absolute value than every positive real.

When $r \notin \mathbb{Z}$, the fractional part $r - \lfloor r \rfloor$ lies strictly between 0 and 1, and adding an infinitesimal preserves both inequalities. When $r \in \mathbb{Z}$ and $\varepsilon \geq 0$, the remainder is in $[0, 1)$. When $r \in \mathbb{Z}$ and $\varepsilon < 0$, subtracting one from the proposed floor makes the remainder $1 + \varepsilon$, which also lies in $[0, 1)$. This proves the inequalities.

The ring is discretely ordered. A positive element with nonempty positive support exceeds every real, in particular 1; a positive constant is an ordinary positive integer. Thus 1 is its least positive element, and two distinct elements of $\mathcal{A}_{\text{fin}}$ cannot both satisfy the floor inequalities. $\square$

The boundary correction in (23.3) is essential. For example, $\lfloor -X^{-1} \rfloor_{\mathcal{A}_{\text{fin}}} = -1$, not 0. This correction is also explicitly recorded in ProveIt’s current omnific-floor description [20]; it is not a new correction discovered by this report.

[write] Lemma 23.1 is the collection’s `odg:thm:floor` (omnific-diophantine-geometry report) read in the Puiseux field. Under 03’s identification of X with ω (section 17), $\mathcal{P}$ is a subfield of $\mathbf{No}$; p , r and ε are the positive-growth part, the real constant and the infinitesimal tail of `odg:eq:floor`; and $\mathbf{Oz} \cap \mathcal{P} = \mathcal{A}_{\text{fin}}$, because the positive-growth

part of a Puiseux series is finite. Formula (23.3) is therefore the same formula, and its negative-infinitesimal correction is the case $r \in \mathbb{Z}$, $\varepsilon < 0$ there. `odg:thm:floor` is proved in Lean for every surreal (`Algebra/SurrealNumbers/Surreal/Foundations/OmnificFloor.lean: omnificFloor_spec, existsUnique_omnific_integerPart, omnificFloor_of_integer_negative_tail`; row “Full `odg:thm:floor`” of docs/FORMALIZATION.md). The Lean development does not mention $\mathcal{P}$ or $\mathcal{A}_{\text{fin}}$, so the lemma as stated here, with the finiteness of p and the integer-part property of $\mathcal{A}_{\text{fin}}$ in $\mathcal{P}$, is not formalized. Source 03 credits the Surreal README for the correction.

Theorem 23.2. *The nonnegative cone $\mathcal{A}_{\text{fin}, \geq 0}$ is an uncountable model of Robinson arithmetic plus open induction in the ordered semiring language.*

Proof. A discretely ordered ring gives the elementary semiring and predecessor axioms on its nonnegative cone. In particular, if $x > 0$ then $x - 1 \geq 0$. Uncountability follows already from the distinct elements aX , $a > 0$ real.

We prove open induction directly, using the familiar integer-part direction of Shepherdson’s theorem [6, 7]. Let $W \subseteq \mathcal{A}_{\text{fin}, \geq 0}$ be a nonempty set defined by a quantifier-free formula with parameters from $\mathcal{A}_{\text{fin}}$. Interpret the same formula in the real closed field $\mathcal{P}$. Its truth set is a finite union of intervals and points: the finitely many roots of the polynomials appearing in the formula partition the line into regions of constant sign.

Intersect these finitely many cells with $[0, \infty)$ and choose the first one meeting $\mathcal{A}_{\text{fin}}$. A point cell contributes its own element. For an interval cell with lower endpoint $b \in \mathcal{P}$, the integer-part property gives the least integer-part element at or above b , or the least one strictly above b when the endpoint is excluded. This element is nonnegative; if it were above the interval, that interval would contain no element of $\mathcal{A}_{\text{fin}}$ at all. For a cell beginning at 0, the same argument starts with 0 or 1. Thus W has a least element.

If a quantifier-free induction instance failed while its base and successor steps held, the nonempty quantifier-free set of counterexamples would have a least member m . It would satisfy $m > 0$, so $m - 1 \in \mathcal{A}_{\text{fin}, \geq 0}$ would satisfy the formula. The successor step would then force the formula at m , a contradiction. $\square$

[write] The proof of [theorem 23.2](#) is the Shepherdson route of the collection’s `dsn:cor:openinduction (definable-surreals-and-omnific-integers report: open induction for the integer part  $I_A$  of a real closed field  $K_A$ )` and of the paragraph citing Shepherdson’s theorem in the subsection “Sums of squares and induction” of the `omnific-diophantine-geometry report (open induction for the omnific integers of a divisible Hahn workspace)`. Here it is applied to $\mathcal{A}_{\text{fin}}$ inside the real closed field $\mathcal{P}$; uncountability and the semiring axioms are 03’s. `dsn:cor:openinduction` is pending in docs/FORMALIZATION.md. As 03 says, $\mathcal{A}_{\text{fin}, \geq 0}$ is a Shepherdson-type model, not a new model of arithmetic.

23.1 Why the Puiseux field itself has Borel codes

In the ambient coefficient space $\mathbb{R}^{\mathbb{Q}}$, the field (23.1) is exactly the union, over $n \geq 1$ and $k \in \mathbb{Z}$, of the closed sets

$$\{a : a_q = 0 \text{ whenever } q \notin \frac{1}{n}\mathbb{Z} \text{ or } q < k/n\}.$$

It is therefore F_σ and standard Borel. The convolution and unique graph arguments of Theorem 20.2 make its field operations Borel. Every such support has increasing order type at most ω .

This gives a useful three-way contrast. Finite principal parts have finite support; the real closed Puiseux field has a uniform support bound ω ; the full rational-exponent Hahn field has supports of unbounded countable type. Real closedness alone does not prevent a Borel coefficient presentation. Full arbitrary-support closure is what makes Corollary 20.4 apply.

24 A Polish finite-normal-form presentation

24.1 The topology and its Borel structure

Write an element of $\mathcal{A}_{\text{fin}, \geq 0}$ uniquely as $(S, n, (a_q)_{q \in S})$, where all $a_q \neq 0$. If $S \neq \emptyset$, positivity means $a_{\max S} > 0$ and the integer n is unrestricted. If $S = \emptyset$, we require $n \geq 0$.

For finite nonempty $S \subseteq \mathbb{Q}_{>0}$ and $n \in \mathbb{Z}$, let

$$U_{S,n} = \{a \in (\mathbb{R} \setminus \{0\})^S : a_{\max S} > 0\}.$$

For $S = \emptyset$, let $U_{\emptyset,n}$ be a singleton when $n \geq 0$. Put on $\mathcal{A}_{\text{fin}, \geq 0}$ the topological disjoint union

$$(\mathcal{A}_{\text{fin}, \geq 0}, \tau_{\text{FS}}) = \bigsqcup_{S,n} U_{S,n}. \quad (24.1)$$

Support and integer constant are locally constant in this topology.

Proposition 24.1. *The topology τ_{FS} is Polish. Its Borel sets are exactly the Borel sets obtained from the canonical finite-support coefficient coding.*

Proof. Each nonsingleton component is open in a finite-dimensional Euclidean space, and hence Polish. There are countably many components. A countable topological disjoint union of Polish spaces is Polish: choose bounded complete metrics on the components and put distance 2 between distinct components. Taking the union of countable dense sets gives separability.

Every coefficient map is continuous for (24.1). Conversely, an exact support condition consists of nonzero tests at finitely many coordinates and zero tests at countably many others, so is Borel in the coefficient product. On each such Borel piece the coefficient coordinates give exactly the usual finite-dimensional Borel structure. Since there are countably many pieces, the two Borel structures agree. $\square$

This is not the order topology. The order topology on a discretely ordered uncountable arithmetic domain is discrete and therefore not separable. Nor is τ_{FS} the ambient coefficient subspace topology, where convergence need not make the exact support eventually constant. The distinction is responsible for the cancellation discontinuities below.

24.2 A general support-normalization lemma

Lemma 24.2. *Let $U \subseteq \mathbb{R}^d$ be nonempty and open, and let $(p_j)_{j \in J}$ be a finite family of real polynomials. Define a map whose output is the finite list of nonzero coordinates $p_j(u)$, with the target topologized as a disjoint union by exact nonzero support. Let $E = \{j : p_j \not\equiv 0\}$. Its discontinuity set is exactly*

$$\mathcal{Z} = \bigcup_{j \in E} \{u \in U : p_j(u) = 0\}.$$

This set is closed and nowhere dense in U . The map is continuous on countably many closed subsets covering U .

Proof. Outside $\mathcal{Z}$, all nonidentically-zero coordinates stay nonzero on a sufficiently small neighborhood, while the others are always zero. The output support is therefore constant and the coordinate formulas are continuous.

At $u \in \mathcal{Z}$, choose $j \in E$ with $p_j(u) = 0$. A nonzero real polynomial cannot vanish on a nonempty open set. Thus arbitrarily near u there are points where this coordinate is nonzero. Their output support differs from the support at u , so their images lie outside the clopen target component containing the image of u . Continuity fails. This also proves that the finite union of zero sets is nowhere dense; closedness is immediate.

For $Z \subseteq E$ and integer $k \geq 1$, use the closed set

$$F_{Z,k} = \{u \in U : p_j(u) = 0 \ (j \in Z), \quad |p_j(u)| \geq 1/k \ (j \in E \setminus Z)\}. \quad (24.2)$$

These sets cover U : at any point choose the actual zero set and then a large enough k for the finitely many remaining coordinates. On each piece, the output support is constant and the map is continuous. For a zero-dimensional component consisting of one point, the same conclusion holds with empty discontinuity set. $\square$

24.3 The exact arithmetic cancellation loci

Fix an input component $U_{S,n} \times U_{T,m}$. Extend the variable coefficient families by $a_q = 0$ for $q \notin S$ and $b_q = 0$ for $q \notin T$. For addition, the output coefficient at $r > 0$ is

$$A_r = a_r + b_r, \quad (24.3)$$

and its constant is $n + m$. For multiplication, the output constant is nm and the coefficient at $r > 0$ is

$$M_r = nb_r + ma_r + \sum_{p+q=r} a_p b_q. \quad (24.4)$$

Only finitely many exponents need to be considered: $S \cup T$ for addition and $S \cup T \cup (S + T)$ for multiplication.

For multiplication define

$$E_{S,T,n,m} = \{r \in S \cup T \cup (S + T) : M_r \not\equiv 0\},$$

where identity means polynomial identity in the independent input coefficient variables after fixing the integers n, m . This removes absent terms and zero-factor components correctly.

Theorem 24.3 (Exact cancellation theorem). *On the Polish presentation (24.1), addition and multiplication are Borel. Their discontinuity sets are given componentwise by*

$$D_+ \cap (U_{S,n} \times U_{T,m}) = \bigcup_{r \in S \cap T} \{a_r + b_r = 0\}, \quad (24.5)$$

$$D_\times \cap (U_{S,n} \times U_{T,m}) = \bigcup_{r \in E_{S,T,n,m}} \{M_r = 0\}. \quad (24.6)$$

Globally, both sets are closed and nowhere dense. Both operations are continuous on countably many closed pieces. In particular, both are continuous at every point of the common dense open set $\mathcal{A}_{\text{fin}, \geq 0}^2 \setminus (D_+ \cup D_\times)$.

Proof. On an input component, the actual normal form is obtained from (24.3) or (24.4) by deleting the zero coefficients. The integer output constant is fixed. Apply Lemma 24.2. For

addition, a coefficient occurring in only one input support is constrained to be nonzero, so the possible zero sets are precisely those in (24.5). For multiplication the expected polynomial support is exactly $E_{S,T,n,m}$.

Every input component is clopen, and these components partition the domain. A set whose intersection with each component is closed is globally closed; a set whose intersection with each nonempty open component is nowhere dense has empty interior and, when closed, is nowhere dense globally. This gives the assertions about D_+ and $D_\times$.

The pieces (24.2) are closed relative to a clopen component, hence closed in the full domain. Taking them over countably many components, zero patterns, and integers k gives countable closed covers for both operations. To see Borelness directly, the inverse image of an open set is a countable union of sets open relative to these closed pieces. Such sets are Borel (indeed F_σ in a metric space). $\square$

Example 24.4 (Addition: cancellation below the leading term). For real δ sufficiently near 0, set

$$x = X^2 + X, \quad y_\delta = X^2 + (-1 + \delta)X.$$

All these elements are positive and remain in fixed input components. But

$$x + y_\delta = 2X^2 + \delta X.$$

At $\delta = 0$ the output support loses exponent 1, so the outputs do not converge to $2X^2$ in τ_{FS} . Leading coefficients of positive inputs cannot cancel each other; the lower coefficient is what fails here.

Example 24.5 (Multiplication: a vanished middle coefficient). Let $x = X + 1$ and $y_\delta = (1 + \delta)X - 1$. Near $\delta = 0$ both inputs are positive, their supports and integer constants are fixed, and

$$xy_\delta = (1 + \delta)X^2 + \delta X - 1.$$

The middle term disappears at $\delta = 0$. Thus $(X + 1, X - 1) \in D_\times$. Multiplication by 0, on the other hand, is constant and continuous; this explains the exclusion of identically zero coefficient polynomials.

Remark 24.6 (A measurability qualification). Countable closed-piece continuity is the statement proved here. We do not identify it without further hypotheses with the Baire-one class of maps into this disconnected target. Nor does the dense-open conclusion say that an algebraic cancellation can be ignored by exact computation: deleting a single zero coefficient changes the canonical component.

25 Glazer's obstruction survives every compatible Polish recoding

We use precisely the following published result.

Theorem 25.1 (Glazer, Corollary 1 of [1]). *There is no model of Robinson arithmetic plus !Open on an uncountable Polish space in which addition is continuous and multiplication is Borel.*

Theorem 25.2 (No removal of the addition obstruction). *No Polish topology on the set $\mathcal{A}_{\text{fin}, \geq 0}$ with the same Borel sets as τ_{FS} makes addition continuous. More generally, there is no Polish presentation Borel-isomorphic to the canonical one in which transported addition is continuous.*

Proof. By Theorem 23.2, the algebraic structure is an uncountable model of the required arithmetic. By Theorem 24.3, multiplication is Borel in the canonical Borel structure. Changing the Polish topology without changing its Borel sets leaves multiplication Borel on the product; finite products of standard Borel spaces have the same product Borel structure. Continuous addition would contradict Theorem 25.1. Transporting the operations through a Borel isomorphism changes none of these hypotheses. $\square$

Corollary 25.3. *The distinction between global continuity and continuity off a closed nowhere dense set is genuine for this explicit model: the latter is attained for both operations, while the former is impossible for addition under any Borel-compatible Polish presentation.*

This is a sharp distinction between an empty discontinuity set and an explicit closed nowhere dense one. It is not a proof that our particular zero loci have minimal cardinality, minimal dimension, or a minimal descriptive rank among all recodings.

Glazer's Theorem 2 also excludes continuous multiplication for uncountable Borel models of the stronger theory $\mathbf{Q} + I\Delta_0 + \text{Exp}$ [1]. We have proved only open induction for $\mathcal{A}_{\text{fin}, \geq 0}$. The multiplication obstruction from that stronger result is therefore *not* transferred to every compatible topology on $\mathcal{A}_{\text{fin}, \geq 0}$. Multiplication fails to be continuous in our displayed topology by Theorem 24.3; whether a different compatible Polish topology can make multiplication continuous remains a separate question here.

26 Where the two parts meet [merge]

The real model of Part I is one stratum of the cone $\mathcal{A}_{\text{fin}, \geq 0}$ of section 24. Read ω as X and let

$$\iota : M_{\mathbb{R}} \longrightarrow \mathcal{A}_{\text{fin}}, \quad \iota(r, n) = rX + n,$$

which is 05's map $\iota_{\mathbb{R}}$ of theorem 9.1 in 03's notation. Put

$$C_{\mathbb{R}} = \{a \in \mathcal{A}_{\text{fin}, \geq 0} : \text{supp } a \subseteq \{1\}\} = \bigcup_{n \in \mathbb{Z}} U_{\{1\}, n} \cup \bigcup_{n \geq 0} U_{\emptyset, n}.$$

Proposition 26.1 (The real model as a stratum [merge]). (i) ι is an injective ordered additive map onto $C_{\mathbb{R}}$, fixing 0 and 1. The set $C_{\mathbb{R}}$ is clopen in $(\mathcal{A}_{\text{fin}, \geq 0}, \tau_{\text{FS}})$ and closed under addition, but not under multiplication.

- (ii) On $\{(r, n) \in M_{\mathbb{R}} : r > 0\}$, ι is a homeomorphism onto $\bigcup_n U_{\{1\}, n}$, for the topology of $M_{\mathbb{R}}$.
- (iii) At the standard numerals the two topologies differ: every ordinary integer $n \geq 0$ is isolated in τ_{FS} , but $(0, n)$ is not isolated in $M_{\mathbb{R}}$. Thus τ_{FS} induces on $C_{\mathbb{R}}$ a topology strictly finer than the one transported from $M_{\mathbb{R}}$.
- (iv) Addition is continuous on $(C_{\mathbb{R}}, \tau_{\text{FS}})$. Consequently $(C_{\mathbb{R}}, \tau_{\text{FS}})$ is another uncountable, locally compact Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous addition; it is not perfect, its isolated points being exactly the standard numerals.

Proof. (i) An element of $\mathcal{A}_{\text{fin}}$ with support in $\{1\}$ is $n + aX$; by (24.1) it lies in $\mathcal{A}_{\text{fin}, \geq 0}$ when $a > 0$, or $a = 0$ and $n \geq 0$. These are exactly the values of ι . Addition is coordinatewise and the order is decided by the leading coefficient, as in (3.1). The set $C_{\mathbb{R}}$ is a union of components of the topological sum (24.1), hence open, and its complement is the union of the other components, hence also open. Finally $X \cdot X = X^2 \notin C_{\mathbb{R}}$.

(ii) Both $U_{\{1\},n}$ and the part $r > 0$ of the slice of integer coordinate n in $M_{\mathbb{R}}$ are the open ray $(0, \infty)$ with its Euclidean topology, and ι is the identity on the coefficient.

(iii) $U_{\emptyset,n}$ is a singleton component of (24.1). In $M_{\mathbb{R}}$, $(1/j, n) \rightarrow (0, n)$ by (5.1).

(iv) On a component $U_{S,n} \times U_{T,m}$ with $S, T \subseteq \{1\}$, (24.5) is a union over $r \in S \cap T \subseteq \{1\}$ of the sets $\{a_1 + b_1 = 0\}$, which are empty because both leading coefficients are positive. So $C_{\mathbb{R}}^2$ does not meet D_+ , and addition maps $C_{\mathbb{R}}^2$ into $C_{\mathbb{R}}$. A clopen subspace of a Polish space is Polish, the components are points and open rays, and the algebraic structure is that of $M_{\mathbb{R}}$, so corollary 4.11 applies. $\square$

The boundary. Glazer’s Corollary 1 concerns a model of $Q+I\text{Open}$ with Borel multiplication. The ring cone $\mathcal{A}_{\text{fin}, \geq 0}$ is such a model (theorems 23.2 and 24.3), so no Borel-compatible Polish topology on $\mathcal{A}_{\text{fin}, \geq 0}$ makes its addition continuous (theorem 25.2). The stratum $C_{\mathbb{R}}$ is not such a model: as an additive monoid it is $M_{\mathbb{R}}$, which carries no unital semiring multiplication at all (corollary 10.5), and the multiplication of $\mathcal{A}_{\text{fin}, \geq 0}$ leaves it. On $C_{\mathbb{R}}$ addition is continuous for τ_{FS} itself; the discontinuities of addition on $\mathcal{A}_{\text{fin}, \geq 0}$ lie on the cancellation loci (24.5), which need two inputs whose supports share an exponent below their leading ones, and so lie outside $C_{\mathbb{R}}^2$. Thus Part I and theorem 25.2 delimit, inside one ring, where Corollary 1 applies: the additive stratum admits continuous addition for both topologies, while the ring, with its Borel multiplication and open induction, admits it for none.

The Baire model. The image $\iota_B(M_B)$ of theorem 9.1 lies in 03’s rational Hahn ring $\mathcal{H}_{\mathbb{Q}}$ of (17.1): its supports are contained in $\{1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup \{0\}$ and have reverse order type at most $\omega + 1$. Its elements with infinitely many nonzero coefficients lie neither in $\mathcal{A}_{\text{fin}}$ nor in the Puiseux field $\mathcal{P}$, whose denominators are bounded. Each coefficient of $\iota_B(a, n)$ is a_j , n or 0, a continuous function on M_B , so the composite of 05’s homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$ (theorem 6.1) with ι_B is a coefficient-observable standard Borel parametrization of $\iota_B(M_B)$ in the sense of definition 17.1. This is consistent with theorem 21.2, whose bound can be taken to be $\omega + 2$ here, while the full field $\mathbb{R}((t^{\mathbb{Q}}))$ lies on the non-Borel side of theorem 20.2. The Baire model is one of the restricted support families that section 27 calls for.

A correction to the placement record [write]. The placement commit 3d2177df4 and the intake dossier said that 03’s topology τ_{FS} restricts on the stratum to the topology of 04 and 05. Parts (ii) and (iii) of proposition 26.1 give the exact statement: the two topologies agree on the elements with positive real coefficient and differ at the standard numerals, which τ_{FS} isolates. Both make addition continuous on the stratum, so the conclusion drawn there stands.

27 Consequences for exact notation systems and formalization

27.1 A practical separation of three design goals

The results separate three tasks that can otherwise be conflated. *Arithmetic* asks whether operations can be performed on valid inputs. *Validation* asks whether an arbitrary code is a valid well-based normal form. *Topological presentation* asks whether the operations respect a complete separable topology on the names or values. A positive answer to one is not a positive answer to the others.

In the finite-support model, every code carries an explicit finite list, and arithmetic is polynomial coefficient arithmetic followed by exact zero tests. The canonical domain is Polish, the operations are Borel, and their failure of continuity is exactly the support-changing cancellation calculated above. No general theorem of computability over arbitrary real coefficients is claimed: a real coefficient supplied by an unrestricted approximation oracle need not have decidable equality with zero.

For full Hahn fields on scattered exponent groups, support validation is a Borel predicate. This is not the same as decidability by a finite algorithm. The Π_3^0 examples already involve quantification over infinitely many rows. For dense rational exponents the obstruction is stronger: no standard Borel system of fully observable names covers every valid series.

A useful exact system must therefore limit its admissible support families, relax coefficient observability, or accept a nonstandard measurable domain. The Puiseux field demonstrates that a severe-looking support restriction may still permit real closedness. Theorem 21.2 says every observable Borel system necessarily has some uniform countable support bound, even when that bound is not built into the notation explicitly.

27.2 No paradoxes are needed

All main results are in ZFC and concern sets. For countable Γ , the ambient coefficient space $\mathbb{R}^\Gamma$ is a set of cardinality continuum. Its valid-code subset is a set whether or not it is Borel. The rational Hahn ring $\mathcal{H}_\mathbb{Q}$ and its finite subring $\mathcal{A}_{\text{fin}}$ are also sets. The construction never forms a set of all surreal numbers or quantifies over all class maps as a set.

The optional identification with Conway normal forms can be read as a set-indexed definable embedding into the proper class **No**, or carried out inside a sufficiently large set-sized normal-form universe. Standard Replacement handles the image of a fixed set of codes. No global choice, large cardinal, or class-sized Polish topology is used.

27.3 A formalization plan with an explicit dependency boundary

A Lean development can be organized into independent interfaces rather than starting from the entire proper-class surreal hierarchy.

- Layer 1. Finite algebra and normalization.** Use finitely supported rational-exponent coefficient maps with an integer constant. Establish convolution, leading-exponent sign, discreteness, and the coefficient identities (24.3)–(24.4). The exact cancellation examples are small executable tests at this layer.
- Layer 2. Series and arithmetic.** Use a set-sized Hahn or Puiseux field, prove the principal-part floor with its negative-infinitesimal case, and derive quantifier-free induction via the one-variable sign-cell argument. Newton–Puiseux is a named external dependency until its real-closedness interface is formally supplied.
- Layer 3. Topological support components.** Formalize the countable disjoint union, the polynomial normalization lemma, and the closed-piece covers. None of this requires descriptive ranks or proper-class surreals.
- Layer 4. Descriptive-set-theoretic structure.** Formalize the fixed Kleene–Brouwer embedding, Hausdorff-sum support criterion, analytic boundedness, and observable-field equivalence. The classical Hausdorff characterization and Glazer theorem must be proved or explicitly listed as hypotheses; importing them informally is not a checked proof.

This is a proposed architecture, not a claim that these layers have been implemented or that all required library APIs currently exist. The provided Python artifact checks finite rational instances, not Lean terms or infinitary assertions.

28 Further research questions and concrete next targets

The following questions are proposed here. Unless explicitly attributed, “open” means unresolved in this report, not certified as absent from the entire literature.

Question 28.1 (Continuous multiplication on the finite model). Does $\mathcal{A}_{\text{fin}, \geq 0}$ admit a Polish topology with the canonical Borel sets in which multiplication is continuous?

Theorem 25.2 settles the analogous addition question negatively. Glazer’s stronger multiplication theorem cannot be invoked without establishing its extra arithmetic hypotheses. A positive solution would isolate a concrete asymmetry between the two operations; a negative solution should identify an obstruction valid already for this particular open-induction model.

Question 28.2 (How small can the discontinuity set be?). Among all Borel-compatible Polish presentations of $\mathcal{A}_{\text{fin}, \geq 0}$, what restrictions hold on the discontinuity set of addition? Can it be countable? Can it be contained in a prescribed small-dimensional or otherwise canonical set?

The present model gives a nonempty closed nowhere dense set. Glazer excludes only its complete disappearance, not every possible reduction in complexity. This is a natural refinement of the topological theorem that can first be studied on explicit finite-support components.

Question 28.3 (Exact complexity beyond finite lexicographic rank). For a fixed countable scattered order L , compute the least Borel level of $\text{WO}(L)$ directly from an economical Hausdorff decomposition. Which Wadge degrees occur among these support predicates?

The qualitative dichotomy is settled by Theorem 18.4, while Theorem 19.1 shows that simply increasing finite order rank need not increase Borel level. A satisfactory answer should distinguish ordinal support capacity from descriptive complexity rather than treating them as interchangeable ranks.

Question 28.4 (Borel, algebraically rich subfields of the rational Hahn field). Classify Borel subfields $F \subseteq \mathbb{R}((t^{\mathbb{Q}}))$ containing all rational monomials and stable under specified truncations or algebraic-root operations. Which countable support bounds can such fields attain?

The Puiseux field is one real closed example with supports of type at most ω . The full Hahn field is excluded. An interesting target is a well-described hierarchy of truncation-closed real closed Borel subfields between them, together with provably sharp support bounds at each stage. No claim is made that every ordinal cutoff is preserved by inversion.

Question 28.5 (How much coefficient observability is necessary?). If only a selected family of coefficient combinations or truncations is required to be Borel, which versions of Theorem 20.2 survive?

The current proof uses all countably many coefficient maps to obtain one Borel map into the full product. This pinpoints a possible escape route and suggests investigating minimal observation families that still recover the binary support cylinder in Theorem 22.1.

Question 28.6 (Certified support languages). Construct useful exact languages whose valid supports carry explicit well-foundedness certificates, and determine their closure under arithmetic, root extraction, and named truncations.

The repository already contains effective restricted omnific calculi [21]. The present results suggest comparing them by their Borel parameter spaces and support bounds as well as by decidable equality. A certificate verifier for a fixed language is not a total validator for all raw rational-exponent supports; Theorem 22.1 forbids conflating those objectives.

Question 28.7 (A minimal arithmetic hypothesis for the presentation barrier). Which concrete consequences of open induction suffice to rule out continuous addition in the finite principal-part model or related integer parts?

A proof specialized to finite Puiseux principal parts might replace the full open-induction interface with explicitly stated root-approximation or division properties. This would clarify the boundary between algebraic normal-form obstructions and arithmetic induction.

Question 28.8 (Formalized transfer from support theory to arithmetic). Can the observable Hahn-field criterion and the finite-support cancellation theorem be formalized as reusable results, with the classical Hausdorff and Glazer inputs exposed as exact dependency interfaces?

The suggested first milestone is the support-sum criterion (18.1) together with the normalization Lemma 24.2. Both are modular and elementary enough to be reviewed independently before the larger descriptive-set-theoretic machinery is introduced.

[merge] Question 28.1 is the multiplicative counterpart, on $\mathcal{A}_{\text{fin}, \geq 0}$, of question 14.4. The question on the size of the discontinuity set of addition concerns all of $\mathcal{A}_{\text{fin}, \geq 0}$; on the clopen stratum $C_{\mathbb{R}}$ of proposition 26.1 that set is empty.

29 Conclusions of Part II

The intersection with Glazer’s research produces a precise hierarchy rather than a blanket impossibility theorem. Full Hahn support is Borel exactly on scattered countable exponent orders. Dense alphabets carry a coanalytic-complete validation problem, and no standard Borel family with observable coefficients can exhaust supports of unbounded countable type. This obstruction persists inside one fixed interval of omnific integers.

At the same time, a real closed Puiseux subfield and its finite principal-part integer part have explicit Borel presentations. The latter even admits a Polish topology where both arithmetic operations are continuous off closed nowhere dense sets, with exact polynomial equations describing every failure. Glazer’s theorem makes the remaining addition discontinuity unavoidable under all compatible Polish recordings. The positive construction and the negative result constrain one another: Borel arithmetic is attainable, globally continuous addition is not, and arbitrary well-based support introduces a separate obstruction before topological continuity can even be asked in the canonical standard-Borel setting.

A Worked examples and an audit of the hypotheses (source 05)

A.1 A negative constant term is compatible with positivity

In $M_{\mathbb{R}}$, take

$$x = (\frac{3}{2}, -17), \quad y = (\frac{1}{4}, 6).$$

Both are positive, and

$$x + y = (\frac{7}{4}, -11).$$

Division of x by 5 gives

$$q_5(x) = (\frac{3}{10}, -4), \quad r_5(x) = 3,$$

because $-17 = 5(-4) + 3$. The quotient is positive because its real coordinate is positive. A definition that incorrectly demanded a nonnegative integer coordinate for every element would break the predecessor and division axioms.

A.2 A nonstandard finite-candidate calculation

Let $L = (\frac{1}{2}, -5)$, $U = (\frac{2}{3}, 0)$, and require

$$y \equiv (\frac{7}{13}, 2) \pmod{3G}.$$

The real coefficient of a congruence parameter is irrelevant because $\mathbb{R}$ is divisible. Among $L, L+1, L+2$, the candidate $L+1$ has integer coordinate $-4 \equiv 2 \pmod{3}$, so it is a witness. Every candidate lies below U , since its leading coefficient is $1/2 < 2/3$. No iteration a nonstandard number of times is used: the finite search is justified by full invariance on $3G$ -cosets.

A.3 A Baire convergence that order cannot detect continuously

Let e_j be the rational unit vector at coordinate j . Then

$$(e_j, 0) \rightarrow (0, 0), \quad (e_j, 0) > n\mathbf{1} \quad \text{for every fixed standard } n.$$

The initial coefficients eventually agree with zero to every prescribed finite length. In omnific notation this sequence is

$$\omega, \quad \omega^{1/2}, \quad \omega^{1/3}, \quad \dots,$$

which converges to zero in the *specified coefficient topology*, not in the arithmetic order topology. This is a concrete illustration, not a claim about an unspecified topology on surreal numbers.

A.4 Hypothesis checklist

Potential pitfall	Resolution in the proofs
Only the division axioms are checked	Theorem 4.9 , corollary 4.11 , and proposition 4.12 establish the complete theory and every additive induction instance.
An open ray is given an incomplete metric	Section 5 provides a compatible logarithmic metric on each open ray and a complete metric on the whole sum.

Potential pitfall	Resolution in the proofs
The intended topology is silently replaced by the order topology	Both topologies are explicitly distinguished; the arithmetic order topology is discrete.
Standard-shift invariance is used with a nonstandard quotient	Example 4.4 exhibits the failure; lemma 4.7 uses invariance on mG -cosets instead.
The all-zero rational branch is lost	The nonnegative-cone tree in (6.2) explicitly includes it.
Computing an output tag secretly requires deciding equality of streams	For a negative output integer coordinate, a negative input supplies a finite bound on the first positive coefficient.
A type-theoretic “computable model” is confused with a countable one	The presentation is Type-2 on all Baire sequences; not every element is computable, and order has only a limit decision procedure.
The omnific image is assumed closed under multiplication	It is only an additive fragment. Theorem 10.4 rules out even an alternative semiring multiplication on the same carrier.
A proper class is treated as a Polish space	Only explicit set-sized images and a sufficiently large set-sized birthday cutoff are used.
Existing repository proofs are credited with new theorems	The formalization roadmap distinguishes the inspected integer theorems from the additional nonstandard and topological obligations.
Numerical testing is mistaken for a proof	The exact test scope is documented in section 13 ; no formal verification claim is made for this article.
An affirmative proof is mistaken for certified historical priority	The source question and version are identified; independent priority and refereeing remain separate from the mathematical proof.

B Reproduction and the scope of the executable checks (source 04)

The package requires a standard LaTeX installation for the article and Python 3.10 or later for the companion code. The Python program uses only the standard library and exact fractions.Fraction arithmetic. From the package directory, run

```
python3 verification.py --output verification_results.json
pdflatex -interaction=nonstopmode -halt-on-error polish_presburger.tex
pdflatex -interaction=nonstopmode -halt-on-error polish_presburger.tex
```

A Makefile supplies equivalent targets. The delivered run used seed 20261003 and passed 75,079 assertions. Its breakdown is:

Check category	Assertions
Rational, integer, and pairing codes	6,468
Finite group and order identities	9,955
Exact quotient and remainder	2,151
Boundary witness transfer	21,037
Finite candidate comparison	1,200
External periodicity counterexample	2,321
Positive-pivot matrix tests	18,000
Monus integer-coordinate obstruction	400
Baire-code round trips	97
Baire quotient and remainder prefixes	9,700
Baire addition prefixes	3,750
Total	75,079

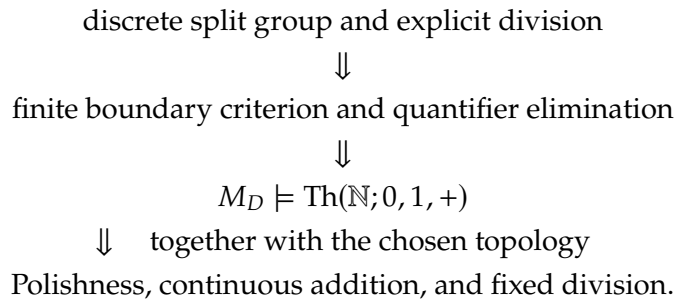
The finite group tests use rational coefficient vectors and signed integer coordinates. The boundary tests examine witnesses from a specified finite pool and compare them with the exact candidate set. They do not exhaust an infinite group. The matrix tests use finite initial pieces of matrices satisfying the positive-pivot conditions. The stream tests check finite prefixes of deterministic infinite oracles, including long zero runs. The monus tests verify exact coordinate obstructions along finite portions of the sequences used in the proof, not topological convergence by numerical approximation.

The model-theoretic, infinite-matrix, and topological statements rest on the proofs in the manuscript. Tests are useful for detecting sign errors, incorrect negative quotients, wrong coordinate offsets, and mismatched coding branches, but a successful run is not a substitute for those proofs.

[write] The commands above use source 04's delivery names. In this report its program is `code/04-polish-presburger-verification.py`, its recorded run data `data/04-polish-presburger-verification_results.json`, its Makefile `code/04-polish-presburger-Makefile` (which still names `polish_presburger.tex`, `verification.py` and `verification_results.json`), and its manuscript is merged into `this article.tex`. The program's default output is `verification_results.json` in the working directory, so it is rerun with an explicit output path outside the report (README). The intake reran it on a copy with Python 3.14.4: 75,079 assertions passed, and the record was identical up to line endings.

C A compact proof-dependency map (source 04)

The answer can be read independently of the optional extensions:



For the real example, [section 5.4](#) completes the argument at that point. For literal Baire space, add only the two chart codings in [section 6.3](#). The computation of the monus locus,

the embedding classification, the growth-order obstruction, and the surreal realization are subsequent results; none is a hidden prerequisite for the affirmative construction.

D Source audit, contribution ledger, and reproducibility (source 03)

D.1 Repository material inspected

The repository branch was resolved to the commit printed in Section 16. The following guides were inspected in selected ranges at that revision:

```
Algebra/SurrealNumbers/README.md
Algebra/SurrealNumbers/docs/README.md
Algebra/SurrealNumbers/docs/
foundations-and-computation/omnific-notations/README.md
```

The first guide supplies the context of the omnific ring, its least positive element, and the floor boundary correction. The notations guide reports the existing Kleene–Brouwer support-validity classification and carefully states the limited scope of its finite checks. The present article does not certify all statements in either guide.

The external mathematical inputs are Newton–Puiseux, the classical Hahn-field construction, Hausdorff’s characterization of countable scattered orders, standard descriptive-set-theoretic facts, and Glazer’s Corollary 1. Analytic boundedness and the needed Kleene–Brouwer reduction are proved in the text. Shepherdson’s theorem is contextualized and its needed direction is also proved directly.

D.2 Contribution ledger

Statement	Status in this report
General coanalytic support hardness	Prior classical mechanism; already recorded for omnific notations in ProveIt. Reproved to make the reductions explicit.
Scattered/support dichotomy	Proved from Hausdorff theory and the fixed tree reduction; global priority not asserted.
Observable full-Hahn criterion	Proved, including Borel field operations on the scattered side and exclusion of even repeated observable names on the nonscattered side.
Exact $\mathbb{Z}^d$ Borel levels	Proved with explicit continuous reductions and a separate sharp support-order bound.
Uniform observable support bound	Classical analytic boundedness specialized to coefficient families; a proof is supplied.
Fixed-interval binary localization	Explicit application of the support reduction, not advertised as a new foundational lower-bound method.
Principal-part integer part and floor	Shepherdson-type classical model; negative-infinitesimal correction also already in ProveIt.

Statement	Status in this report
Exact cancellation loci	Proved for the stated Polish finite-normal-form topology, including closed-piece continuity and zero-factor cases.
No compatible continuous addition	Direct application of Glazer after all model and Borel hypotheses have been verified.
Formal verification	No Lean proof delivered. Python checks only finite exact-rational identities, order tests, floor cases, and finite embedding data.

D.3 Finite verification artifact

The archive contains `verify.py`, a Python 3.10+ standard-library-only program, and the actual result `verification.json`. The recorded run used Python 3.13.5, deterministic seed 2601003, and passed **84,388 assertions**. These include 6,000 ring-law checks, 2,000 closure checks, 2,000 positive-cone checks, 999 discreteness checks, 7,500 floor-interval checks, 63 negative-infinitesimal boundary checks, 25 cancellation checks, 65,792 rational Kleene–Brouwer embedding checks on 256 nodes, and 9 coherence and branch-approximation checks.

The finite embedding is an initial segment of one fixed midpoint construction: nodes are enumerated by the weight $\sum_i (s_i + 1)$, and each new node is inserted between its previously assigned neighbors. This checks the direction of the order reversal and the rational band. It does not test the well-foundedness of arbitrary infinite trees.

To reproduce:

```
python verify.py --output verification.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is internal, so no BibTeX run or network access is required. The finite checks are useful for catching sign, support, and boundary mistakes; they neither prove descriptive completeness nor replace the mathematical proofs or an independent review.

[write] In this report source 03’s program is `code/03-borel-presentations-verify.py`, its recorded run data `data/03-borel-presentations-verification.json`, and its provenance record data `data/03-borel-presentations-provenance.json`; the record still describes 03’s own 23-page PDF, which is not shipped. Its manuscript is merged into this `article.tex`. The intake reran the program on a copy with Python 3.14.4: 84,388 assertions passed, and the record differed from the delivered one only in the interpreter version and line endings.

E Provenance of this report

Sources. Three manuscripts of batch 86 of the repository’s intake, all dated October 3, 2026, all AI-assisted (03: “prepared with ChatGPT”), arrived together in commit 31fdc6571 and were placed in commit 3d2177df4. Each inspected the repository at its own pin; the pins are 28 to 35 commits before the placement.

Source	Archive	Pin	Contributes
05	polish-presburger-glazer	a11efab09	base of Part I: sections 2 to 13 , its questions, sections A and 15
04	Polish_Presburger_Glazer_ProveIt	3ad5f878c	every passage marked [04] in Part I ; sections 2.3, 4.3, 4.5, 5.2, 5.4, 6.3, 7.3, 8.1, 10.1, 11.2, 11.3 and 13.4 , its questions, sections B and C
03	glazer-proveit-research	c5b4881fa	Part II except the notes and section 26 ; section D

Written for this report: [section 1](#) (except the quoted status statements), [section 13.5](#), the merged questions of [section 14.1](#), [section 26](#), the notes marked [write] or [merge], and these appendices.

Where the merge chose.

- *Base.* Source 05, for the weaker hypotheses of its Polish criterion and Δ_2^0 theorem ([section 1.2](#)).
- *Printed once.* The main theorem (05's [theorem 3.1](#), with 04's statement as [remark 3.2](#)); the division algorithm ([proposition 4.2](#) = 04's Lemma 2.2); full Presburger arithmetic of the cone ([corollary 4.11](#) = 04's Corollary 3.5); the predecessor and truncated-subtraction loci ([proposition 8.2](#) and [theorem 8.3](#) = 04's Proposition 7.4 and Theorem 7.3); the omnific realization ([theorem 9.1](#) = 04's Proposition 10.1, same proof); the no-semiring theorem and its corollary ([theorem 10.4](#) and [corollary 10.5](#) = 04's Theorem 9.3 and Corollary 9.4, whose model-specific proof coincides with 05's); the real endomorphisms ([theorem 11.1](#) = 04's Theorem 8.2).
- *Second proofs from 04.* [Proposition 4.2](#) (uniqueness by coordinates), [lemma 4.7](#) (as [theorem 4.8](#)), [theorem 4.9](#) (as [theorem 4.10](#)), [theorem 6.1](#) (as [theorems 6.3 and 6.4](#)), [theorem 7.1](#) (as [theorem 7.2](#)), [proposition 8.2](#), [theorem 8.3](#) (with the nowhere-density clause), [theorem 10.4](#), [theorem 11.1](#).
- *Both printed.* The order obstructions [theorem 7.4](#) (05) and [theorem 7.5](#) (04), whose hypotheses are incomparable ([remark 7.7](#)); the counterexamples [example 4.4](#) (05) and [proposition 4.6](#) (04); the Polish criterion [theorem 5.2](#) (05) and the transfer theorem [theorem 5.4](#) (04), which alone gives continuous division for general D .
- *Questions.* Four questions asked by both 05 and 04 are merged ([section 14.1](#)); 05's question on canonical surreal cutoffs and topologies and 04's on coefficient topologies inside surreal arithmetic overlap only partly and are printed separately. 03's eight questions are printed as delivered. Part I has seventeen questions, Part II eight.
- *Not printed separately.* 04's title page, abstract (its non-claims are in [section 1.5](#)), reading routes, Definition 2.1 (05's setup), the opening paragraph of its Section 9 (Glazer's Corollary 1, in [section 1.1](#)), and the statements and proofs listed under "printed once"; 05's and 03's title pages and abstracts (their status statements are in [section 1.5](#)). The merged abstract replaces them.
- *Renamed symbols.* Listed in [section 1.3](#); no normalization was changed.
- *Added.* [Proposition 26.1](#) and the discussion in [section 26](#); the remark [remark 7.7](#); Glazer's speculation ([section 1.1](#)), which no source mentions; the notes citing the collection's

labels in [Part II](#).

Intake checks. The intake ran the three delivered suites on copies (Python 3.14.4, standard library only): 03 passed 84,388 assertions in 11 groups, 04 passed 75,079 assertions in 13 categories, 05 passed 17,930 checks in 18 families. Their records agree with the shipped ones up to interpreter strings and line endings. The three delivered checksum manifests verified and were not shipped. It also checked by hand that $M_{\mathbb{R}}$ is a countable clopen sum of rays and a model of $\text{Th}(\mathbb{N}; 0, 1, +)$, the Polish clause of [theorem 11.2](#), [theorem 7.5](#), and the worked examples of [section A](#); no false statement was found. For this report the following references were checked against their publishers' records on October 3, 2026: Glazer's arXiv record (one version, no journal reference); Tserunyan's notes (the file at the cited address is dated November 26, 2025); Paran and Vo (*Israel J. Math.* 273 (2026), no. 2, 979–1000, online December 11, 2025); Enayat, Hamkins and Wcisło (*Fund. Math.* 256 (2022), 171–193); Jeřábek (*Math. Logic Quart.* 65 (2019), no. 1, 108–115); Haase (*ACM SIGLOG News* 5(3) (2018), 67–82); Alvir and Rossegger (published in *J. Symb. Log.* 85 (2020), 1079–1101, added here). The other classical references were not re-checked.

Correction. [Section 26](#) corrects the placement record's description of how 03's topology restricts to the real model.

F Limitations and non-claims of the sources

Every limitation stated by a source is kept. They are grouped by source; the items written for the merge come last.

Source 05.

- 05.1 The construction answers Glazer's Question 2 *as written*; independent bibliographic priority has not been established. The literature search verified the question in the source PDF and checked the arXiv record; it did not establish the absence of an earlier solution.
- 05.2 Classical Presburger elimination, the $\mathbb{Z}$ -group facts and elementary topological-group facts are not claimed as new; the lexicographic algebra is classical.
- 05.3 The manuscript is AI-assisted, not independently refereed and not Lean-verified; no new Lean or Rocq formalization is included; the proposed Lean modules are a plan, not existing files.
- 05.4 The executable tests are finite instances and stream prefixes; they do not quantify over arbitrary elements of $\mathbb{Q}^{\mathbb{N}}$, and finite prefixes do not decide equality of infinite streams.
- 05.5 No answer to Glazer's Question 1 is claimed; no uncountable Polish model of Peano arithmetic; no continuous arithmetic order on these nondiscrete spaces; no terminating decision of equality of arbitrary infinite sequences (comparison is approximated with at most one change).
- 05.6 The omnific maps are additive and order-preserving, not ring embeddings; the topologies are specified coefficient topologies, not a canonical topology on $\mathbf{Oz}$; no topology on a proper class is used; all constructions are set-sized in ZFC.

- 05.7 Neither model admits any unital semiring multiplication with the given addition and unit; nothing contradicts Glazer's obstruction for stronger arithmetic, and the no-semiring argument does not replace it.
- 05.8 The repository's integer Cooper lemmas are correct on Int; the existence of the integer eliminator does not certify the theorem for a new carrier; the original integer theorem need not be weakened or retracted.
- 05.9 The questions are proposed directions, not certified as open in the literature; the user's repository was not modified.

Source 04.

- 04.1 The construction answers Question 2 as printed; historical priority has not been established; the literature check did not establish that the construction has never appeared elsewhere.
- 04.2 The manuscript is unrefereed; no new theorem is represented as machine-checked; Lean/Rocq certification is not supplied.
- 04.3 The classical $\mathbb{Z}$ -group characterization and quantifier elimination are not claimed as new.
- 04.4 The difference between one-step periodicity and residue invariance is a porting requirement, not a reported bug in the integer proof.
- 04.5 Baire coordinates encode the arithmetic structure and are not base digits; the stream API cannot decide totality of an arbitrary oracle, and partial, negative or noninteger entries are not supported.
- 04.6 The source audit was limited to the named files at its pin; no repository files were modified and no repository-wide build was performed; the formalization stages are proposals, not assertions about existing declarations.
- 04.7 The tests are finite, exact and partial; the model-theoretic, infinite-matrix and topological statements rest on the proofs.
- 04.8 No third-party font files or paper copies are included.

Source 03.

- 03.1 A research draft, not refereed and not a Lean proof; global priority and independent correctness review are not established.
- 03.2 The repository's coanalytic support obstruction and the negative-infinitesimal floor correction are credited, not claimed as new; the integer-part model is of Shepherdson type, not new; analytic boundedness and the Hausdorff characterization are classical.
- 03.3 The full-Hahn nonpresentation results need observable coefficients and coverage of the whole field; they do not exclude unobservable Borel presentations or Borel real closed subfields.
- 03.4 Only addition is shown impossible to make continuous; Glazer's multiplication obstruction (his Theorem 2) is not transferred, since only open induction is proved for $\mathcal{A}_{\text{fin}, \geq 0}$; continuous multiplication under another compatible topology is open.

- 03.5 The zero loci are not claimed minimal in cardinality, dimension or descriptive rank; countable closed-piece continuity is not identified with the Baire-one class.
- 03.6 The dichotomy is per fixed L , not about the class of scattered orders, nor uniform in a code of L ; the Hausdorff decomposition used need not be computable; the uniform support bound need not be computable from a code.
- 03.7 No general computability over arbitrary real coefficients is claimed.
- 03.8 All results are set-sized in ZFC; no class-sized Polish topology, global choice or large cardinal is used.
- 03.9 The Python checks are finite (they do not test well-foundedness of infinite trees, completeness, or the external theorems); the formalization plan is proposed only; no repository files were changed and the whole repository was not audited.

The merge.

- M.1 The affirmative answer concerns Question 2 as printed. Glazer speculated the opposite; this report draws no conclusion about his Question 1 or about any variant of Question 2 with further hypotheses.
- M.2 [Proposition 26.1](#) and the observations of [section 26](#) are elementary consequences of the sources' theorems; they are not presented as new results about Glazer's problem.
- M.3 Citing a collection label does not certify that label: of those cited, only `odg:thm:floor` is formalized.
- M.4 Nothing in this report is formalized. The pointer from the formal project `Logic/PresburgerArithmetic` and the placement in the `Surreal` collection confer no formal status.

References

- [1] Elliot Glazer, *A Topological Tennenbaum Theorem*, arXiv:2311.13699v1, submitted November 22, 2023 (dated November 20, 2023 on its first page); one version, no journal reference as of October 3, 2026. Theorem 1 is on page 1, Theorem 2 and Corollary 1 on page 3; Questions 1 and 2 and the speculation after Question 2 are on page 8. <https://arxiv.org/abs/2311.13699>.
- [2] Ali Enayat, Joel David Hamkins, and Bartosz Wcisło, *Topological models of arithmetic*, arXiv:1808.01270; *Fundamenta Mathematicae* 256 (2022), 171–193. DOI: [10.4064/fm928-1-2021](https://doi.org/10.4064/fm928-1-2021).
- [3] Christoph Haase, *A survival guide to Presburger arithmetic*, *ACM SIGLOG News* 5(3) (2018), 67–82. DOI: [10.1145/3242953.3242964](https://doi.org/10.1145/3242953.3242964). [Author's manuscript](#).
- [4] Emil Jeřábek, *Rigid models of Presburger arithmetic*, *Mathematical Logic Quarterly* 65 (2019), no. 1, 108–115. Preprint arXiv:1803.05797v2, 2018. DOI: [10.1002/malq.201800019](https://doi.org/10.1002/malq.201800019).
- [5] Philip Ehrlich and Elliot Kaplan, *Number Systems with Simplicity Hierarchies II*, arXiv:1512.04001v1, 2015. <https://arxiv.org/abs/1512.04001>.
- [6] J. C. Shepherdson, *A non-standard model for a free variable fragment of number theory*, *Bulletin de l'Académie Polonaise des Sciences, Série des Sciences Mathématiques, Astronomiques et Physiques* 12 (1964), 79–86.
- [7] J. Glivická and P. Glivický, *Shepherdson's theorems for fragments of open induction*, arXiv:1701.02001, 2017. <https://arxiv.org/abs/1701.02001>.
- [8] A. S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics 156, Springer, 1995. <https://doi.org/10.1007/978-1-4612-4190-4>.
- [9] A. Tserunyan, *Introduction to Descriptive Set Theory*, lecture notes, version dated November 26, 2025. Particularly Sections 1, 13, and 16. https://www.math.mcgill.ca/atserunyan/Teaching_notes/dst_lectures.pdf.
- [10] R. Alvir and D. Rossegger, *The complexity of Scott sentences of scattered linear orders*, arXiv:1810.11423; *Journal of Symbolic Logic* 85 (2020), 1079–1101. See Section 1.3 for the Hausdorff hierarchy. <https://arxiv.org/abs/1810.11423>.
- [11] B. H. Neumann, *On ordered division rings*, *Transactions of the American Mathematical Society* 66 (1949), 202–252. <https://doi.org/10.1090/S0002-9947-1949-0032593-5>.
- [12] V. Puiseux, *Recherches sur les fonctions algébriques*, *Journal de Mathématiques Pures et Appliquées, Série 1*, 15 (1850), 365–480. https://www.numdam.org/item/JMPA_1850_1_15__365_0/.
- [13] E. Paran and T. N. Vo, *A skew Newton–Puiseux Theorem*, *Israel Journal of Mathematics* 273 (2026), no. 2, 979–1000; first published online December 11, 2025. The introductory classical theorem is the fact used here. <https://doi.org/10.1007/s11856-025-2859-2>.
- [14] Vladimir Reshetnikov, *ProveIt: Presburger arithmetic is decidable*, `Logic/PresburgerArithmetic/README.md`, repository snapshot `a11efab09a2d08f692a09829cb6d3440ddd1795a` (source 05's pin). [Pinned repository file](#).
- [15] Vladimir Reshetnikov, *ProveIt: the arithmetic heart of Cooper elimination*, `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Cooper.lean`, lines 1–200 inspected at source 05's pin. [Pinned source file](#).
- [16] Vladimir Reshetnikov, *ProveIt: Surreal*, `Algebra/SurrealNumbers/README.md`, lines 1–230 inspected at source 05's pin; see the formalization discussion of normal forms, omnific integers, least positivity, and standard congruences. [Pinned repository file](#).
- [17] Vladimir Reshetnikov, *ProveIt repository. Presburger arithmetic is decidable: project README*. `Path Logic/PresburgerArithmetic/README.md`, at commit `3ad5f878c26d2d82c092190af2f3edc50d6feb06` (source 04's pin). [Pinned source](#).

- [18] Vladimir Reshetnikov, Provelt repository. *Cooper.v*: the constructive integer finite-residue core. Path `Logic/PresburgerArithmetic/Coq/Cooper.v`, at source 04's pin. Inspected definitions include `periodic`, `periodic_shift`, and `periodic_has_residue`. [Pinned source](#).
- [19] Vladimir Reshetnikov, Provelt repository. *Surreal*: project README, especially the omnific normal-form and constant-term description. Path `Algebra/SurrealNumbers/README.md`, at source 04's pin. [Pinned source](#).
- [20] V. Reshetnikov, *ProveIt: Surreal-number project and research-report guides*, repository material at commit `c5b4881fa603339c7ff90f5e1098ede0651243d8` (source 03's pin). Paths: `Algebra/SurrealNumbers/README.md` and `Algebra/SurrealNumbers/docs/README.md`. <https://github.com/VladimirReshetnikov/ProveIt>. These are project descriptions, not independent certification of every listed result.
- [21] V. Reshetnikov, *Notations for Omnific Integers: Holonomic towers, shielded and rational-omega calculi, and the complexity of equality*, research-report README in ProveIt, at source 03's pin. Path: `Algebra/SurrealNumbers/docs/foundations-and-computation/omnific-notations/README.md`. The guide records prior Π_1^1 support-validity results and their provenance. <https://github.com/VladimirReshetnikov/ProveIt>.